KINEMATIC AND DYNAMIC MODELLING OF GRINDING PROCESSES

by

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## ABSTRACT

The use of abrasive tools in grinding and similar manufacturing processes continues to increase in the production of high surface quality or difficult to process materials used in especially aviation, automotive and biomedical industry. The complicated geometric structures of the abrasives used in these processes show significant differences with the tools in other machining processes. Instability in material removal operations has been one of the critical obstacles in manufacturing, hindering productivity as well as resulting in unfavorable workpiece quality. Abrasive processes are often associated with finishing operations, aimed to give workpiece a final geometry and surface condition which makes chatter even more critical in grinding. For these reasons, it is quite time-consuming, costly, and in some cases impossible to achieve the desired quality and performance with conventional trial and error methods in abrasive processes. Process models based on analytical and experimental methods constitute the aim and goal of this thesis as they can be used effectively in the analysis of these processes and in selecting the most appropriate process conditions to increase the performance of abrasive processes. In this study, a new simulation method named geometric-kinematic model has been developed for grinding. The geometric-kinematic model provides the prediction of grinding forces and surface roughness of the workpiece by simulating the micro-interactions of the abrasive particles and the workpiece surface. Using the milling analogy and the normal distribution of the individual grits on the wheel surface, determination of active grits hence the chip thickness calculation per grit is also possible. A time-domain simulation is constructed employing the regenerative effect by utilizing the dynamic chip thickness calculation. The stability regions are determined through time domain simulations and analytical model predictions. The simulation results are compared and verified by experimental data.



**Keywords:** Surface grinding, kinematic force model, stability model





## 摘要

在研磨和类似制造过程中使用磨削工具持续增加，特别是在航空、汽车和生物医学行业，用于生产高表面质量或难以加工的材料。这些过程中使用的磨削工具的复杂几何结构与其他加工过程中的工具存在显著差异。材料去除操作中的不稳定性一直是制造业的关键障碍之一，阻碍了生产效率，并导致工件质量不佳。磨削工艺通常与精加工操作相关联，旨在赋予工件最终几何形状和表面条件，这使得研磨中的振颤问题更加关键。由于这些原因，在磨削工艺中，使用传统的试错方法实现所需质量和性能既耗时又昂贵，在某些情况下甚至不可能。基于分析和实验方法的工艺模型构成了本论文的目标，因为它们可以有效地用于分析这些工艺，并选择最合适的工艺条件以提高磨削工艺的性能。在本研究中，开发了一种名为几何运动学模型的新型模拟方法用于研磨。几何运动学模型通过模拟磨削颗粒与工件表面的微观相互作用，预测磨削力和工件表面粗糙度。利用铣削类比和轮面上单个磨粒的正态分布，可以确定有效磨粒，从而计算每颗磨粒的切屑厚度。通过利用动态切屑厚度计算中的再生效应，构建了时域模拟。通过时域模拟和分析模型预测确定了稳定性区域。模拟结果通过实验数据进行比较和验证。



关键词：表面磨削，运动学力模型，稳定性模型

## ÖZET

Taşlama ve benzeri üretim süreçlerinde aşındırıcı malzemelerin kullanımı, özellikle havacılık, otomotiv ve biyomedikal endüstrisinde gerekli olan yüksek yüzey kalitesi veya işlenmesi zor malzemeler üretiminde artmaya devam etmektedir. Bu süreçlerde kullanılan aşındırıcı takımların karmaşık geometrik yapıları, diğer üretim yöntemlerine göre büyük farklılık göstermektedirler. Talaş kaldırma operasyonlarındaki kararsızlık, üretim hacmine etki eden kritik engellerden birisidir ve elverişsiz iş parçası kalitesine neden olur. Taşlama işlemleri genelde son yüzey oluşturma işlemleriyle ilişkilendirilir ve son yüzey oluşturma oluşturma işleminde iş parçası üzerinde gerçekleşicek takım tırmalası kötü bir yüzey pürüzlülüğüne neden olucaktır. Bu nedenlerden ötürü, taşlama operasyonlarında geleneksel deneme yanılma yöntemleriyle istenilen kalite ve performans elde edilmesi oldukça zaman alıcı, masraflıdır hatta bazı durumlarda mümkün değildir. Bu tezin amacı ve hedefi, analitik ve deneysel yöntemlere dayanan süreç modelleri oluşturup, bu süreçlerin analizinde etkili bir şekilde kullanılabilmek ve aşındırıcı operasyonların performansını artırmak için en uygun süreç parametrelerinin seçilip kullanılmasıdır. Bu çalışmada taşlama için geometrik-kinematik model adlı yeni bir simülasyon yöntemi geliştirilmiştir. Geometrik kinematik model, taşlama taşı üzerindeki aşındırıcı parçacıklarının ve iş parçası yüzeyinin mikro etkileşimlerini simüle ederek taşlama kuvvetlerinin ve iş parçasının yüzey pürüzlülüğünün tahminini sağlar. Freze analojisi ve taşlama taşı yüzeyinde tek tek aşındırıcı parçacıkların normal dağılımı kullanarak aktif parçaların belirlenmesi, dolayısıyla aşındırıcı başına talaş kalınlığı hesaplanması da mümkündür. Dinamik talaş kalınlığı hesaplaması ve rejeneratif etki kullanılarak bir zaman kümesi simülasyonu oluşturulmuştur. Kararlılık bölgeleri, zaman kümesi simülasyonları ve analitik model tahminleri aracılığıyla belirlenir. Simülasyon sonuçları deney veriler ile karşılaştırılarak doğrulanır.



**Anahtar Kelimeler:** Satıh taşlama, kinematik kuvvet modeli, kararlılık modeli





## ÖZET

磨削及其他生产过程中使用磨削材料，特别是在航空航天、汽车和生物医学行业中需要高表面质量或难加工材料的制造中，持续增加。这些过程中使用的磨削工具组的复杂几何结构与其他生产方法相比存在巨大差异。磨削操作中的不稳定性是影响生产规模的关键障碍之一，会导致不合适的工件质量。磨削操作通常与最终表面形成过程相关联，而在最终表面形成过程中，工具的磨削会在工件上产生不良的表面粗糙度。由于这些原因，在磨削操作中，通过传统的试错方法获得所需质量和性能非常耗时、昂贵，甚至在某些情况下不可能。本论文的目标和目的是基于分析和实验方法建立过程模型，有效地用于这些过程的分析，并选择并使用最合适的工艺参数来提高磨削操作的性能。在这项研究中，开发了一种名为几何-运动学模型的新仿真方法用于磨削。几何运动学模型通过模拟磨削工具上磨削颗粒与工件表面的微观相互作用，来预测磨削力和工件表面粗糙度。通过铣削类比和磨削工具表面上磨削颗粒的单独正态分布，可以确定有效部件，从而计算每个磨削颗粒的磨削厚度。通过动态磨削厚度计算和再生效应，建立了时间集仿真。稳定性区域通过时间集仿真和解析模型预测来确定。仿真结果通过与实验数据对比进行验证。



Anahtar Kelimeler: Satıh taşlama, kinematik kuvvet modeli, kararlılık modeli

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## 1 INTRODUCTION

Grinding is one of the many manufacturing methods and it can be considered as the oldest one. The first grinding operations in the world was done in the early ages of the society [1]. After mankind started to make tools they realized they needed to sharpen their tools to assure the durability and continue the usage without failure. First attempts were sharpening the tools by rubbing them with stones. Since then grinding is used widely around the world.

Grinding is included in abrasive processes under machining operations. Like other machining operations it includes material removal. This material removal cannot be considered as cutting and regarded as abrasive operations. Abrasive operations basically mean in the medium of contact of two materials, the harder material removes particles from the surface of the softer material. In the world, 20-25% of all expenses regarding the machining operations are coming from grinding [2]. The material removal is done with abrasive particles. These abrasive particles are bonded in various ways to the grinding wheel [3]. Grinding wheels come in different types and shapes. The type of abrasives is a key factor which includes aluminum oxide, SiC, CBN and diamond grits [4].

Abrasive technology is generally used in finishing operations. The aim is to give the workpiece the final geometry and surface profile. Mostly after all production steps are completed, the materials are ground and then sent to inspection. There are some other cases in which the grinding is used. If the workpiece material is comparably soft and easy to machine, a production engineer will most likely choose the turning or milling operations as roughing and primary shaping step but if the workpiece is a hard to machine material such as Ti, Ni or Cr alloys [5], which are mostly used in aerospace industry, the engineer would choose grinding as roughing operation. Abrasive operations are known for their increased performance on hard to machine materials [6]. First, a direct cutting method is not used in grinding and usually the depths of cut are low. This enables the abrasive wheel to form particles of the workpiece to remove them. Moreover, in grinding the cutting speeds are generally high and the heat can be absorbed by the wheel more comparing to standard cutting operations especially when



## 1 简介

磨削是众多制造方法中的一种，也可以被认为是其中最古老的方法。世界上最早的磨削操作是在社会早期完成的[1]。当人类开始制造工具时，他们意识到需要磨利工具，以确保其耐用性并继续使用而不损坏。最初的尝试是通过用石头摩擦工具来磨利它们。从那时起，磨削在世界范围内被广泛使用。

磨削被包含在切削加工的磨削工艺中。与其他切削加工一样，它包括材料去除。这种材料去除不能被视为切割，而应被视为磨削操作。磨削操作基本上是指在两种材料的接触中，较硬的材料从较软材料的表面去除颗粒。在世界范围内，20-25%的切削加工总费用来自磨削 [2]。材料去除是通过磨削颗粒完成的。这些磨削颗粒以各种方式粘合到磨削轮上[3]。磨削轮有不同类型和形状。磨料类型是一个关键因素，包括氧化铝、碳化硅、立方氮化硼和金刚石砂粒[4]。

磨削技术通常用于精加工工序。其目的是使工件获得最终几何形状和表面轮廓。通常在所有生产步骤完成后，材料会被磨削，然后送去检验。还有一些其他情况下会使用磨削。如果工件材料相对较软且易于加工，生产工程师很可能会选择车削或铣削作为粗加工和初步成型步骤，但如果工件是难加工材料，如钛、镍或铬合金[5]，（这些材料大多用于航空航天工业），工程师则会选择磨削作为粗加工步骤。磨削操作以其在难加工材料上的高效率而闻名[6]。首先，磨削不使用直接切削方法，且通常切削深度较低。这使得磨削轮能够形成工件颗粒并将其去除。此外，磨削时的切削速度通常较高，且热量可以被轮子更多地吸收，与标准切削操作相比，尤其是在

CBN wheels are used [7]. Furthermore, the grinding wheel can be more durable and the tool life is increased [8].

使用CBN砂轮[7]。此外，磨削轮可以更耐用，工具寿命也会延长[8]。

### 1.1 Literature Survey on the Mechanics of Grinding

### 1.1 研磨力学文献综述

As for other cutting operations, empirical and analytic models have been used in modeling of grinding. Until 1980s the models mostly relied on linear regressions and empirical-physical approaches [9]. Empirical models rely on equations which consist of many grinding parameters such as depth of cut, feed rate, spindle speed, workpiece velocity etc. The important parameters are identified or calibrated after doing many tests on specific conditions. These equations are used to predict process forces, surface roughness, temperature, wear and vibrations [10]. The advantage of empirical models is that they are easy to implement, can be considered shop floor friendly and sometimes practical if the production engineer will consider only a few operations for a considerable amount of time. Although these models are mostly very accurate, generation of them is very time consuming and their modularity is very low. In case of any changes such as. change of tool, workpiece or any process parameter, the model will most probably be useless new sets of experiments will have to be conducted.

至于其他切削加工，经验模型和分析模型已被用于研磨建模。直到20世纪80年代，这些模型主要依赖线性回归和经验-物理方法[9]。经验模型基于包含许多研磨参数（如切削深度、进给速度、主轴转速、工件速度等）的方程。在特定条件下进行大量试验后，会识别或校准重要参数。这些方程用于预测工艺力、表面粗糙度、温度、磨损和振动[10]。经验模型的优势在于易于实现，可被视为车间友好型，并且如果生产工程师只考虑少数操作相当长的时间，有时也很实用。尽管这些模型大多非常准确，但它们的生成非常耗时，模块化程度很低。在任何更改的情况下，例如刀具、工件或任何工艺参数的变化，模型很可能变得无用，必须进行新的实验集。

Grinding process is very hard to model analytically due to its stochastic nature [11]. The grits on the wheel come in random shapes and sizes which means that the cutting edge is not defined. In reality, everything is in a constant state of change in grinding as during the operation the wear may become so rapid that the process parameters may change even in the first passes of grinding [12]. However, through some simplifications and assumptions one can model the grinding process analytically as can be found in the keynote paper by Tönshoff et al. [13].

磨削过程由于其随机性，很难进行解析建模 [11]。砂轮上的磨粒形状和大小都是随机的，这意味着切削刃没有明确的定义。实际上，磨削过程中一切都在不断变化，因为在操作中磨损可能变得非常快，以至于磨削过程参数甚至在最初的几次磨削中就会发生变化 [12]。然而，通过一些简化和假设，人们可以解析地建模磨削过程，正如Tönshoff等人发表的主旨论文中所发现的 [13]。

The analytical modelling initially requires a definition of geometric properties on the contact surfaces of the grinding wheel and the workpiece. The identification process of cutting geometry relies on the measurements done on the grinding wheel before the grinding process. These measurements are crucial as unlike turning or milling, the abrasive grits on the grinding wheel don’t have well defined cutting edges. The geometric properties of the wheel can be categorized into two, namely the macro and micro properties [14]. The macro properties mostly related to the unbalance and run-

解析建模最初需要定义砂轮与工件接触表面的几何特性。切削几何的识别过程依赖于磨削前对砂轮进行的测量。这些测量至关重要，因为与车削或铣削不同，砂轮上的磨粒没有明确的切削刃。砂轮的几何特性可分为两类，即宏观特性和微观特性[14]。宏观特性主要与不平衡、运行、夹紧误差或过度磨损相关。另一方面，砂轮的微观特性可以列为磨粒的密度、分布、形状等。

out, clamping errors or excessive wear. The micro properties of the wheel on the other hand can be listed as density, distribution, shape etc. of the grits.

The measurements of the wheel regarding its macro properties can be done straightforwardly with tactile methods. In [15], a tungsten carbide tip coupled with a oscillating pin is applied on the grinding wheel surface. Any defects made on the surface can be successfully observed with this method if small forces and oscillations are applied. Buchholz [16] used pneumatic sensors to measure the macro features. The concept is the same as the method described in [15]. The pneumatic sensor is coupled with a pressure plate where constant air pressure is applied on the plate and motions are observed. Moreover, it is also possible to measure the macro structure with AE sensors [17]. These sensors provide a non-contact method with an increased sample rate. Again, the sensors need to be coupled with a single-point dresser tool.

The measurements on the wheel surface regarding the micro sense first tried with inductive wheel loading sensors [18]. This is a thermoelectrical method and requires very complex test setup. A winding, core and a stray field are used with a magnetic head. Magnetic head records the electrical activity on the surface of the wheel. Furthermore, the other way of identifying micro structure of the wheel is using light sensors. The first attempt was done by Piegert [19]. This is a scattered light-sensor and works with a signal processing module. Nowadays with the advancement of measurement technologies, the reflection and laser sensors are used to determine the grit properties on the wheel surface [14].

After the observations and identifications of basic grit properties, the next step is to create a wheel topography model to use it in the analytical process models.

Generally, the individual grits are carefully analyzed, and the geometric properties of the grits are identified. The results are to be used in the computer simulations of the grinding process [20,21]. The surface of the grinding wheel is scanned, and the geometric properties are determined for grit density, height, width, rake angle, oblique angle, edge radius and distance between grits. A straightforward way is to model the grits is to assume that they are in a simple geometrical shape such as a sphere, a cone or sometimes more complex shapes such as cuboids or tetrahedrals [22,24]. The other way, which is computationally more expensive, is to model each single grit in the simulation so that each geometric property stored in arrays of the data rather than



出现，夹紧误差或过度磨损。另一方面，车轮的微观特性可以列为磨粒的密度、分布、形状等。

关于车轮的宏观特性测量，可以通过触觉方法直接进行。在[15], 中，将碳化钨触头与振荡销一起施加在磨轮表面上。如果施加较小的力和振荡，这种方法可以成功观察到表面上的任何缺陷。Buchholz [16] 使用气动传感器来测量宏观特征。其概念与[15]中描述的方法相同。气动传感器与一个压力板耦合，该板上施加恒定的气压，并观察运动。此外，还可以使用AE传感器[17]来测量宏观结构。这些传感器提供一种非接触式方法，并具有更高的采样率。同样，传感器需要与单点修整工具耦合。

关于车轮表面的微观特性测量，最初尝试使用感应轮加载传感器[18]。这是一种热电方法，需要非常复杂的测试设置。使用绕组、铁芯和杂散磁场，配合磁头。磁头记录车轮表面的电活动。此外，识别车轮微观结构的其他方法是使用光传感器。Piegert [19]首次进行了尝试。这是一种散射光传感器，配合信号处理模块工作。如今随着测量技术的进步，使用反射和激光传感器来确定车轮表面的砂粒特性[14]。

在完成基本磨料特性的观测和识别后，下一步是创建一个轮廓模型，用于分析过程模型。

通常，会仔细分析单个砂粒，并识别砂粒的几何特性。这些结果将用于砂削过程的计算机模拟 [20,21]。砂轮表面被扫描，并根据砂粒密度、高度、宽度、前角、倾斜角、边缘半径和砂粒间距确定其几何特性。一种直接的方法是假设砂粒呈简单的几何形状，例如球体、圆锥，有时甚至是更复杂的形状，如长方体或四面体 [22,24]。另一种方法虽然计算成本更高，但会在模拟中为每个单个砂粒建模，以便将每个几何特性存储在数据数组中，而不是应用一个通用的形状。在砂轮表面上识别所有砂粒是一项费力的挑战，但可以通过使用从较小砂粒测量集中获得的随机分布来简单地建模砂粒。随机分布的选择在文献中进行了讨论和应用 [20,21]。

applying one general shape. The identification of all grits on the wheel surface is a tiresome challenge, but one can simply model the grits using random distribution that is obtained from the measurements of smaller set of grits. The selection of random distribution is discussed and applied in the literature [25,28].

After the wheel topography is modeled, observing the kinematics of the grinding process is one of the most effective methods of modelling the chip removal mechanism. In the literature there are many works about the grinding kinematics. McDonald et al. [29] developed a computationally cheap method which involves determining peak surface values of the grits. The data is gathered through 3D measurements of the wheel surface. The active grits are also calculated through the kinematic simulations of the grits with the interaction on the workpiece surface. Barrenetxea et al. [30] developed a model to predict the tribo-thermo-mechanical behavior of the grinding process. The idea was to apply the controlled kinematic chip removal through a time domain simulation method. The method of creating a standardization of geometrical shapes of the grits is applied in [31]. Some elemental grit shapes are chosen as cuboids, triangle prisms, triangular pyramid variants and dodecahedrons, and cutting edges are generated by slicing with planes on the predefined elemental grit shapes to represent variety of grits on the wheel surface. Orthogonal cross section of the trajectories of the grits on the workpiece is created to calculate the chip thickness in the process. Uhlmann et al. [32] proposed a variety of methods to model the grinding kinematics for different types of abrasive operations such as conventional surface grinding, oscillation surface grinding and tilt surface grinding.

Surface roughness is another key factor in grinding as it is generally used in finishing operations. The final geometry and dimensions are crucial factors in abrasive processes. In [33] a simplified predictive surface roughness model is developed in which the stochastic nature of the grinding is considered. The prediction is verified by using the found relationship between the roughness and chip thickness. In [34], the effect of process parameters on surface roughness is analyzed. The influence of grinding wheel parameters such as grain density, grain size or process parameters such as rotational speed, feed rate and depth of cut are analyzed on the grinding force. An optimization of input parameters is proposed for the increased workpiece surface roughness. Single-grit wheel experiments are conducted by Ma et al. [35] and showed that the grit-wise effect



应用一个通用形状。在砂轮表面上识别所有砂粒是一项费力的挑战，但可以通过使用从较小砂粒测量集中获得的随机分布来简单地建模砂粒。随机分布的选择在文献中进行了讨论和应用[25,28]。

在轮磨削形貌建模后，观察磨削过程的运动学是模拟切屑去除机制最有效的方法之一。在文献中有很多关于磨削运动学的研究。McDonald 等人[29] 开发了一种计算成本较低的方法，该方法涉及确定磨粒的峰值表面值。数据通过轮磨削表面的 3D 测量获得。通过磨粒与工件表面相互作用的运动学模拟，也计算了主动磨粒。Barrenetxea 等人 [30] 开发了一个模型来预测磨削过程的摩擦热力学行为。该想法是通过时域模拟方法应用受控的运动学切屑去除。在 [31] 中应用了创建磨粒几何形状标准化的方法。选择了一些基本磨粒形状作为长方体、三角棱柱、三角锥变体和十二面体，并通过在预定义的基本磨粒形状上用平面切割来生成切削刃，以表示轮磨削表面上的各种磨粒。创建了磨粒在工件上轨迹的正交截面，以计算过程中的切屑厚度。Uhlmann 等人[32] 提出了多种方法来模拟不同类型的磨削运动学，例如常规表面磨削、振荡表面磨削和倾斜表面磨削。

表面粗糙度是磨削过程中的另一个关键因素，因为它通常用于精加工操作。最终几何形状和尺寸是磨削过程中的关键因素。在 [33] 中，开发了一个简化的预测表面粗糙度模型，其中考虑了磨削的随机性。通过使用粗糙度与切屑厚度之间的关系进行验证。在 [34]，中，分析了工艺参数对表面粗糙度的影响。分析了磨削轮参数（如颗粒密度、颗粒尺寸）和工艺参数（如转速、进给速度和切削深度）对磨削力的影响。提出了优化输入参数以提高工件表面粗糙度的方法。Ma 等人进行了单粒度轮实验 [35] 并表明，颗粒对表面粗糙度的影响基于论文中描述的两个角度，即倾斜角和偏转角。

on surface roughness is based on the two angles described in the paper as inclining and deflecting angles.

对表面粗糙度的影响基于论文中描述的两个角度，即倾斜角和偏转角。

Grinding force is a key factor to determine the strategy for the process. Force can predict if the operation can be done under the machine capabilities. It can lead to prediction of surface roughness, temperature, vibrations, wear etc. Empirical force models are applied on the literature [36,37]. These early empirical models consist of mathematical relations between input and output parameters. This means that the grinding force is expressed with respect to the cutting conditions such as grinding depth, spindle speed, feed rate with calibrations constants. Even though these empirical models can predict grinding forces accurately, they rely on high number of time consuming tests. Malkin [38] proposed a model consisting of specific energy constant which depends on chip formation, ploughing and sliding forces. Single grit models are proposed to closely study the grit formation and material properties of the workpiece. This way the force can be modeled with respect to strain and strain rate hardening of the workpiece material [39]. An analytical model was presented in [40] to predict the surface roughness and force by using Johnson-Cook material and dual zone contact models.

磨削力是决定工艺策略的关键因素。力可以预测操作是否在机床能力范围内完成。它还可以预测表面粗糙度、温度、振动、磨损等。经验力模型应用于文献[36,37]。这些早期的经验模型由输入和输出参数之间的数学关系组成。这意味着磨削力是相对于切削条件（如磨削深度、主轴转速、进给速率）以及校准常数来表达的。尽管这些经验模型可以准确预测磨削力，但它们依赖于大量耗时测试。
Malkin [38] 提出了一种包含比能常数（该常数取决于切屑形成、切削力和滑动力）的模型。提出了单砂粒模型，以密切研究砂粒形成和工作材料的材料特性。通过这种方式，力可以根据工件材料的应变和应变率硬化进行建模[39]。在[40]中提出了一种解析模型，通过使用Johnson-Cook材料和双区接触模型来预测表面粗糙度和力。

## 1.2 Literature Survey on the Dynamics of Grinding

## 1.2 研磨动力学文献综述

Instability in material removal operations has been one of the critical obstacles in manufacturing, hindering productivity as well as resulting in unfavorable workpiece quality. Abrasive processes are often associated with finishing operations, aimed to give workpiece a final geometry and surface condition which makes chatter even more critical in grinding.

材料去除过程中的不稳定性一直是制造业的关键障碍之一，它阻碍了生产效率，并导致工件质量不佳。磨削等研磨工艺通常与精加工操作相关联，旨在赋予工件最终几何形状和表面条件，这使得颤振在研磨中更加关键。

As a process related self- excitations are one of the significant problems in grinding operations because the geometrical accuracy and the surface finish are the crucial anticipations of the process [41], several models have been developed to cope with this problem.

作为一种与过程相关的自激振动是研磨操作中的一个重要问题，因为几何精度和表面光洁度是过程[41], 的关键预期，因此已经开发了多种模型来应对这个问题。

The vibration phenomenon of the grinding has been carefully investigated and distinguished into three categories; self-excited vibrations due to workpiece surface

磨削的振动现象已被仔细研究和区分成三类；由于工件表面引起的自激振动

regeneration, wheel surface regeneration and forced vibrations. Even though the contribution of the wheel regeneration can be considerable on conventional wheels, majority of the vibration originate from the workpiece surface regeneration since the wear is relatively negligible on CBN wheels [42].

磨削的振动现象已被仔细研究并分为三类：自激振动、工件表面再生引起的振动和砂轮表面再生引起的振动；即使砂轮再生的贡献在传统砂轮上可能相当可观，但大多数振动起源于工件表面再生，因为CBN砂轮上的磨损相对可以忽略不计[42]。

A time-domain model for the surface grinding is presented in [43] whereas dynamic simulation is constructed to see the workpiece-wheel interaction in an enhanced way. The workpiece is discretized into elements by using the z-buffer approach. This approach provides more advanced way to observe the volume engaged by the wheel and thus modelling the equivalent chip thickness removed from the workpiece. By pairing this dynamic material engagement mechanism with a specific-energy based force model, the time-domain simulations are constructed with a Simulink model.

在[43] 中，介绍了一种用于表面磨削的时域模型，同时构建动态仿真以更清晰地观察工件-砂轮的相互作用。工件通过使用z缓冲方法离散化为多个元素。这种方法提供了更先进的方式来观察砂轮接触的体积，从而对从工件上去除的等效切屑厚度进行建模。通过将这种动态材料接触机制与基于比能的力模型相结合，使用Simulink模型构建了时域仿真。

Additional time-domain dynamic model is introduced in [44]. The work focuses on the simulation of the cylindrical plunge grinding governed by common grinding parameters. A predictive model for chatter boundaries is developed together with the calculation of the growth rates. Some crucial effects are considered, mostly non-linear attributes such as distributed forces along the contact length, tool and workpiece vibration and the delay between the consecutive time discretization. The simulation program is able to model the dynamic forces, vibrations, stability regions and the surface profiles of the workpiece.

在[44]中，介绍了额外的时域动态模型。研究重点在于模拟由常见磨削参数控制的圆柱 plunging 磨削过程。开发了一种预测颤振边界的模型，并计算了增长速率。考虑了一些关键效应，主要是非线性属性，如沿接触长度的分布力、工具和工件的振动以及连续时间离散化之间的延迟。仿真程序能够模拟动态力、振动、稳定性区域以及工件的表面轮廓。

The effect of the stochastic nature of the grinding on the dynamic response of the process is discussed in [45] where excitation is calculated by distributed individual grits. The emphasis is put on the grinding contact especially the distribution of grits and the unbalance of the wheel. It is presented that the time domain simulations are essential for modelling the nonlinearities in grinding as the operation itself is considered highly complicated

研磨的随机性对过程动态响应的影响在 [45] 中讨论，其中通过分布的独立磨粒计算激励。重点放在研磨接触上，特别是磨粒的分布和轮子的不平衡。提出时域仿真对于建模研磨中的非线性至关重要，因为该操作本身被认为是高度复杂的

The employment of milling analogy while modeling grinding operations is used carefully in [40] where a surface roughness and thermo-mechanical force models are developed. Instead of a cutter with defined geometry, numerous grits are created with randomly distributed shapes. Each grit is treated separately, all the information is stored in an array and the theory and formulations are repeated for each grit. Applying the

在模拟磨削操作时，采用铣削类比的方法在[40] 中得到了谨慎的应用，其中开发了对表面粗糙度和热力-机械力模型。与具有定义几何形状的刀具不同，会创建大量形状随机分布的砂粒。每个砂粒都被单独处理，所有信息都存储在数组中，并且针对每个砂粒重复理论和公式。应用了


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Johnson-Cook Material model and dual zone contact model presented in [46] the grinding forces can be predicted with low discrepancies relative to the experimental outputs.

It is denoted in [42], that the types of vibrations in the grinding process can be categorized into two: forced vibration and self-excited vibration. The main cause for the forced vibrations is the unbalance of the grinding wheels. In such cases the source can be hydraulic devices or any other external source as well. In this type of vibrations, the source can easily be located through frequency measurement. It is useful to mention that the gyroscopically induced self-excited vibrations are also studied in the literature [47]. However, the self-excited vibrations are mainly due to the regenerative chatter in. The regenerative effect in abrasive processes is similar to other cutting processes: During the cut, as a result of the phase between the vibration waves in the successive passes the chip thickness will be varying. The phase shift between these waves will determine the stability of the process based on the chip thickness variation.

However, in grinding processes the source of regeneration mechanism is based on two effects. The regeneration may be effective on either the wheel or the workpiece surface [42]. This part makes the stability analysis of the grinding more complex compared to cutting processes. The regenerative effect on the workpiece surface is related to the surface quality and the effect on the wheel is related to the wear resistance. The wheel regeneration propagates slower than the workpiece regeneration. The tool regeneration can be seen even in the dressing process. These two regenerations can be modeled with double delay systems.

Altintas and Weck [48] mentions Inasaki's model [42] by applying "n" intersection areas on the surface of the workpiece rather than just one. The paper investigates the limiting phase criterion which is derived from the Nyquist stability criterion. It is stated that when the phase of the transfer function goes below the limiting phase curve the system will have the means to start a regenerative chatter. If the phase curve cuts the limit curve in the first section, there is the risk of workpiece induced chatter. If the intersection occurs in the second section of the limit curve only, the chatter is caused by the side of the wheel.



[46] 中提出的Johnson-Cook材料模型和双区接触模型，磨削力可以与实验输出结果相比，预测出较低的偏差。

在 [42]，中表明，磨削过程中的振动类型可分为两种：强制振动和自激振动。强制振动的主要原因是砂轮的不平衡。在这种情况下，振动源可以是液压设备或其他外部源。对于这种类型的振动，可以通过频率测量轻松定位振动源。值得一提的是，文献中也研究了陀螺引起的自激振动[47]。然而，自激振动主要是由再生颤振引起的。磨削过程中的再生效应与其他切削过程相似：在切削过程中，由于连续进给中振动波之间的相位差，切屑厚度将发生变化。这些波之间的相位差将根据切屑厚度的变化决定工艺的稳定性。

然而，在磨削过程中，再生机制的产生基于两种效应。再生作用可能对砂轮或工件表面[42]有效。这使得与切削过程相比，磨削的稳定性分析更为复杂。工件表面的再生效应与表面质量有关，而砂轮的再生效应与耐磨性有关。砂轮的再生传播速度比工件的再生慢。即使在修整过程中也能看到刀具的再生。这两种再生可以用双延迟系统进行建模。

Altintas 和 Weck [48] 提到了 Inasaki 的模型 [42]，他们通过在工件表面应用“n”个交点区域，而不是仅一个，来应用该模型。该论文研究了从奈奎斯特稳定性准则推导出的极限相位准则。文中指出，当传递函数的相位低于极限相位曲线时，系统将具备启动再生颤振的条件。如果相位曲线在第一象限与极限曲线相交，则存在工件诱导颤振的风险。如果交点仅发生在极限曲线的第二象限，则颤振是由砂轮侧引起的。

Double-delay systems are developed to investigate the wear on the wheel and the workpiece further [49]. The dynamic model includes the two inherent delayed force fluctuations caused by the both surfaces due to regeneration mechanism of the both surfaces. The inspection of the double regenerative grinding process is done by handling the differential equations written for both the undulations due to the wheel surface and the workpiece surface. After the development of the model, a sensitivity analysis is proposed to detect the chatter of the doubly inherent structure. With this sensitivity analysis the change of parameters in an optimal way is proposed to cope with the instability in grinding operations.

开发了双延迟系统，以进一步研究砂轮和工件的磨损 [49]。动态模型包括由于两个表面的再生机制而产生的两个固有延迟力波动。通过处理为砂轮表面和工件表面振动的微分方程，对双再生磨削过程进行了检查。在模型开发后，提出了一种敏感性分析，以检测双重固有结构的颤振。通过这种敏感性分析，提出了以最佳方式改变参数的建议，以应对磨削操作中的不稳定性。

In transverse grinding, the wheel moves along a slender workpiece introducing unique grinding dynamics. The work presented in [50] offers a new dynamic modelling for this unique interaction in grinding. The plunge grinding operations are relatively less complex as they contain one region of contact. On the other hand, the transverse dynamics must be analyzed in two different contact regions.

在横向磨削中，砂轮沿细长工件移动，引入了独特的磨削动力学。[50] 中提出的工作提供了一种针对这种独特磨削交互作用的新型动态建模。浅磨削操作相对简单，因为它们只包含一个接触区域。另一方面，横向动力学必须在两个不同的接触区域进行分析。

## 1.3 Objective & Scope

## 1.3 目标与范围

In this work, the main aim of the research is to increase the performance of grinding operations by use of mathematical models, develop measurement and monitoring methods and select the best strategy for grinding, concerning the surface roughness, maximum MRR and product quality.

在本工作中，研究的主要目的是通过使用数学模型来提高磨削操作的性能，开发测量和监控方法，并针对表面粗糙度、最大MRR和产品质量选择最佳的磨削策略。

The scope of this thesis is to develop measurement methods for grinding tools, models to mimic the grinding operation down to particle level, model the process forces both for the static and dynamic case and speculate chatter free grinding conditions all which can be achieved by implementing analytical models.

本论文的范围是开发磨削工具的测量方法，建立模拟磨削操作的模型直至颗粒级，模拟静态和动态情况下的过程力，并推测无振颤磨削条件，所有这些都可以通过实现分析模型来达到。

Abrasive machining has mostly been used for finishing operations, but since the aviation industry brought the requirement of using difficult to cut materials, grinding came to use for roughing operations due to its increased performance on these types of materials. The complicated geometric structures of the abrasives used in these processes show significant differences from the other machining processes. For these reasons, it is quite time-consuming and expensive to sustain a decent result and

磨削加工主要应用于精加工工序，但自从航空航天工业提出使用难加工材料的需要以来，由于磨削在这些材料上的性能提升，它逐渐被用于粗加工工序。这些过程中使用的磨料的复杂几何结构与其他加工过程存在显著差异。因此，通过经验建模方法在磨削过程中维持良好的结果和性能非常耗时且昂贵。基于解析和实验方法的工艺模型构成了本项目的目标和宗旨，因为它们可以有效地用于这些过程的分析，并选择最合适的工艺条件。在本论文的范围内，通过实验和解析方法详细研究了磨料几何形状、磨削运动学、力和振动，并开发了工艺模型。通过这种方法，所需的实验数量可以减少到最低，并能做出高度精确的估计。此外，开发和实验验证的工艺模型被用于确定实现预期质量所需的最短时间和最低成本的条件，并在工业部件上进行了应用演示。

performance with empirical modelling methods in grinding processes. Process models based on analytical and experimental methods constitute the aim and goal of this project as they can be used effectively in the analysis of these processes and in selecting the most appropriate process conditions. Within the scope of the thesis, abrasive grit geometry, grinding kinematics, forces and vibrations were studied in detail by experimental and analytical methods and process models were developed. With this approach, the required number of experiments can be reduced to the lowest and highly accurate estimations can be made. In addition, developed and experimentally validated process models were used to determine the process conditions required to achieve the desired quality at the lowest possible time and cost, and applications on industrial components were demonstrated.

In order to achieve the described aims of the thesis, a measurement method and procedure for the abrasive grits has been developed. Using the measurements, the abrasive wheel topography has been modeled. Applying milling analogy and the geometrical random distribution of the grits on the grinding wheel, the simulation can predict the dynamic forces and vibrations. A geometric-kinematic model is developed in order to predict the grit trajectories during the grinding process. The active grits are calculated and the chip thickness per active grit is computed with the trajectory simulation of the wheel. The dynamic model is created with the inclusion of the regenerative effect of the workpiece surface. The aim of this model is to model the dynamic behavior of the surface grinding and predict the region of grinding parameters which cause instability in the closed loop time domain system. Applying milling cutter analogy in grinding allows the use of the dynamic milling equations presented in [51,52]. These models will be used to plan and schedule the best available performance of grinding operations based on the required process conditions.

## 1.4 Lab equipment

In this subtitle, the lab equipment that is used during this thesis is presented.



磨削加工主要应用于精加工工序，但自从航空航天工业提出使用难加工材料的需要以来，由于磨削在这些材料上的性能提升，它逐渐被用于粗加工工序。这些过程中使用的磨料的复杂几何结构与其他加工过程存在显著差异。因此，通过经验建模方法在磨削过程中维持良好的结果和性能非常耗时且昂贵。基于解析和实验方法的工艺模型构成了本项目的目标和宗旨，因为它们可以有效地用于这些过程的分析，并选择最合适的工艺条件。在本论文的范围内，通过实验和解析方法详细研究了磨料几何形状、磨削运动学、力和振动，并开发了工艺模型。通过这种方法，所需的实验数量可以减少到最低，并能做出高度精确的估计。此外，开发和实验验证的工艺模型被用于确定实现预期质量所需的最短时间和最低成本的条件，并在工业部件上进行了应用演示。

为了实现论文所述的目标，已开发出一种用于磨料颗粒的测量方法和流程。利用这些测量数据，磨轮表面形貌已被建模。通过应用铣削类比和磨料在砂轮上的几何随机分布，该模拟可以预测动态力和振动。开发了一种几何-运动学模型，用于预测磨削过程中磨料颗粒的运动轨迹。计算了有效磨料颗粒，并通过砂轮的轨迹模拟计算了每个有效磨料颗粒的切屑厚度。动态模型在考虑工件表面再生效应的情况下创建。该模型的目的是模拟表面磨削的动态行为，并预测在闭环时域系统中导致不稳定的磨削参数区域。通过应用磨削中的铣刀类比，可以使用[51,52]中提出的动态铣削方程。这些模型将用于根据所需的工艺条件，规划和调度磨削操作的最佳性能。

## 1.4 实验室设备

在本字幕中，展示了本论文实验期间使用的实验室设备。

### 1.4.1 Grinding CNC Machine

The grinding CNC machine used in the experiments is Chevalier Smart B818 CNC Profile & Forming Grinding CNC machine. %5 oil based coolants are used in the experiments. The machine is shown in Figure 1.




Figure 1: Grinding CNC Machine

The machine is a 3 axis CNC capable of moving in x, y and z axes. It is sufficient for academic use and tests where mass production is not the main concern. The dynamometer couple is placed on the magnetic bed for force measurements during grinding tests. A couple of fixtures are produced in order to enable the use of different shaped of workpieces to be clamped on the dynamometer. There are 2 spindles on the machine with low and high-speed capabilities. The high-speed spindle can reach 40000 rpm. Where the low speed one has the maximum speed of 8000 rpm. The bearings inside the spindle will lose their efficiency and there is a chance to break when the spindle is used below the minimum speed. Furthermore, the spindle will lose its torque and power efficiency as the speed gets slower. It is advised that the speed should be chosen between the 70% of its max speed to 100% of its maximum speed as the torque and power curves gets low in the lower speeds. Since it is high speed spindle, accordingly smaller grinding wheels should be used.  For conventional wheels at full rpm it is recommended to use maximum 20mm diameter wheels and for CBN wheels



1.4.1 磨削CNC机床

实验中使用的磨削CNC机床是Chevalier Smart B818 CNC轮廓与成型磨削CNC机床。实验中使用了%5油基冷却液。该机床如图1所示。




图1：磨削CNC机床

这台机器是一款3轴CNC机床，能够在x、y和z轴上移动。它足以满足学术使用和测试的需求，而无需考虑大规模生产。在磨削测试中，测力计耦合装置放置在磁力工作台上进行力测量。为了能够将不同形状的工作件夹持在测力计上，制作了几个夹具。这台机器上有两个主轴，分别具有低速和高速能力。高速主轴的最高转速可达40000 rpm，而低速主轴的最大转速为8000 rpm。当主轴在最低转速以下使用时，内部的轴承会降低效率，甚至有断裂的风险。此外，随着转速变慢，主轴的扭矩和功率效率也会下降。建议主轴转速应选择在其最大转速的70%到100%之间，因为在低速时扭矩和功率曲线会变得较低。由于这是高速主轴，因此应使用较小的磨削砂轮。对于常规砂轮，在满转速时建议使用最大直径为20mm的砂轮，而对于CBN砂轮

the wheel diameter can go up to maximum of 30mm diameter to satisfy the required grinding speed. For larger tools it is recommended to decrease the spindle speed rather than 40k to maintain a cutting speed of around 35m/s.

The machine has a dresser unit installed next to the magnetic table and it is ready to be used if a dresser wheel is purchased. It is noticed during the experiments that the dresser unit creates unpredictable noise signals even when it is not running. These noises were observed on the thermocouple measurements as thermocouples are highly sensitive to magnetic current and the temperature measurement was not very successful. Due to these problems, the dresser unit was unplugged from the PLC board of the machine during this study and thus not used in the experiments.

### 1.4.2 *μ*surf Explorer Nanofocus

Nanofocus is a light sourced microscope which is used in measuring of the grinding wheels. There are a number of lenses available for the measurement of individual grits or multiple of them. The wheel measurement is the fundamental part of modelling abrasive operations as the grinding mechanism is quite complex compared to other machining operations. The measurements in nanofocus is done by light fringes. The discretization is done along z axis layer by layer such as the way the rapid prototyping works. The light reflections may be problematic on conventional grinding tools as generally the aluminum and SiC abrasive grits may shine too much and introduce noise to the measurement. A coating procedure may be applied to increase the measurement quality in that case. On the other hand, measurements done on CBN tools can be conducted easily. The Nanofocus comes with a software which enables 3D scanning of the grits. The device is shown in Figure 2.



砂轮直径可以增加到最大30mm，以满足所需的磨削速度。对于较大的工具，建议降低主轴转速，而不是保持在40k，以维持约35m/s的切削速度。

机器旁边安装了梳妆台单元，如果购买梳妆台轮子，就可以使用。实验中发现，即使梳妆台单元不运行，也会产生不可预测的噪声信号。这些噪声在热电偶测量中观察到，因为热电偶对磁电流高度敏感，温度测量并不成功。由于这些问题，在本研究中，梳妆台单元被从机器的PLC板上拔下，因此未在实验中使用。

### 1.4.2 *μ*surf 探针纳米聚焦

Nanofocus 是一种用于测量砂轮的光源显微镜。有多种镜头可用于测量单个磨料颗粒或多个颗粒。砂轮测量是建模磨削过程的基础部分，因为与其他加工操作相比，磨削机制相当复杂。纳米聚焦测量是通过光条纹完成的。离散化是沿 z 轴逐层进行的，就像快速原型制造的方式一样。在传统磨削工具上，光反射可能存在问题，因为通常铝和 SiC 磨料颗粒可能会过于闪耀，从而引入测量噪声。在这种情况下，可以应用涂层工艺来提高测量质量。另一方面，在 CBN 工具上的测量可以轻松进行。Nanofocus 配备了能够实现磨料颗粒 3D 扫描的软件。该设备如图 2 所示。



Figure 2: Nanofocus

It is recommended to use 2 types of lenses on grinding wheels, namely 20x and 50x lenses. The 20x lens can cover an area of approximately 0.64mm$^2$ which will probably include around 30 abrasive grits. where a sample picture is shown in Figure 3. This lens is used to count the grits and determine the grit density; one of the key elements in process modeling. Red regions indicate the light scan.

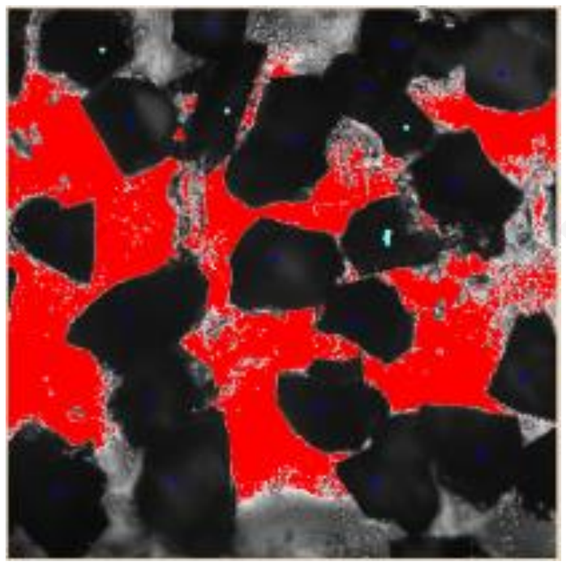


Figure 3: 20x measurement on Nanofocus





图2：纳米聚焦

建议在砂轮上使用 2 种镜头，即 20x 和 50x 镜头。20x 镜头可以覆盖约 0.64mm$^2$ 的区域，可能包含约 30 个磨料颗粒，其中样本图片如图 3 所示。该镜头用于计数磨料颗粒并确定磨料密度；这是过程建模中的一个关键要素。红色区域表示光扫描。

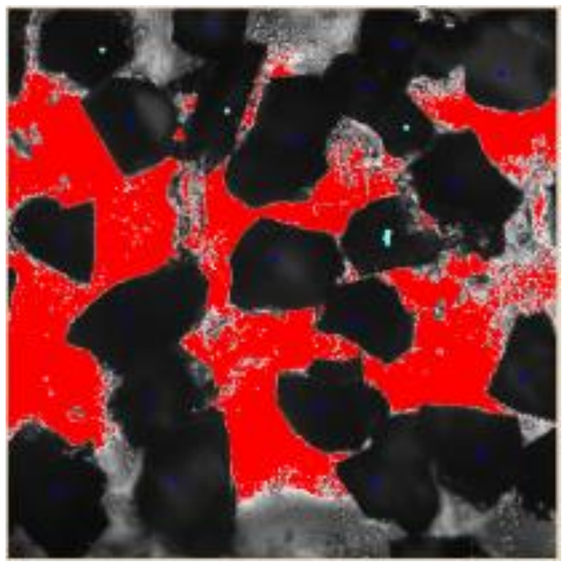


图3：纳米聚焦20x测量

The 50x lens, on the other hand, provides just one grit per one scan most of the time. However, this enables a closer look to the abrasive grit offering identification of cutting edge of the grits and some certain process inputs such as oblique angle, rake angle, edge radius etc. A sample of the 50x lens scan is shown in Figure 4.

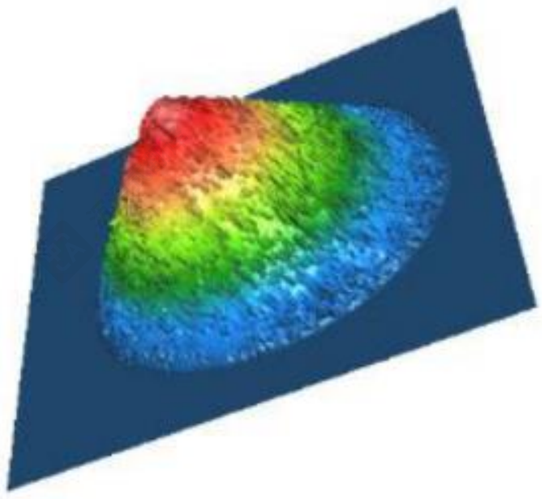


Figure 4: 50x lens capture

### 1.4.3 Kistler Dynamometer

A 3-axis Kistler 9129AA table type dynamometer is used in the experiments. The focus is given on the forces of z and x axes. Although the forces in the y-direction exist, they are mostly neglected in grinding operations as they are negligibly low.

### 1.4.4 Talysurf Surface Profilometer

For surface roughness measurements on the workpiece, a Talysurf surface profilometer was used. The device has a diamond probe $2\mu$ and provides a sufficient measuring sensitivity for grinding operations. Although the accuracy is quite high with this device,



另一方面，50倍镜头每次扫描通常只提供一种磨料粒度。然而，这能够更近距离地观察磨料粒度，从而识别粒度的切削刃以及某些特定的工艺输入，如倾斜角、前角、边缘半径等。50倍镜头扫描的样品如图4所示。

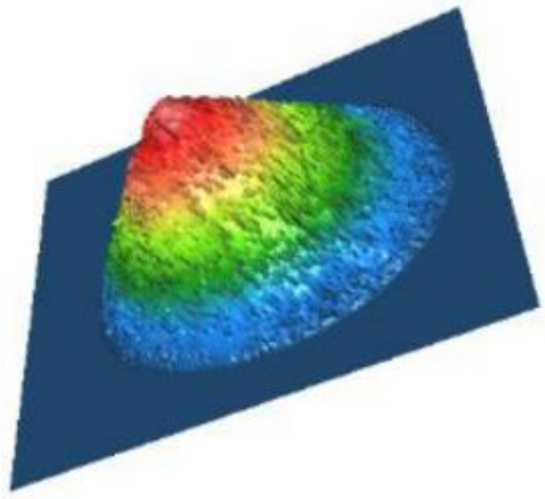


图4：50倍镜头拍摄

### 1.4.3 Kistler测力计

实验中使用了三轴Kistler 9129AA台式测力计。重点在于z轴和x轴的力。虽然y方向存在力，但在磨削操作中通常被忽略，因为它们可以忽略不计。

### 1.4.4 Talysurf表面轮廓仪

在工件表面粗糙度测量中，使用了泰勒斯普表面轮廓仪。该设备配备钻石探针 $2\mu$ ，为磨削操作提供了足够的测量灵敏度。虽然该设备的精度相当高，

it lacks mobility and when a surface roughness measurement is needed, the workpiece is dismounted and carried to the measurement room and mounted again.

但它缺乏移动性，当需要测量表面粗糙度时，工件需要拆卸并带到测量室，然后重新安装。

## 1.5 Layout of the Thesis

## 1.5 论文布局

The organization of the thesis is given as follows:

论文的组织结构如下：

Chapter 2 is about Measurement and Modelling of Abrasive Tools. The steps are given:

第2章是关于磨料工具的测量与建模。步骤如下：

- “C” number calculation, where the grit number per $mm^2$ is calculated
- Individual grit observation, where the geometric properties of the grits are identified.
- Axial discretization of the wheel & Wheel creation, where the wheel is modeled with normal distribution.

- “C”数值计算，其中计算每毫米$^2$ 的磨料颗粒数
- 单个磨料颗粒观察，其中识别磨料颗粒的几何特性。
- 轮轴离散化与轮子创建，其中轮子使用正态分布进行建模。

Chapter 3 is about the Kinematic Model of Abrasive Processes. The steps are given:

第三章是关于磨削过程的运动学模型。步骤如下：

- Calculation and printing of the kinematic trajectories of the grits.
- Identification of active grits using the intersections of the grits.
- Surface roughness in perpendicular to cutting direction and force prediction is completed using the geometric and kinematic model

- 计算并打印磨料的运动轨迹。
- 通过磨料的交点识别有效磨料。
- 垂直于切削方向的表面粗糙度和力预测是使用几何和运动学模型完成的。

Chapter 4 is about Dynamic Force Model of Abrasive Processes. The steps are given:

第四章是关于磨削过程的动态力模型。步骤如下：

- Calculation of dynamic chip thickness and dynamic forces using the geometric and kinematic model.
- Vibration and surface roughness in cutting direction prediction using the dynamic model.
- Time domain simulations.

- 使用几何和运动学模型计算动态芯片厚度和动态力。
- 使用动态模型预测切削方向的振动和表面粗糙度。
- 时域仿真。

Chapter 5 is about Stability of Abrasive Processes. The steps are given:

第五章是关于磨削过程的稳定性。步骤如下：

- Stability prediction using time domain simulations and analytic methods.
- Verification of the proposed models with the experiments
- Stability conditions under varying wheel parameters.

- 使用时域仿真和分析方法进行稳定性预测。
- 通过实验验证所提出的模型。
- 不同轮胎参数下的稳定性条件。

Chapter 6 is about Discussion, Conclusion and Suggestions for Future Research

第六章是关于讨论、结论和未来研究建议

## 2 MEASUREMENT AND MODELLING OF ABRASIVE TOOLS

The most fundamental difference between grinding and other cutting operations regarding the material removal is the uncertainty and random nature of abrasive process chip formation. In grinding, the cutting edges are undefined whereas the geometrical properties of turning and milling tools can easily be obtained.

This chapter is about determining the geometric properties of abrasive particles to be used in simulations and experiments. Determination of these geometric features before applying the analytic models is one of the most important stages of grinding process modeling. The geometric values assigned to these measurements will be used for representing each particle. Information on the geometric properties of the tools in the industrial cutting operations is provided by the toolmakers, but this is not possible in grinding wheels and subsequent measurements are necessary.

Due to the manufacturing method of the grinding wheels the topography on the wheel surface will have random structure [41]. In measurements it is seen that the abrasive grits are randomly distributed over the surface of the wheel. This is a result of the way that the wheels are produced. First the abrasive particles are produced and gathered together. They are passed through a sieve to assure that the grits on the wheel have a certain dimensional range. The B number on the wheel represents the grit size which corresponds the size of the sieve that was used. After the sifting process, the abrasive grits are assembled on the grinding wheel. There two types of grinding wheels with respect to the type of assembly of the grits on the wheel; bonded and plated. The bonded types are the wheels that have multiple layers of abrasive grits on the surface of the tool on top of each other. The bond material provides the adhesion force between the layers of abrasive grits. Multiple layers of the abrasive grits provide dressing option for the grinding wheels when they are worn. The plated wheels are covered with single layer of abrasive grits and cheaper to produce comparing to the bonded ones as less overall material is needed. They are used without dressing due to existence of only single layer of abrasive grits. Thus, when the plated wheel is worn it is replaced.



## 2 研磨工具的测量与建模

磨削与其他切削操作在材料去除方面的最根本区别在于磨削过程中磨粒形成的不确定性和随机性。在磨削中，切削刃是不确定的，而车削和铣削刀具的几何特性则很容易获得。

本章内容是关于确定用于模拟和实验的研磨颗粒的几何特性。在应用解析模型之前确定这些几何特征是研磨过程建模的最重要阶段之一。分配给这些测量的几何值将用于表示每个颗粒。工业切削操作中工具的几何特性信息由工具制造商提供，但在研磨轮中无法做到这一点，因此后续测量是必要的。

由于砂轮的制造方法，砂轮表面的形貌将具有随机结构 [41]。在测量中可以看到，磨料颗粒在砂轮表面随机分布。这是砂轮生产方式的结果。首先，磨料颗粒被制造并聚集在一起。它们通过筛子筛选，以确保砂轮上的磨料颗粒具有特定的尺寸范围。砂轮上的 B 数代表磨料颗粒尺寸，对应所用筛子的尺寸。筛选过程后，磨料颗粒组装在砂轮上。根据磨料颗粒在砂轮上的组装类型，砂轮分为两种：粘结型和电镀型。粘结型砂轮是工具表面上有多层磨料颗粒重叠的砂轮。粘结材料提供磨料颗粒层之间的粘附力。多层磨料颗粒为砂轮在磨损时提供了修整选项。电镀砂轮覆盖着一层磨料颗粒，与粘结型相比更便宜生产，因为所需总材料更少。由于只有一层磨料颗粒，它们无需修整使用。因此，当电镀砂轮磨损时，它被更换。




Figure 5: Grinding wheels for different applications

It is also convenient to categorize the grinding wheels into two with respect to their abrasive grit materials. Grinding wheels which have Aluminum oxide or SiC abrasive grits are called conventional wheels whereas the wheels that have CBN or diamond particles are called super abrasive wheels. Super abrasive wheels are usually preferred for processing high alloy materials, such as nickel alloys, as they provide increased performance as a result of elevated hardness, thermal condition and abrasion resistance. Conventional wheels are cheaper and generally selected in low cost operations. Within the scope of this thesis, the overall focus is on CBN wheels whereas some applications of the conventional wheels are also presented.

To represent the surface of the grinding wheel and the randomness of the abrasive grits, electron microscope measurements were done on a CBN grinding wheel as shown in Figure 6 and Figure 7. As it can be seen, the distribution of the abrasive grits on the surface of the wheel is random and this results in a material removal with undefined cutting edges. Hence, proposing a modeling method for abrasive tools is crucial for process model development.






图5：不同应用的磨轮

根据磨料颗粒材料，也将砂轮分为两种。具有氧化铝或 SiC 磨料颗粒的砂轮称为传统砂轮，而具有 CBN 或金刚石颗粒的砂轮称为超硬砂轮。超硬砂轮通常用于加工高合金材料，如镍合金，因为它们由于硬度、热状态和耐磨性的提高而提供了更高的性能。传统砂轮更便宜，通常在低成本操作中选用。在本论文的范围内，总体重点是 CBN 砂轮，但也介绍了传统砂轮的一些应用。

为了表征磨轮的表面和磨料颗粒的随机性，对图6和图7所示的CBN磨轮进行了电子显微镜测量。如图所示，磨轮表面的磨料颗粒分布是随机的，这导致材料去除时具有未定义的切削刃。因此，为磨料工具提出建模方法是开发工艺模型的关键。

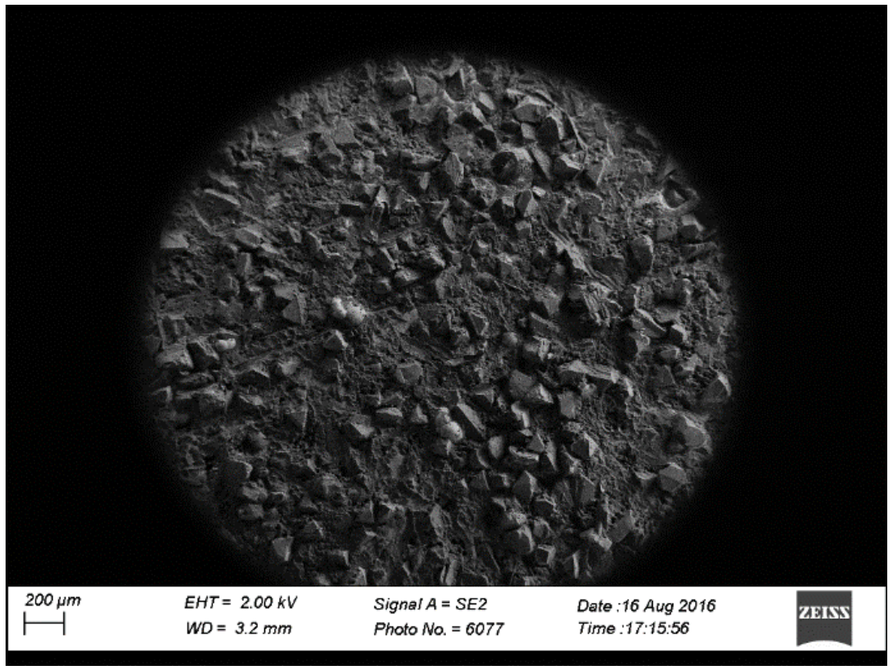



Figure 6: SEM picture of a CBN wheel surface I

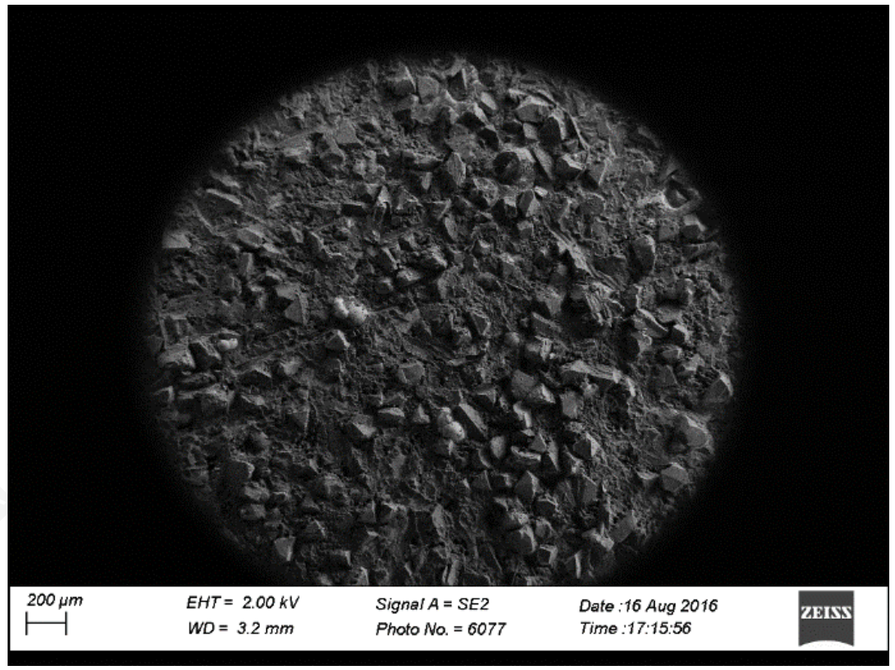



图6：CBN轮面SEM照片I

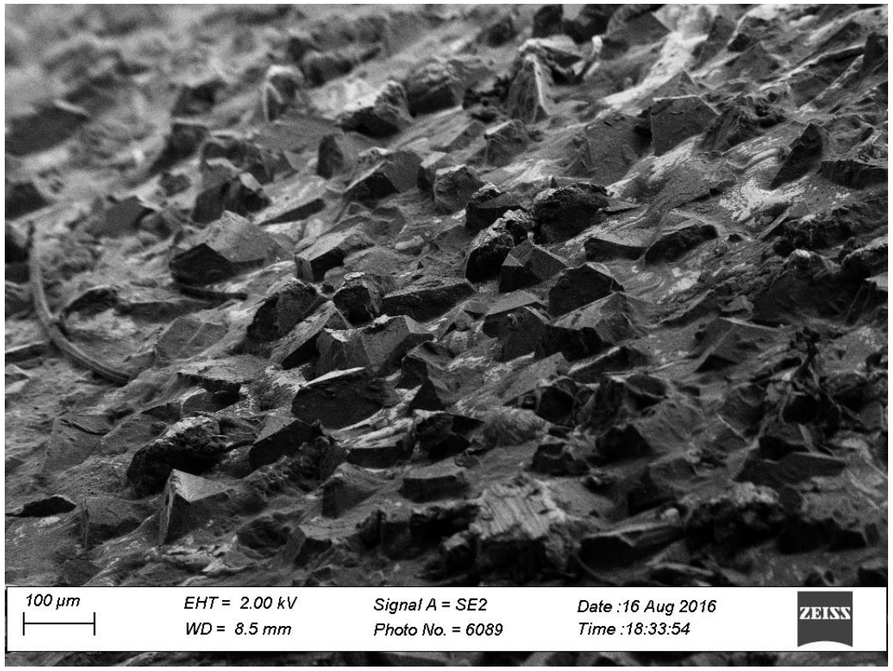



Figure 7: SEM picture of a CBN wheel surface II

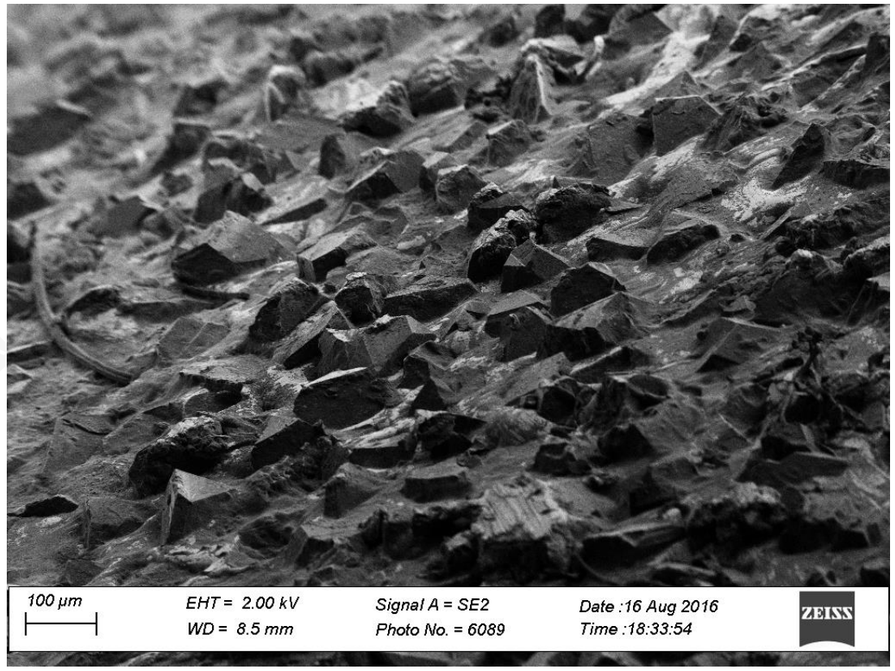



图7：CBN轮面SEM照片II



17

| Wheel no | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Type | Mounted point | Mounted point | Mounted point | Mounted point | Mounted point | Mounted point |
| Grit Material | CBN | CBN | CBN | CBN | CBN | $Al_2O_3$ |
| Diameter (mm) | 10 | 5 | 30 | 10 | 10 | 20 |
| Grit size | B126 | B64 | B151 | B126 | B126 | B120 |
| Bond Type | Electroplated | Electroplated | Electroplated | Electroplated | Electroplated | Vitrified |

Table 1: Grinding wheel specifications

As a result of the stochastic nature of the grinding wheels it is convenient to model the wheel using Gaussian distribution created from a sub set of data which are obtained through the measurements from the surface of the wheel. The fundamental information for process modelling is as follows:

- Number of abrasive grits per $mm^2$ (Analogous to teeth number in milling)
- Grit dimensions (height, width, width of cut)
- Process related geometric properties such as oblique angle, rake angle and edge radius …etc.

In this chapter, a wheel model is proposed by using the fundamental information obtained from optical measurements. The grit data then processed and fit into a gaussian distribution thus creating the virtual wheel. The measurements are done on 6 different wheels and the specifications are given in Table 1. First 5 of the wheels are CBN and preferably have different number of grits sizes. $6^{th}$ wheel is selected as a mounted point conventional wheel (Aluminum oxide). The wheels are shown in Figure 8, from the left to right the wheels numbers are 1,2,3,4,5 and 6 respectively. The measurement and modeling of the before mentioned wheels will be presented in the following sections to provide a database for future process modeling applications.



| 轮号 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Type | 安装点 | 安装点 | 安装点 | 已挂载点 | 已挂载点 | 已挂载点 |
| Grit 材料 | CBN | CBN | CBN | CBN | CBN | $Al_2O_3$ |
| 直径 (mm) | 10 | 5 | 30 | 10 | 10 | 20 |
| 磨料粒度 | B126 | B64 | B151 | B126 | B126 | B120 |
| Bond Type | 电镀 | 电镀 | 电镀 电镀 电镀 | | | 玻璃化 |

表1：砂轮规格

由于砂轮具有随机性，因此方便使用高斯分布对砂轮进行建模，该分布由从砂轮表面测量获得的部分数据子集创建。工艺建模的基本信息如下：

- 每毫米磨料颗粒数$^2$ (类似于铣削中的齿数)
- 磨料颗粒尺寸(高度、宽度、切削宽度)
- 处理相关的几何属性，如倾斜角、刻面角和边缘半径……等等。

在本章中，通过光学测量获得的基本信息，提出了一种轮模型。然后对砂粒数据进行处理，使其拟合高斯分布，从而创建虚拟轮。测量是在6个不同的轮上进行的，其规格如表1所示。前5个轮是CBN砂轮，并且最好具有不同数量的砂粒尺寸。$6^{th}$ 轮被选为安装点传统砂轮（氧化铝）。这些轮子显示在图8中，从左到右的轮子编号分别是1、2、3、4、5和6。上述轮子的测量和建模将在下一节中介绍，以提供未来工艺建模应用的数据库。




Figure 8: Measured grinding wheels. From left to right 1) 10mm B126 CBN 2) 5mm B64 CBN 3)30mm B150 CBN 4) 10mm B126 CBN dressed 5) 10mm B126 CBN worn 6) 20mm B120 $Al_2O_3$




图8：测量的磨轮。从左到右 1) 10mm B126 CBN 2) 5mm B64 CBN 3)30mm B150 CBN 4) 10mm B126 CBN 磨钝 5) 10mm B126 CBN 磨损 6) 20mm B120 $Al_2O_3$

## 2.1 C Number Identification

C number is defined as the number of grits that reside on an area of $1mm^2$ on the surface of the grinding wheel which is analogous to number of cutting teeth in milling. C number should be the initial parameter to be identified. C number of the wheel can be found by using the 20x lens on the $\mu$surf Nanofocus confocal microscope. Measurements were made at different points of the tool by scanning in the X and Y axes and determining the abrasive grits. In this method, the particles lying above a certain height of the Z axis are counted by using the microscope and confocal approach systems. When the confocal option is used, the highest points of the abrasive particles can be identified marked with red dots. Areas marked with red marks are perceived by the device as areas where the light reflects more, that is closer to the lens. Abrasive grits have been determined with this system, which is made at various points of the tool. The points are colored as blue.



## 2.1 C编号识别

C数定义为在磨削轮表面$1mm^2$ 面积上存在的砂粒数量，这类似于铣削中的切削齿数。C数应作为初始参数进行识别。轮子的C数可以通过在 $\mu$surf纳米聚焦共聚焦显微镜上使用20倍物镜来找到。通过在X轴和Y轴上扫描工具的不同点并确定砂粒来进行测量。在这种方法中，通过显微镜和共聚焦系统计数位于Z轴某个高度以上的颗粒。当使用共聚焦选项时，可以通过红色圆点识别出磨削颗粒的最高点。标有红色标记的区域被设备感知为光反射更多、更接近物镜的区域。该系统已确定磨削砂粒，这些砂粒在工具的各个点上制作。这些点被着色为蓝色。

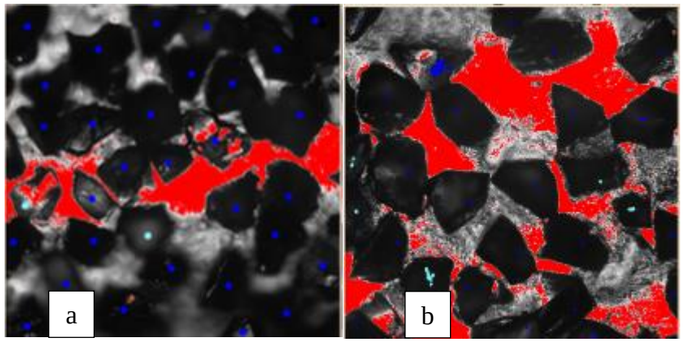



Figure 9: C number identification a) tool #1 b) tool #3

A sample measurement is given in Figure 9. As seen, the grits can be counted with nanofocus. The area of the measurement corresponds to 0.64mm$^2$ for the 20x lens. The numbers are found and then a multiplied with a ratio to match the 1 mm$^2$ area. The C number is used to identify the total number of grits on the whole surface of the grinding wheel.

## 2.2 Individual Grit Measurement

Individual grit measurements are necessary for the identification of process modeling related parameters such as oblique angle, rake angle and edge radius [1,11,12]. For this identification 50x lens in the Nanofocus is used to capture the 3D and 2D scanning of the individual grits. The software provided with this measurement device namely $\mu$surf analysis, provides an on-screen measurement tool on the scanning obtained from the microscope. The identification of the geometrical properties of all the grits on the grinding wheel is a very tiresome approach thus a normal distribution method [40] for creating the virtual wheel is used based on the mean and standard deviation of all measured geometric properties of the individual grits. The sample space for generation of Gaussian distribution is selected to be 50 measurements a little bit more than the magical number 30.. A 3D and 2D measurement sample of an individual grit is presented in Figure 10.



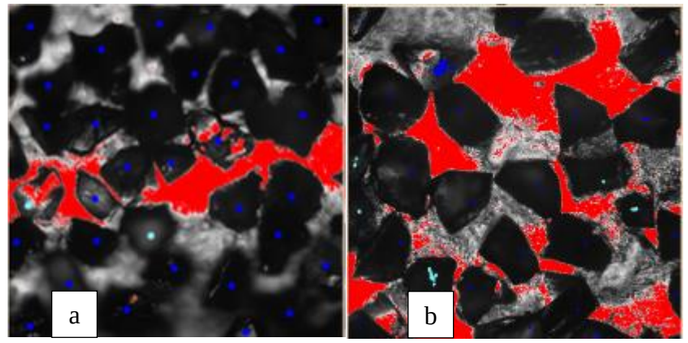



图9：C颗粒识别 a) 工具#1 b) 工具#3

图9中给出了一个样品测量示例。可以看到，颗粒可以通过纳米聚焦计数。测量区域对应于20倍物镜的0.64mm$^2$。找到这些数字后，再乘以一个比例以匹配1mm$^2$的面积。C数值用于识别整个砂轮表面的颗粒总数。

## 2.2 单个颗粒测量

需要单个磨粒的测量数据，以便识别与工艺建模相关的参数，例如倾斜角、后角和边缘半径 [1,11,12]。为此识别，Nanofocus 中的 50x 物镜用于捕获单个磨粒的 3D 和 2D 扫描。与该测量设备配套的软件，即 $\mu$surf analysis，在显微镜获得的扫描图像上提供屏幕测量工具。在磨轮上识别所有磨粒的几何特性是一种非常繁琐的方法，因此基于单个磨粒所有测量几何特性的平均值和标准差，采用正态分布方法 [40] 创建虚拟磨轮。生成高斯分布的样本空间选择为 50 次测量，略多于神奇的数字 30。单个磨粒的 3D 和 2D 测量样本如图 10 所示。

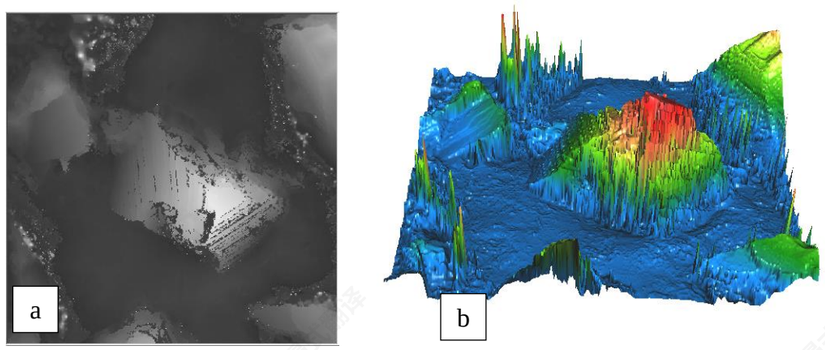



Figure 10: Individual grit measurement with 50x lens a) 2D b) 3D

The Nanofocus software can scan the profiles in x and y directions in the captured scan. The software can project the scans into top and side views. With this feature, the width, height and rake angle of the captured image are measured as follows. In Figure 11a, h1 represents the height, a2 represents the rake angle, and w0 represents the width of the particle. In Figure 11b, the profile of the rounding radius is set by a curve and its radius is measured. In Figure 11b, the radius of the curve is 5 µm. The oblique angle measurement is done by using the same tool that is provided and is shown in Figure 12.

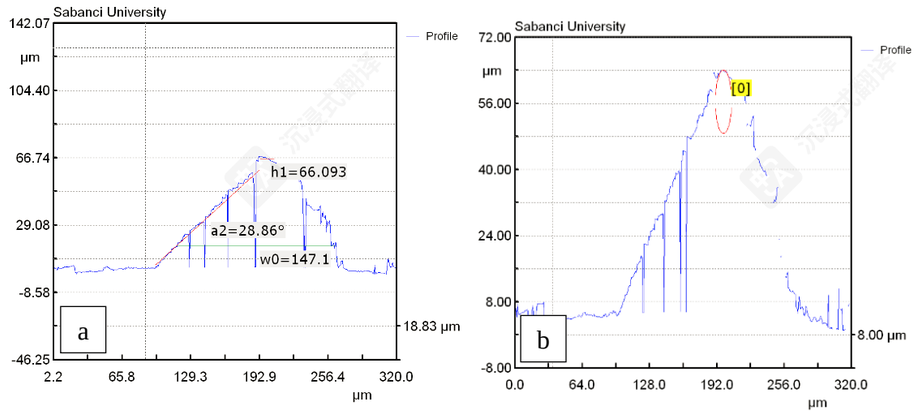



Figure 11: Identification of geometrical properties of individual grits from 2D side view a) dimensions b) edge radius



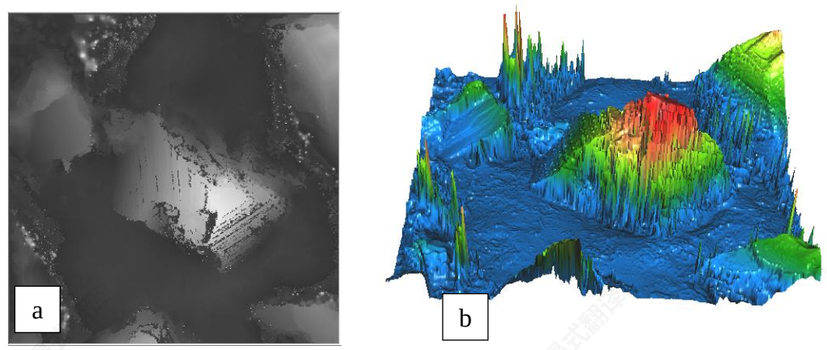



图10：使用50倍镜头进行单个磨粒测量 a) 2D b) 3D

Nanofocus 软件可以在捕获的扫描中沿 x 和 y 方向扫描轮廓。该软件可以将扫描投影到顶视图和侧视图。利用此功能，捕获图像的宽度、高度和后角按如下方式测量。在图 11a 中，h1 表示高度，a2 表示后角，w0 表示颗粒的宽度。在图 11b 中，圆角半径的轮廓由曲线设置，其半径被测量。在图 11b 中，曲线的半径为 5 µm。倾斜角的测量是使用提供的相同工具完成的，如图 12 所示。

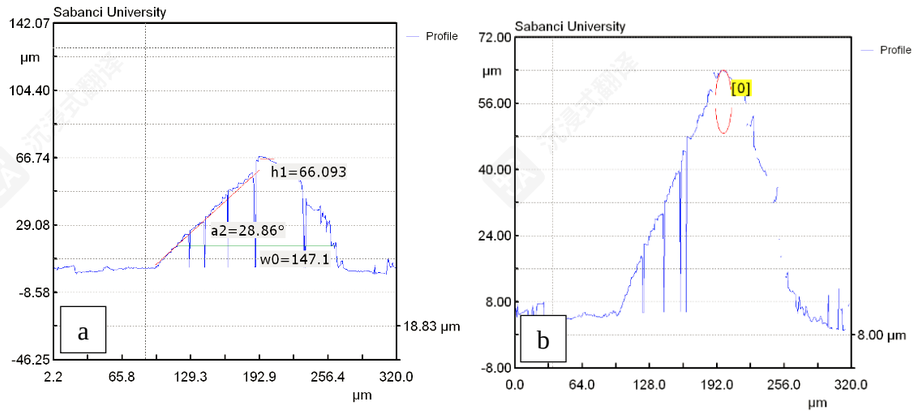



图11：从2D侧视图识别单个磨粒的几何特性 a) 尺寸 b) 边缘半径

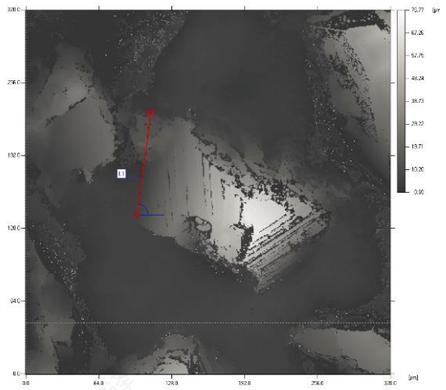


Figure 12: Identification of oblique angle of abrasive grit from 2D top view

However a circular fit does not provide an accurate measurement so later the edge radius scanning are done with SEM.

### 2.3 Virtual Construction of Wheel Surface

Regarding the stochastic nature of the grinding process and the difficulty in modeling the abrasive mechanism it’s crucial that some assumptions must be considered. Essentially, the random values of geometrical properties of the grits are assumed to be Gaussian. It is may be recognized as a solid assumption considering the production steps of the grinding wheels. Another vital assumption would be to presume the distance between all the adjacent grits along the perimeter of the grinding wheel surface are the same. This assumption is believed to be required to apply the milling analogy to the grinding technology. The adjacent distance is identified as an average term of the data obtained from measurements. Furthermore, it is also assumed that the grinding wheel is discretized into elements along the radial direction. Each element is identical, concentric and resembles a ring surrounding the circumference of the wheel. The width of each element is set to have an average value of grit width. Each element is considered to contain one grit and the whole part of it along the wheel width (or axial) direction whereas there may be many along the periphery. One may not find a grit that is divided and shared between two or more elements. The only geometric property of the grits that are randomly chosen will be the height and this will be the crucial factor in



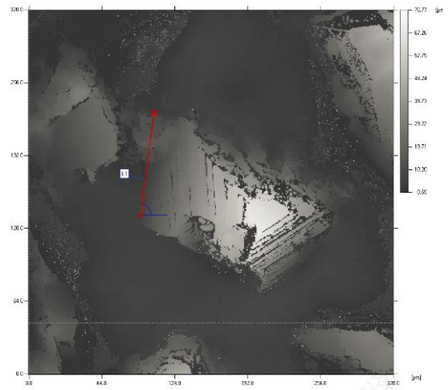


图12：从2D俯视图识别磨料颗粒的倾斜角度

然而，圆形拟合并不能提供精确的测量，因此后来使用扫描电子显微镜（SEM）进行边缘半径扫描。

2.3 轮胎表面的虚拟构建

关于磨削过程的随机性和建模磨削机理的困难，必须考虑一些假设。本质上，假设砂粒的几何属性随机值为高斯分布。考虑到砂轮的生产步骤，这可能被认为是一个合理的假设。另一个重要的假设是假定磨削轮表面所有相邻砂粒沿周长的距离相同。这个假设被认为是为了将铣削类比应用于磨削技术所必需的。相邻距离被识别为测量数据得到的平均值。此外，还假设磨削轮沿径向离散化为元素。每个元素相同、同轴，并类似于环绕车轮圆周的一个环。每个元素的宽度设置为砂粒宽度的平均值。每个元素被认为包含一个砂粒，并在车轮宽度（或轴向）方向上占据其全部部分，而周长方向上可能有多个。可能找不到一个被分成两个或多个元素共享的砂粒。砂粒随机选择的唯一几何属性将是高度，这将是在这项工作中至关重要的因素。

this work. These assumptions are helpful during the modeling in terms of milling analogy. The illustration of the proposed wheel model is presented in Figure 13.

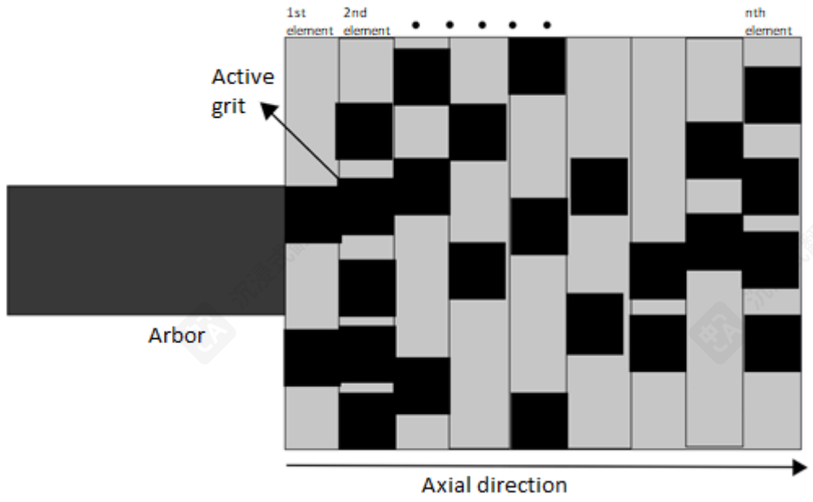



Figure 13: Axial discretization of the wheel

In this figure the grits that are marked with black do not represent all of the grits, instead they refer to the active grits which participate in grinding process. The determination of active grits will be discussed later in the thesis. The C number which refers to the number of grits per $1mm^2$ area does not apply for active grits in the proposed model. It refers to the static number of grits. With the calculation of the mean and standard deviation values from the wheel inspections, a sample space created whose dimension matches the total number of grits along one element. The height amount of an individual grit is selected randomly from this sample space. The total dimension of this sample space is the distributed equivalent of c number over the area of single element. An imaginary wheel surface is created in this sense in which the model knows how many grits are distributed along the axial elements on the wheel and the geometric properties of them.



这些假设在建模时有助于铣削类比。所提出的轮模型示意图如图13所示。

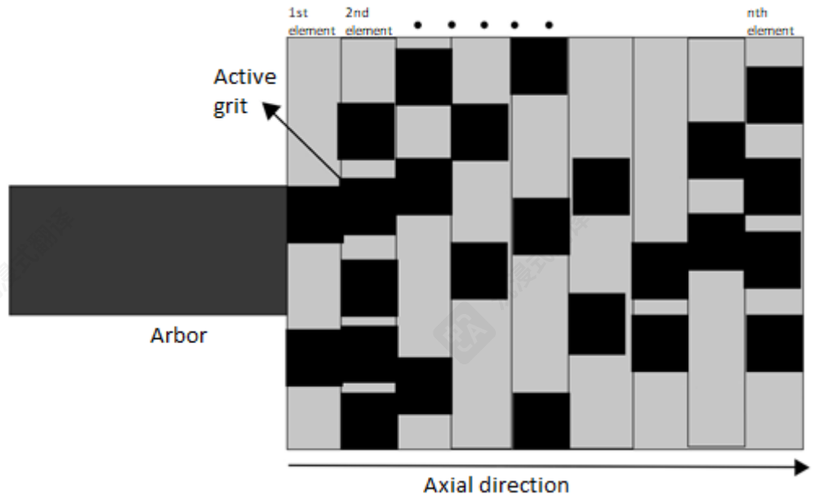



图13：车轮轴向离散化

在图中，标有黑色的磨粒并不代表所有磨粒，而是指参与磨削过程的活性磨粒。活性磨粒的确定将在论文后续部分讨论。C数值，即每$1mm^2$ 面积内的磨粒数量，在所提出的模型中不适用于活性磨粒。它指的是静态磨粒数量。通过计算车轮检查得到的平均值和标准差值，创建了一个样本空间，其维度与单个元件上磨粒的总数相匹配。单个磨粒的高度值从该样本空间中随机选取。该样本空间的总维度是C数值在单个元件面积上的分布等效。从这个意义上来说，创建了一个想象的轮面，模型知道有多少磨粒分布在车轮的轴向元件上，以及它们的几何特性。

## 2.4 Measurement Results

The measurements of 6 different grinding wheels is conducted with the described method. The grit number per 1mm$^2$ area of grinding wheel is counted and then a number of individual grits on the measured. The geometric properties of the measured grits are identified with the beforementioned equipment and software. The results of the wheel measurements are presented in Table 2. Here the same naming convention that is in Table 1 is used.

| Wheel no | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Type | Mounted point | Mounted point | Mounted point | Mounted point | Mounted point | Mounted point |
| Grit Material | CBN | CBN | CBN | CBN | CBN | $Al_2O_3$ |
| Diameter (mm) | 10 | 5 | 30 | 10 | 10 | 20 |
| Grit size | B126 | B64 | B151 | B126 dressed | B126 worn | B120 |
| Bond Type | Electoplated | Electoplated | Electoplated | Electoplated | Electoplated | Vitrified |
| C number | 33 | 97 | 23 | 33 | 31 | 34 |
| Grit width (μm) | 65 | 41 | 73 | 67 | 39 | 63 |
| Edge radius (μm) | 1.15 | 1 | 1.07 | 1 | 1.37 | 5 |
| Mean and std. dev. Of grit height (μm) | 46 & 18 | 30 & 11 | 88 & 17 | 37 & 8 | 25 & 0.2 | 63 & 14 |
| Mean and std. dev. Of width of cut (μm) | 18 & 6 | 9 & 4 | 23 & 11 | 17 & 7 | 16 & 7 | 20 & 6 |
| Average rake angle (degrees) | -52 | -51 | -54 | -53 | -57 | -45 |
| Average oblique angle (degrees) | 24 | 21 | 25 | 31 | 30 | 32 |

Table 2: List of wheel measurement results



## 2.4 测量结果

使用所述方法对6种不同的磨削轮进行了测量。计算了磨削轮每1mm$^2$ 面积内的磨粒数量，然后对测量的单个磨粒进行了计数。使用前述设备和软件识别了测量磨粒的几何特性。车轮测量的结果 presented 在表2中。此处使用了与表1相同的命名约定。

| 轮号 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Type | 安装点 | 安装点 | 已安装点 | 已安装点 | 已安装点 | 安装点 |
| 磨料材料 | CBN | CBN | CBN | CBN | CBN | $Al_2O_3$ |
| 直径 (mm) | 10 | 5 | 30 | 10 | 10 | 20 |
| 磨料尺寸 | B126 | B64 | B151 | B126 磨光 | B126磨损 | B120 |
| 连接类型 | 电镀 | 电镀 | 电镀 | 电镀 | 电镀 vitrified | |
| C 编号 | 33 | 97 | 23 | 33 | 31 | 34 |
| 磨料宽度 (μm) | 65 | 41 | 73 | 67 | 39 | 63 |
| 边缘半径 (μm) | 1.15 | 1 | 1.07 | 1 | 1.37 | 5 |
| 平均值和标准差 磨料偏差 高度 (μm) | 46 & 18 | 30 & 11 | 88 & 17 | 37 & 8 | 25 & 0.2 | 63 & 14 |
| 均值和标准差 宽度开发 切割 (μm) | 18 & 6 | 9 & 4 | 23 & 11 | 17 & 7 | 16 & 7 | 20 & 6 |
| 平均抽水费 角度（度） | -52 | -51 | -54 | -53 | -57 | -45 |
| 平均斜率 角度（度） | 24 | 21 | 25 | 31 | 30 | 32 |

表2：车轮测量结果列表

## 2.5 Summary

In this chapter, a method for grinding wheel measurement and modeling is proposed. As a result of the stochastic nature of grinding tools, it is essential to develop a measurement method and to introduce wheel modeling based on the geometric data that is obtained through measurements. The measurements are used to identify the number of abrasive grits per unit area and the process related geometric dimensions. They are conducted by using a confocal light microscope. The number of abrasive grits are obtained by counting the peak points that are closer to the light source which are highlighted by red color with the instrument. The geometric properties of the individual grits are identified using the related software provided with the measurement device. The x and y profiles are measured from the 2D scans of the grits and the process related parameters are found. Moreover, the wheel model is presented by adding the individual grit data. The grinding wheel surface is discretized into elements along the axial direction. The elements resemble concentric rings of membrane around the periphery of the wheel. With respect to the number of abrasive grits per unit area, the grits are distributed along the discretized elements with each grit having the array of data of the geometric properties.

A set of measurements including 6 different grinding wheels is presented including 5 CBN wheels having different grit sizes and conditions and 1 conventional aluminum oxide wheel. A quick observation states that

- If the grit size is increases, C number decreases.
- The grid width is proportional to B number.
- Edge radii of CBN wheels is smaller than that of the conventional wheels
- The mean and standard deviation of height is getting lower if dressing is applied or the wheel is worn.
- C number may decrease due to wear because of the pull-out mechanism [12].



## 2.5 摘要

在本章中，提出了一种磨头测量和建模的方法。由于磨削工具具有随机性，因此开发测量方法并根据测量获得的几何数据引入磨头建模至关重要。测量用于识别单位面积内的磨料颗粒数和相关的几何尺寸。这些测量通过使用共聚焦显微镜进行。磨料颗粒数是通过计数靠近光源的峰值点获得的，这些峰值点被仪器用红色高亮显示。单个磨料颗粒的几何特性使用与测量设备提供的软件进行识别。从磨料颗粒的二维扫描中测量x和y轮廓，并找到相关的工艺参数。此外，通过添加单个磨料颗粒数据来呈现磨头模型。磨头表面沿轴向离散化为单元。这些单元类似于围绕磨头周边的同心膜环。关于单位面积内的磨料颗粒数，颗粒沿离散化单元分布，每个颗粒具有几何特性的数据数组。

一套包含6种不同磨削砂轮的测量数据被提出，其中包括5个CBN砂轮，具有不同的粒度和状态，以及1个传统的氧化铝砂轮。快速观察表明

- 如果磨料粒度增加，C值会减小。
- 网格宽度与B值成正比。
- CBN砂轮的边缘半径小于传统砂轮。
- 如果进行修整或砂轮磨损，高度的平均值和标准差会降低。

- 由于拔出机制[12]，C值可能会因磨损而减小。

## 3 KINEMATIC MODEL OF ABRASIVE PROCESSES

Some models presented in the literature are insufficient to truly represent the mechanics of abrasive processes accurately. One crucial subject on this inadequacy is the determination of the active grits that participate in grinding. As a result of the stochastic nature of the grinding wheel structures, not all the abrasive grits are active during the process [38]. Throughout the experimental work for this thesis, it has been observed that the identification of the active grits around the grinding surface plays a significant role in the development of prediction of grinding force and workpiece surface roughness. This is significant especially when mounted point grinding wheels are used. In some of the works in the literature the active grit number is treated as a constant number [53] which is identified beforehand from tool measurements. During the studies [54], it was discovered that the percentage of active grits is not a static number, instead it highly depends on the grinding process conditions. In this work, a method to identify the active grit numbers is presented.

In Figure 14, a cross section of the grinding tool is depicted where grit number 1 is active and grit number 2 is passive. As it can be seen the abrasive grits on the surface of the wheel have different height profiles based on the mean and standard deviation of the values obtained from the measurements. This results in a situation where the taller grits compared to the others will indent more into the workpiece surface and the shorter grits will not even contact any material on their trajectory since the path is cleared my more dominant grits in the previous revolutions.



## 3 磨损过程的运动学模型

文献中提出的一些模型不足以准确真实地表示磨削过程的力学机制。关于这种不足的一个关键问题是参与磨削的活性砂粒的确定。由于磨削砂轮结构的随机性，并非所有磨料砂粒在过程中都是活跃的 [38]。在本文的实验工作中，观察到识别磨削表面附近的活性砂粒在磨削力和工件表面粗糙度预测的发展中起着重要作用。这一点在安装点磨削砂轮时尤其重要。文献中的一些工作中，活性砂粒数量被视为一个预先从工具测量中确定的常数 [53] 。在研究 [54]，中发现，活性砂粒的百分比不是一个静态数字，而是高度依赖于磨削过程条件。在这项工作中，提出了一种识别活性砂粒数量的方法。

图14中展示了磨削工具的横截面，其中砂粒1处于激活状态，砂粒2处于非激活状态。从图中可以看出，砂轮表面的磨料砂粒具有不同的高度轮廓，这些轮廓基于测量值得到的均值和标准差。这导致一种情况：与其他砂粒相比，较高的砂粒会更多地压入工件表面，而较低的砂粒在其运动轨迹上甚至不会接触任何材料，因为之前的旋转中更有优势的砂粒已经清除了路径。

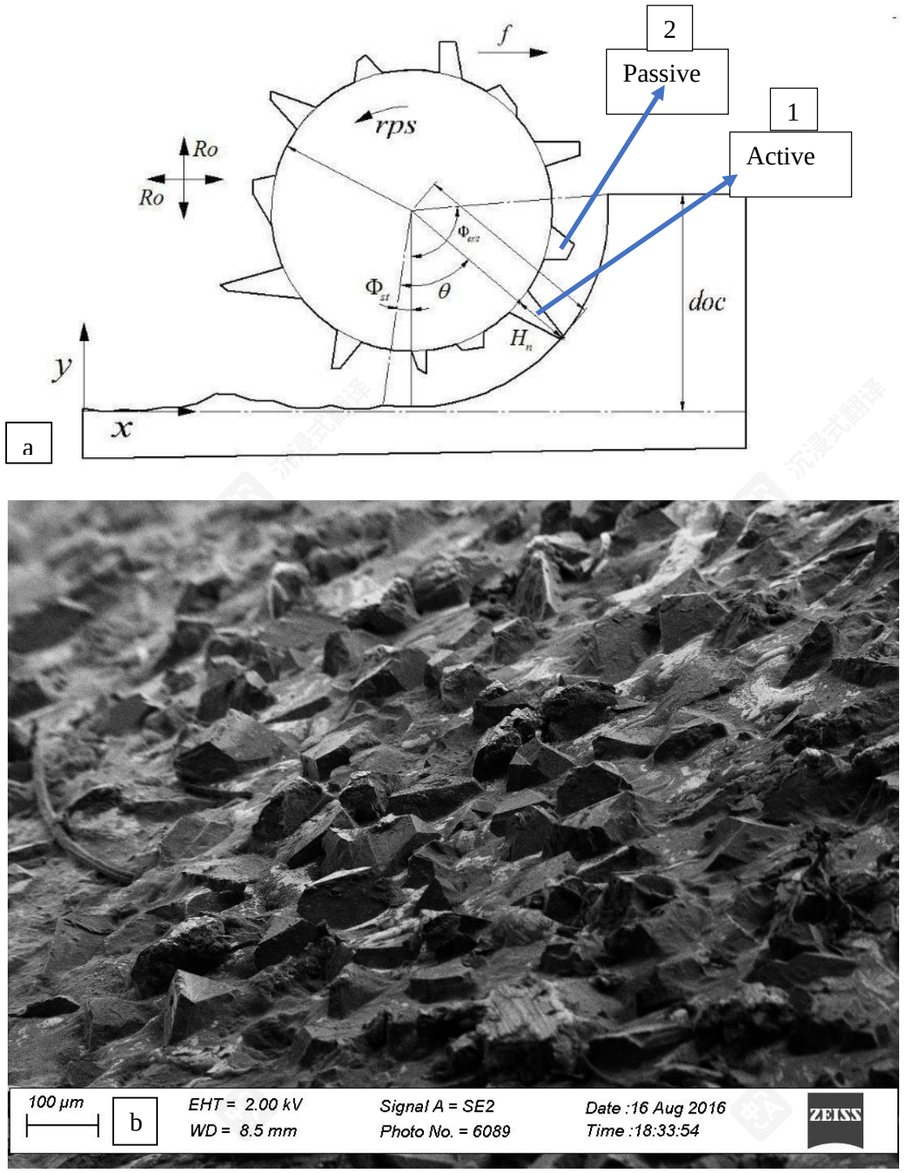



Figure 14: Depiction of active grits (a) virtual wheel (b) real wheel surface

When the algorithm for finding active grits completes the task, the active grits are determined and separated from the rest. For the static model, the virtual wheel is recreated by the active grits and the passive grits are disregarded. The details will be given in the following subsection.



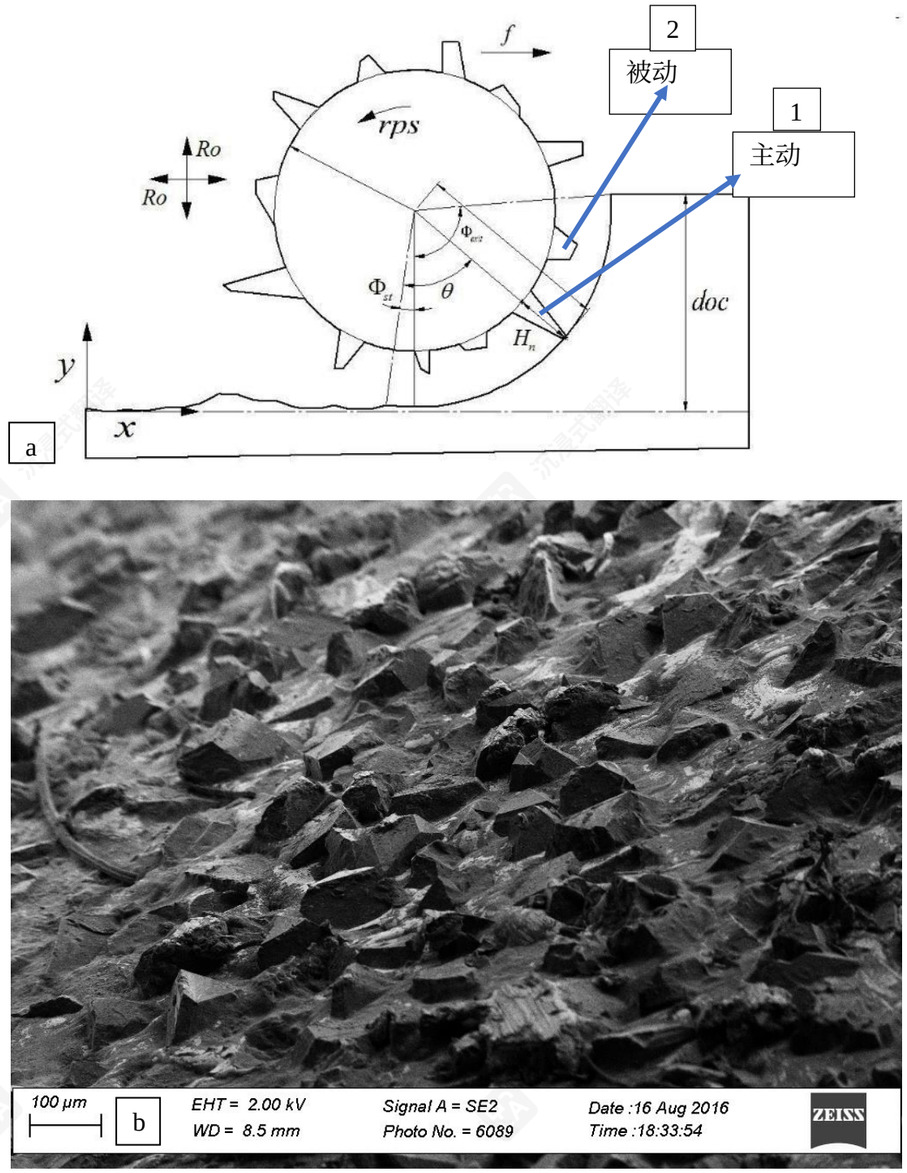



图14：主动砂粒的描绘（a）虚拟轮（b）真实轮表面

当寻找激活砂粒的算法完成任务时，激活砂粒会被确定并与其他砂粒分离。对于静态模型，虚拟砂轮由激活砂粒重新创建，而被动砂粒则被忽略。详细内容将在下一小节中给出。

## 3.1 The Geometric-Kinematic Model

Because of the stochastic nature of the grinding process, all grits on the wheel surface have a varying height profile. The identification of active grits is done through the comparison of grits in terms of their height profile.

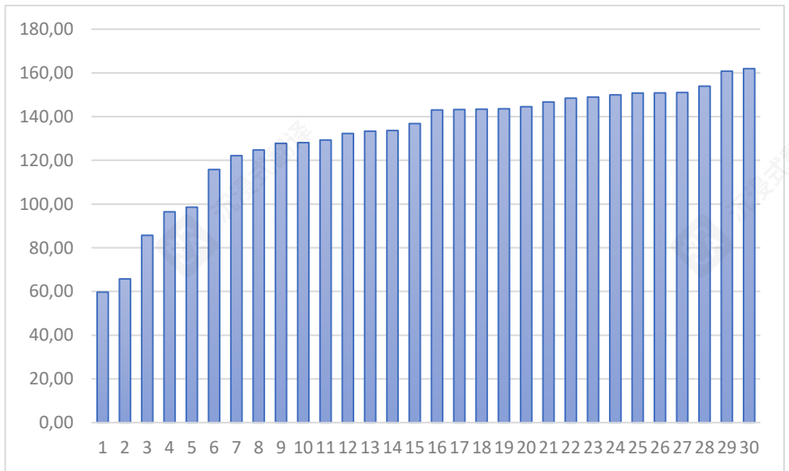



Figure 15: Height distribution of grit measurements

The effect of the workpiece velocity (feed) plays a crucial role on the number of active grits. For instance, imagine a hypothetical grinding operation carried out using a high enough feed velocity and very slow wheel rotational speed. For such a case regardless of the height distribution the number of active grits would be very high. Furthermore, the eccentricity of the grinding wheel will also affect the number of the active grits as it will change the height distribution. This thesis presents a model in which most of the grinding parameters are included in identifying the number of active grits. [55]

### 3.1.1 Electroplated CBN wheel properties

Electroplated CBN tools are wheels that have a single layer of grits which are electroplated through a metallic bond. With these bonds and abrasive CBN grits, these wheels exhibit an enhanced quality of mechanical properties [41]. These abrasive grits



## 3.1 几何-运动学模型

由于磨削过程的随机性，砂轮表面上的所有磨粒都具有不同的高度轮廓。磨粒的识别是通过比较磨粒的高度轮廓来完成的。

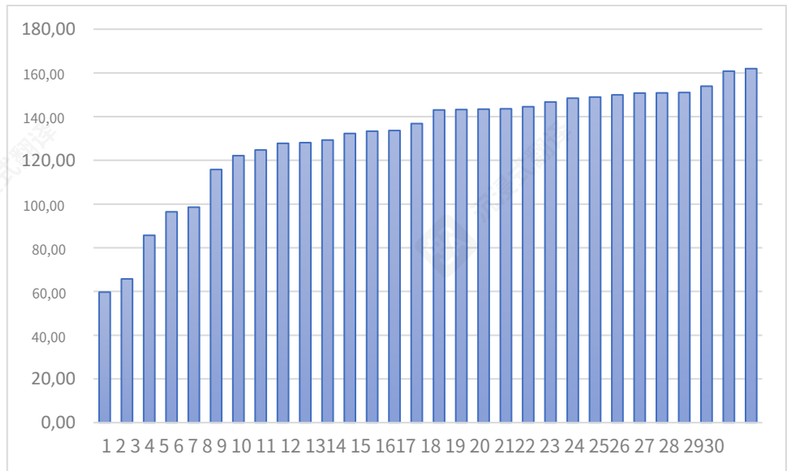



图15：磨粒测量的高度分布

工件速度（进给）的影响对活动磨粒的数量起着至关重要的作用。例如，想象一个假设的磨削操作，使用足够高的进给速度和非常慢的砂轮转速进行。对于这种情况，无论高度分布如何，活动磨粒的数量都会非常高。此外，磨轮的偏心度也会影响活动磨粒的数量，因为它会改变高度分布。本论文提出一个模型，其中包含了大多数磨削参数，用于识别活动磨粒的数量。[55]

### 3.1.1 电镀CBN砂轮特性

电镀CBN刀具是具有单层磨料的轮子，这些磨料通过金属键电镀而成。凭借这些键和磨料CBN磨料，这些轮子表现出更优异的机械性能[41]。这些磨料

have increased toughness and hardness with a high thermal conductivity. These properties permit the use of electroplated CBN wheels in grinding of difficult-to-cut materials [3]. It is crucial to determine the mechanics of abrasive processes using CBN plated wheels which show significantly different characteristics compared to the conventional cutting processes.

The identification of the grit properties is essential for the geometric kinematic model. The geometry of the grits, the distance between them, the number of grits in a $mm^2$ will be the important inputs to the model. The image of the CBN wheel that is used in the tests for force modelling is presented in Figure 16.



Figure 16: 20mm diameter B126 (gritsize) ElectroPlated CBN wheel

The grinding CNC machine used on this work has a high-speed spindle and the experiments were carried out in the speed range of 36000-40000 rpm using smaller wheels. The results of the wheel measurements are given in Table 3. This wheel is different than the conventional wheels as the abrasive grits are CBN material the diameter is significantly low as the use of aluminum oxide wheels with diameters of 200-300 mm is more common.

| Properties | Description |
|---|---|
| Diameter of the wheel | 20 mm |
| Width of the wheel | 10 mm |
| Arbor diameter | 6 mm |
| Grit size | 120 µ |

Table 3: Specifications for the presented wheel in Figure 16.



具有更高的韧性和硬度，以及优异的热导率。这些特性使得电镀CBN轮子适用于加工难加工材料[3]。使用CBN电镀轮进行磨削时，必须确定其磨削过程的力学原理，因为与传统的切削过程相比，这些轮子表现出显著不同的特性。

磨料的特性识别对于几何运动学模型至关重要。磨料的几何形状、它们之间的距离、每毫米$^2$ 内的磨料数量将成为模型的重要输入。用于力建模测试的CBN砂轮的图像如图16所示。



图16：直径20毫米B126（磨料尺寸）电镀CBN砂轮

这台用于此工作的磨削CNC机床配备了高速主轴，实验在36000-40000 rpm的速度范围内使用较小的砂轮进行。砂轮测量的结果见表3。这种砂轮与传统砂轮不同，因为其磨料颗粒是CBN材料，直径显著较低，而使用直径为200-300 mm的氧化铝砂轮更为常见。

| 特性 | 描述 |
|---|---|
| 砂轮直径 | 20毫米 |
| 轮宽 | 10 mm |
| 安装孔直径 | 6 mm |
| 磨料粒度 | 120 µ |

表3：图16所示轮子的规格

Some of the data about the wheel properties can be obtained from the manufacturer, but some properties that are specific to individual wheels cannot be found in the catalogs due to the stochastic nature of the abrasive tools [12]. Those are the individual grit properties such as height, grinding width, oblique and rake angles will influence the grinding mechanics. The grit measurements of the tool shown in Figure 16 are given in Figure 17. The chart illustrates the means and the standard deviations of the measurements together with the variation of the data. These measurements were done on the confocal microscope Nanofocus.

The Nanofocus device has its limits. As it works on a light scan procedure, some reflection errors on the wheel surface may cause deviations in the 3d model creation. If the measurement is aimed at a small viewpoint the values may reside within the noises. The edge radii of the grits will be quite small on grinding tools thus the measurements for edge radii on Nanofocus will not produce accurate results. Therefore, the edge radius measurement has been carried out on an electron microscope. In Figure 18, the images from the SEM machine are provided.

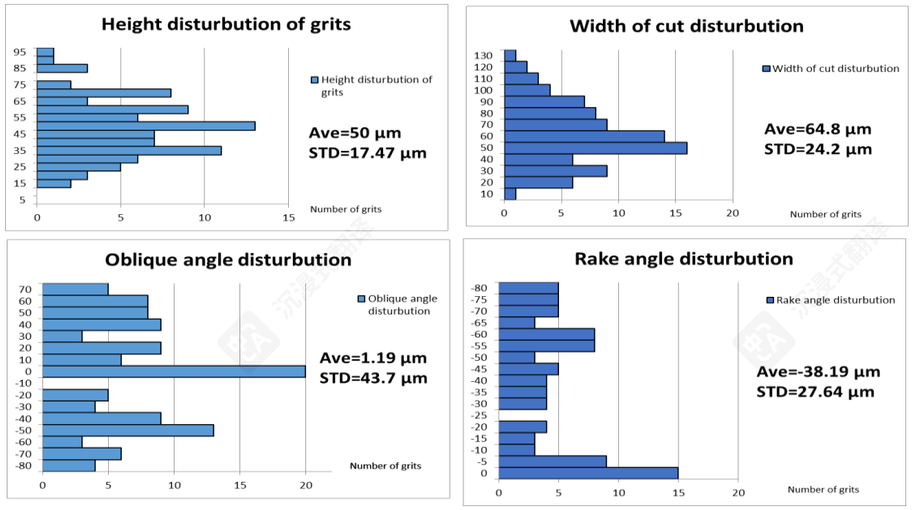



Figure 17: Measurement results of the presented wheel



关于砂轮性能的一些数据可以从制造商处获得，但由于磨料工具的随机性 [12]，特定于单个砂轮的某些性能无法在目录中找到。这些是单个颗粒的性能，如高度、磨削宽度、斜角和前角，它们会影响磨削力学。图16所示工具的颗粒测量结果见图17。该图表说明了测量的均值和标准偏差，以及数据的波动情况。这些测量是在共聚焦显微镜Nanofocus上进行的。

纳米聚焦设备有其局限性。由于它采用光扫描程序工作，轮面的一些反射误差可能导致3D模型创建时出现偏差。如果测量目标为小视场，数值可能存在于噪声范围内。磨削工具上的磨粒边缘半径会非常小，因此在纳米聚焦上进行的边缘半径测量不会产生准确结果。因此，边缘半径测量是在电子显微镜上进行的。图18提供了SEM机器的图像。

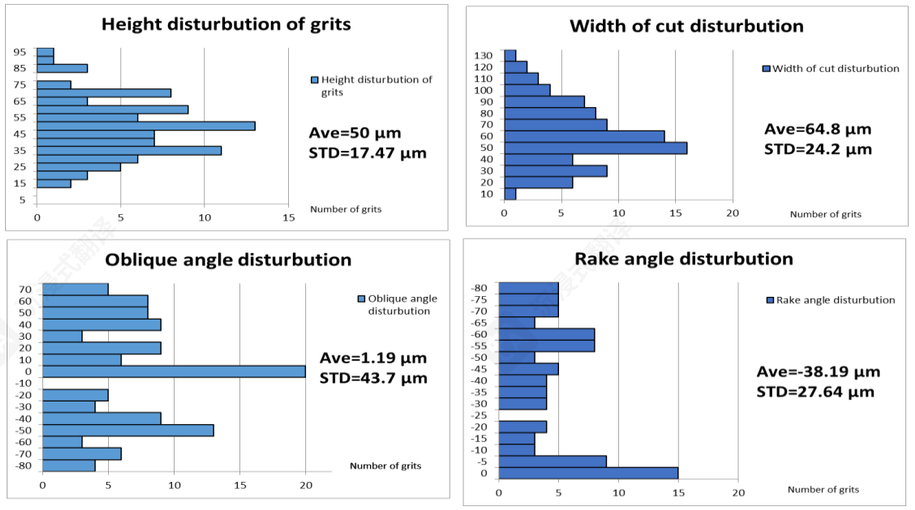



图17：所展示的轮子的测量结果

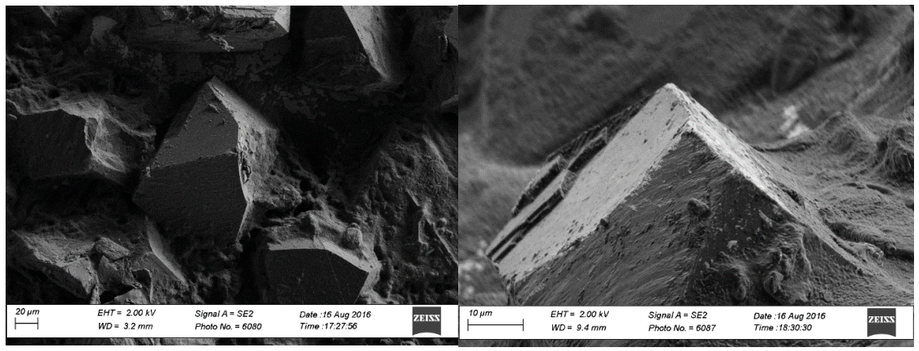



Figure 18: SEM captures of a single grit

It is believed that the measurements obtained from the SEM are more accurate than those of the Nanofocus. It is observed that the nose radius of some of the grits reside on nanoscale. Small edge (1 micrometer or lower) radii is a strong advantage of the CBN tools compared to the conventional wheels as low edge radius lead to less rubbing effect on the grinding zone [56]. This results in a grinding mechanism where while the chip removal is increased the rubbing effect is reduced yielding a more favorable situation regarding the decrease of process forces and temperatures.

### 3.1.2 Determination of Active grits

The principle for determining if one grit is active or not relies on the movement of the grits. The virtual wheel is moved and turned along the surface while the trajectories of the respectful grits are printed, and the intersections of these trajectories are carefully investigated.

The trajectory generation for each grit relies on its radial position. Due to their random heights, the grits will have different radii along the wheel. The model also considers the run-out amount of the wheel and adds it to the respective grit. The coordinates along the trajectory of the grit can be calculated as follows (1-3).

$$x_n = [R_n + Ro.\cos(\theta)\ ]\sin(\theta) + f.t \quad (1)$$

$$y_n = [R_n + Ro.\cos(\theta)](1 - \cos(\theta)) \quad (2)$$

$$R_n = r + H_n + Ro.\cos\ ((n-1).2\pi/N) \quad (3)$$



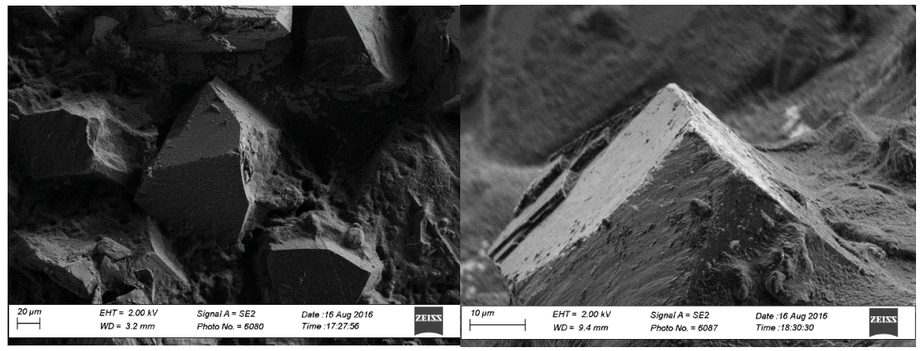



图18：单个磨料的SEM图像

人们认为，从SEM获得的测量结果比纳米聚焦的更准确。观察到一些磨粒的鼻半径位于纳米尺度。与传统砂轮相比，小边缘（1微米或更小）半径是CBN工具的一大优势，因为低边缘半径会导致磨削区摩擦效应减弱[56]。这导致了一种磨削机制，即在增加芯片去除量的同时减少摩擦效应，从而在降低工艺力和温度方面形成更有利的情况。

### 3.1.2 活性磨料测定

确定磨料是否活跃的原理基于磨料的运动。虚拟轮沿表面移动并旋转，同时打印各磨料的轨迹，并仔细研究这些轨迹的交点。

每个砂粒的轨迹生成依赖于其径向位置。由于它们的高度随机，砂粒在轮子上的半径会不同。该模型还考虑了轮子的磨耗量，并将其添加到相应的砂粒上。砂粒轨迹上的坐标可以按如下方式计算（1-3）。

$$x_n = [R_n + Ro.\cos(\theta)\ ]\sin(\theta) + f.t \quad (1)$$

$$y_n = [R_n + Ro.\cos(\theta)](1 - \cos(\theta)) \quad (2)$$

$$R_n = r + H_n + Ro.\cos\ ((n-1).2\pi/N) \quad (3)$$

where $R_n$ is the radius of the respective grit and $R_o$ is the runout amount of the grinding wheel. $H_n$ is the random grit height value, $f$ is the feed, $\theta$ is the immersion angle, $t$ is the time and N is the number of grits on the element. The kinematic trajectories are calculated according to the formulae given above and they are printed in x, y and z domain as shown in Figure 19.

When the measurements on the tool are completed and the virtual wheel is created with normal distribution, the simulation starts as the virtual wheel is turned on the surface of the workpiece. This is a real-time simulation and the trajectories for the active grits are created simultaneously step by step. The steps are discretized in terms of small angle increments and each grit has its respective start and end angles. The grits start to create print their trajectories as the corresponding approach angle is reached during the simulation. As the simulation starts, the first two grit trajectories are printed on the x-z plane as shown in Figure 20. There are 3 cases for the interaction:

i) There is no interaction and the second grit is shorter than the first one
ii) There is no interaction and the second grit is taller than the first one
iii) There is interaction

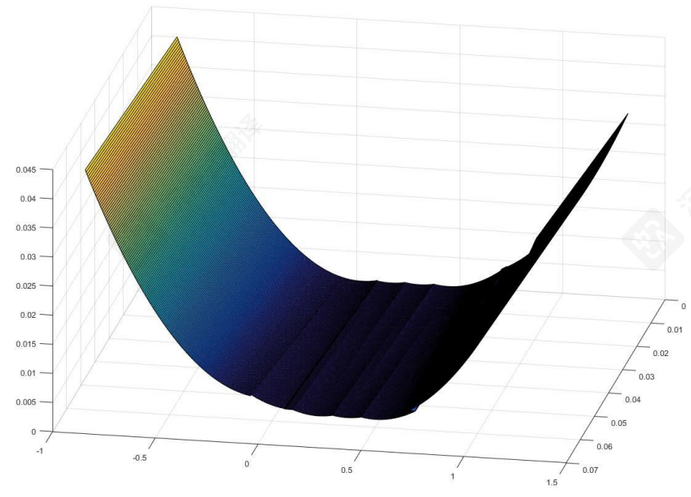



Figure 19: Kinematic trajectories of the grits (all units are in mm)



其中 $R_n$ 是相应砂粒的半径，$R_o$ 是砂轮的磨耗量。$H_n$ 是随机砂粒高度值，f 是进给量，$\theta$ 是浸入角，t 是时间，N 是元件上的砂粒数量。运动学轨迹根据上述公式计算，并按 x、y 和 z 域打印，如图 19 所示。

当工具的测量完成且虚拟轮以正态分布创建后，仿真开始，虚拟轮在工件表面旋转。这是一个实时仿真，主动磨粒的轨迹同时逐步创建。这些步骤以小角度增量离散化，每个磨粒有其对应的起始和结束角度。当仿真开始时，随着对应进给角度的达到，磨粒开始创建并打印其轨迹。仿真开始时，前两个磨粒的轨迹如图20所示打印在x-z平面上。交互有3种情况：

i) 无相互作用，且第二颗砂粒比第一颗短
ii) 无相互作用，且第二颗砂粒比第一颗高
iii) 存在相互作用

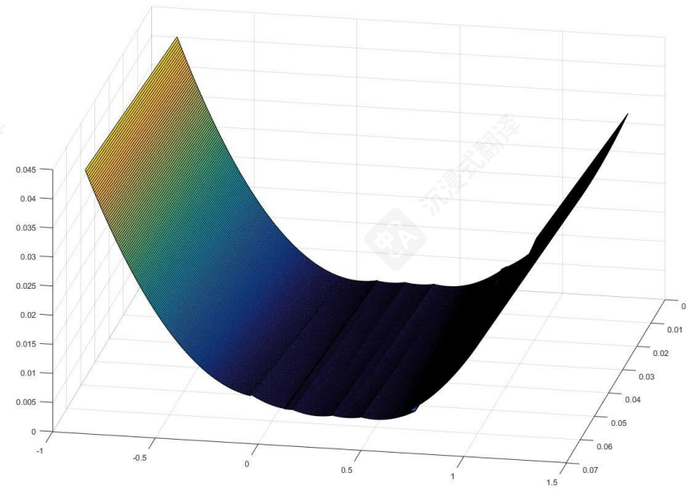



图19：砂粒的运动轨迹（所有单位均为毫米）

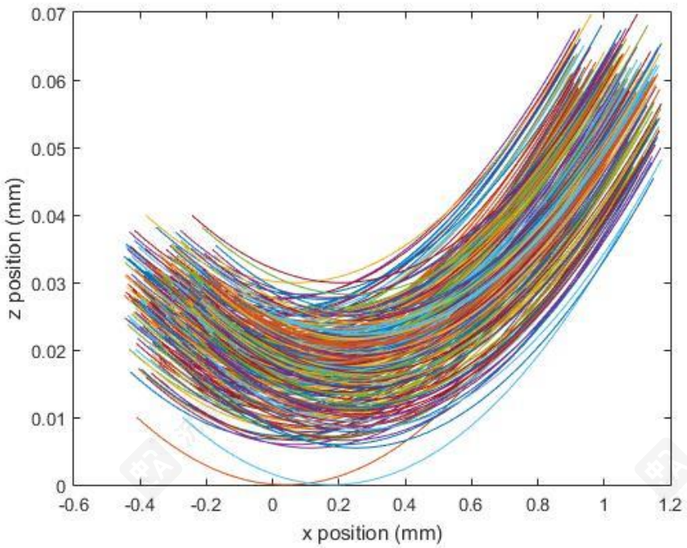



Figure 20: Trajectories on x-z plane

The first case implies that the first grit is dominant to the consecutive one. When the first grit has larger height than the second one it will remove some part of the workpiece material. When the second grit starts its trajectory movement it will not meet with the last surface of the workpiece that is defined by the previous grit. When this case is received in the algorithm it means that the first grit is flagged as active grit and the second grit is flagged as a passive grit.

The second case indicates that the second grit is dominant to the consecutive one. When the first grit starts its trajectory, it will meet with some of the workpiece material and will cut the workpiece. This doesn't guarantee that it is flagged as active because the algorithm waits for the second grit's response. The second grits will also meet with undeformed material on the surface. When the wheel completes its revolution, and comes to the same rotational orientation this time if the second grit isn't reflagged as passive, only then both of the grits are flagged as active. If the second grit is reflagged as passive in the previous program loops, then both grits are reflagged as passive. The trajectories for the depiction of first two cases are given in Figure 21.



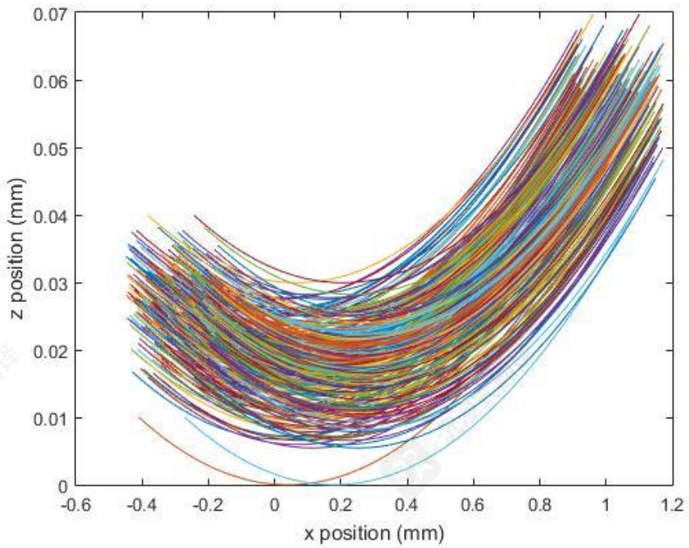



图20：x-z平面上的轨迹

第一种情况意味着第一个磨粒对后续磨粒起主导作用。当第一个磨粒的高度大于第二个磨粒时，它会去除工件材料的一部分。当第二个磨粒开始其轨迹运动时，它不会与由前一个磨粒定义的工件最后表面相遇。当算法接收到这种情况时，意味着第一个磨粒被标记为主动磨粒，第二个磨粒被标记为被动磨粒。

第二个情况表明，第二颗磨料相对于连续磨料占主导地位。当第一颗磨料开始其轨迹时，它会遇到一些工件材料并切割工件。但这并不能保证它被标记为活动状态，因为算法会等待第二颗磨料的响应。第二颗磨料也会遇到表面上的未变形材料。当砂轮完成其旋转时，如果这次第二颗磨料没有被重新标记为被动状态，只有到那时，两颗磨料才会被标记为活动状态。如果在之前的程序循环中第二颗磨料被重新标记为被动状态，那么两颗磨料都会被重新标记为被动状态。前两种情况的轨迹描绘如图21所示。

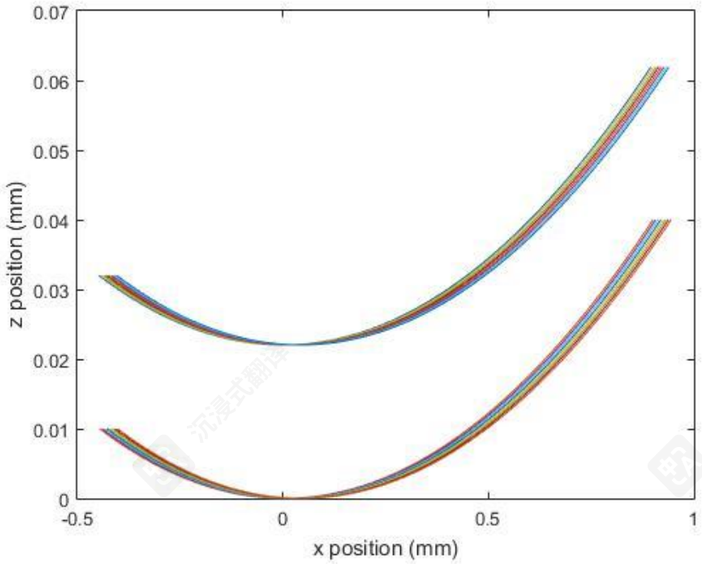



Figure 21: Depiction of case 1 or 2

The third case implies that the consecutive grits have similar heights. When there is a path interaction in the trajectory map. The second grit is flagged as active regardless of the situation of the first grit only if the second grit's trajectory creates a chip thickness that is larger than the minimum uncut chip thickness of the grit. The trajectories for the depiction of the third case is given in Figure 22. The critical chip thickness is defined as:

$$h_{cr} = r_t.[1 - \cos(\alpha)] \quad (3)$$

where $\alpha$ is stagnation angle which is taken as 55 degrees [57] and $r_t$ is edge radius of the grit. Equation 9 has been used in this study as a criterion to update the path generated by active grits. If maximum chip thickness created by a particular grit is less than the critical chip thickness the path is not updated even though the grit is considered as an active one.



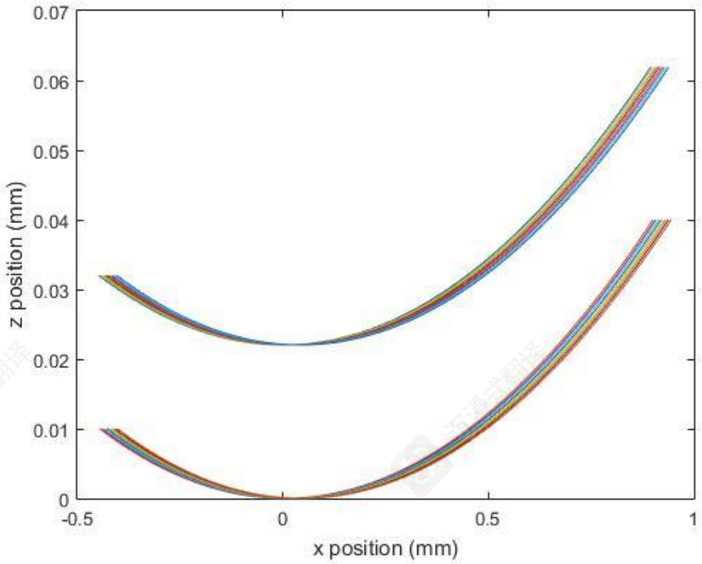



图21：案例1或2的示意图

第三种情况意味着连续磨料的高度相似。当轨迹图中存在路径交互时。只有当第二颗磨料的轨迹产生大于磨料最小未切割切屑厚度的切屑厚度时，第二颗磨料才会被标记为活动状态，而不管第一颗磨料的情况如何。第三种情况的轨迹描绘如图22所示。临界切屑厚度定义为：

$$h_{cr} = r_t.[1 - \cos(\alpha)] \quad (3)$$

$\alpha$ 处为停滞角，取值为55度[57]，$r_t$ 为磨粒的边半径。本研究采用方程9作为更新主动磨粒生成路径的标准。若某磨粒产生的最大切屑厚度小于临界切屑厚度，即使该磨粒被视为主动磨粒，路径也不会更新。

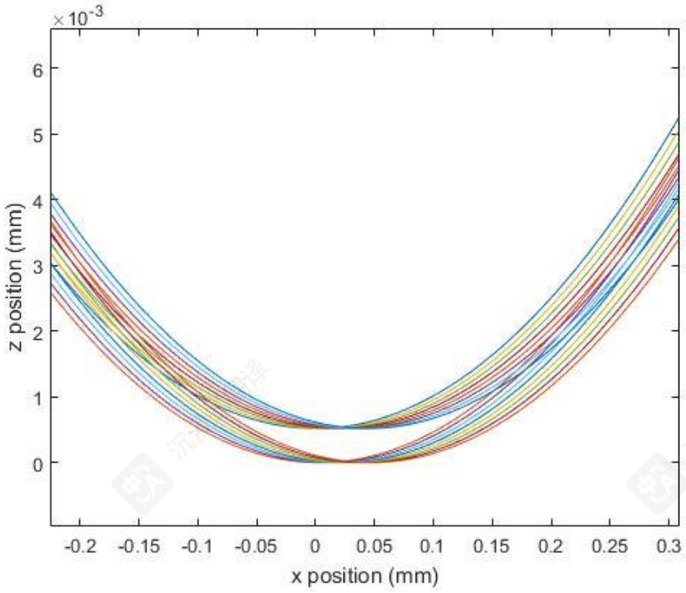



Figure 22: Depiction of case iii

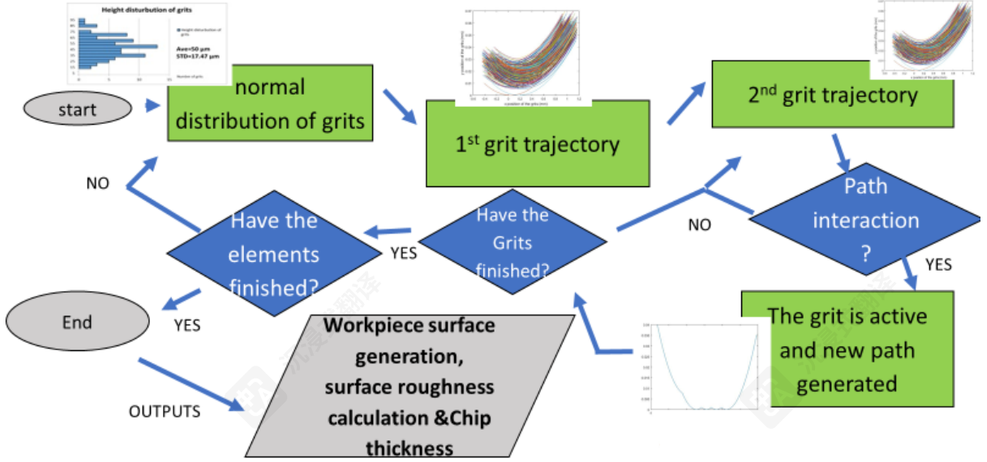



Figure 23: Flowchart of Active grit algorithm

When the trajectories of all the grits of the first element is completed and the active grits of the corresponding element is identified, the algorithm continues with the next element and with the same procedure until the elements are finished. When the elements are finished, the virtual wheel is recreated by omitting the passive grits. A demonstration of the virtual wheel is given in Figure 24.



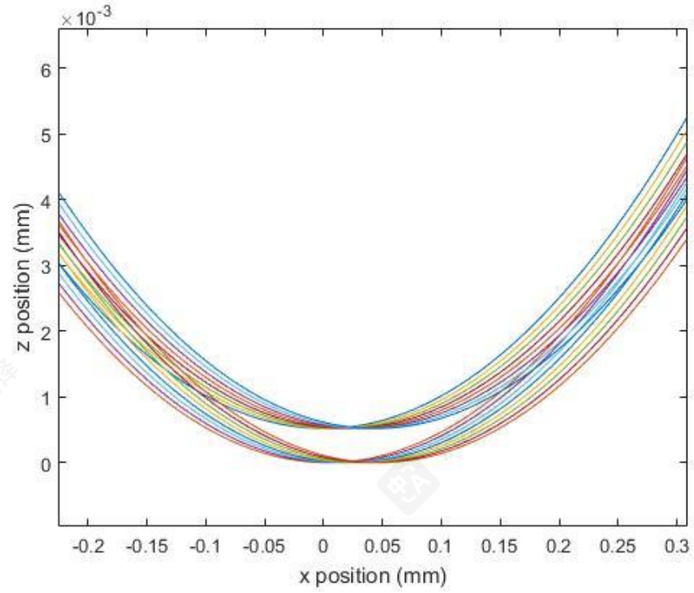



图22：案例iii的示意图

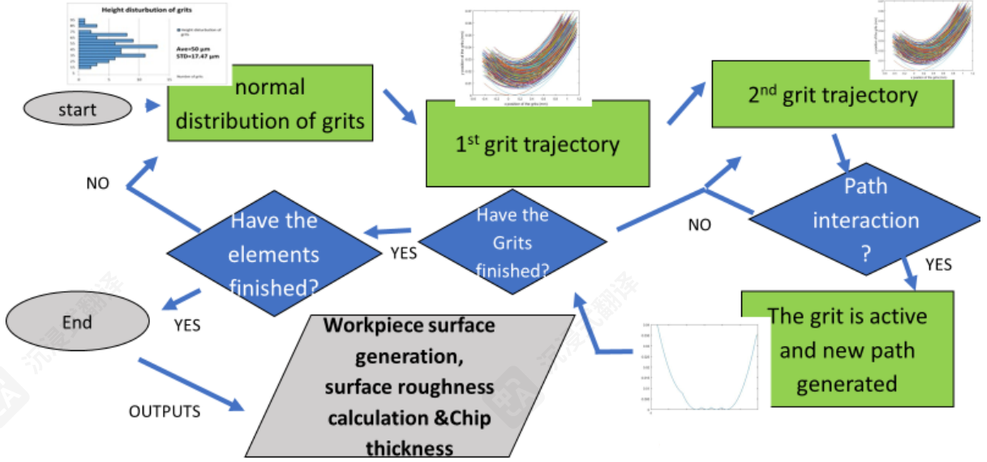



图23：Active grit算法流程图

当第一个元件的所有磨粒轨迹完成且确定相应元件的主动磨粒后，算法继续处理下一个元件，并采用相同步骤，直至所有元件处理完毕。当元件处理完毕后，通过省略被动磨粒重新创建虚拟砂轮。虚拟砂轮的示例如图24所示。

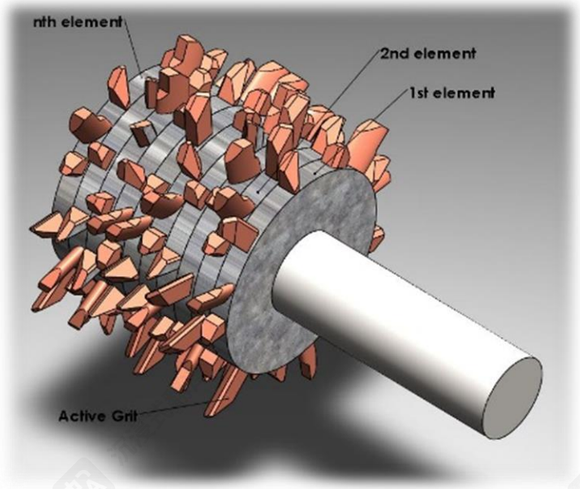



Figure 24: Recreated wheel with active grits only

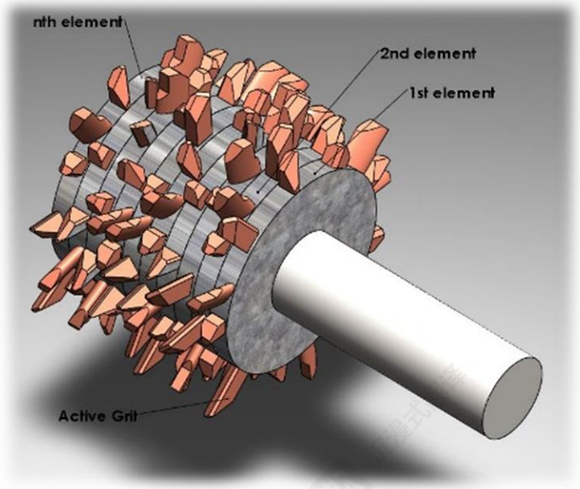



图24：仅含活性磨料的复原图

### 3.2 Surface Roughness in perpendicular to cutting direction

Surface roughness prediction is one of the fundamental aims in process modelling of the abrasive operations since the general use of grinding operations is to give the final geometry and surface quality to the workpiece. The surface roughness of the workpiece perpendicular to cutting direction can be obtained utilizing the geometric-kinematic model developed for this thesis. The model inherently accumulates the data that is required for the surface roughness calculation and it can be calculated instantaneously as the simulation is being performed. Please recall that the geometric-kinematic model is able to print the kinematic trajectories of the abrasive grits as the wheel is being turned and fed to the workpiece from Figure 20. While the trajectories are created for the selected element, the intersection point is investigated again to find the last surface created on the workpiece surface. The trajectories for the half of the revolution of the wheel is presented in Figure 25.

### 3.2 垂直于切削方向的表面粗糙度

表面粗糙度预测是磨削加工过程建模中的一个基本目标，因为磨削加工的普遍用途是给工件赋予最终的几何形状和表面质量。工件垂直于切削方向的表面粗糙度可以通过本论文开发的几何运动学模型获得。该模型本质上积累了表面粗糙度计算所需的数据，并且在模拟过程中可以即时计算。请回忆，几何运动学模型能够打印出砂轮旋转并进给到工件时磨粒的运动轨迹，如图20所示。当为选定元素创建轨迹时，会再次研究交点，以找到工件表面上最后形成的表面。砂轮旋转一半的轨迹如图25所示。


36

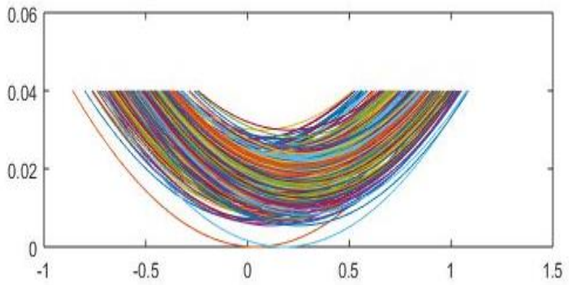



Figure 25: Trajectories for the surface roughness calculation

After the trajectories are printed the trajectories that belong to the passive grits are omitted. The trajectories of the active grits are investigated. The intersections of the consecutive active grits and the intersections between the workpiece surface is identified. The conjugate trajectory is found which has the contact over the workpiece. This trajectory indicated the last surface that is imprinted on the workpiece surface which is illustrated in Figure 26.

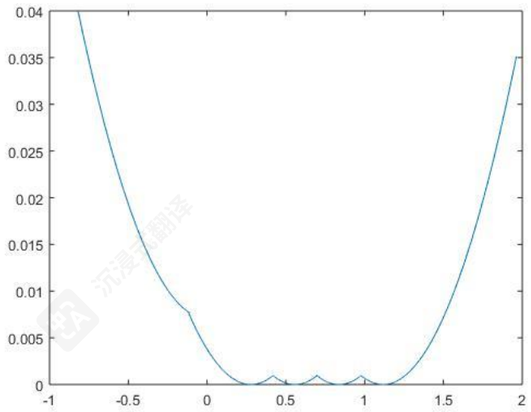



Figure 26: Last surface

The last surface on the workpiece has the information for the surface roughness in the cutting direction inherently and can be easily found the average values of the coordinate points on the trajectory. However, this information is a high underestimate of the



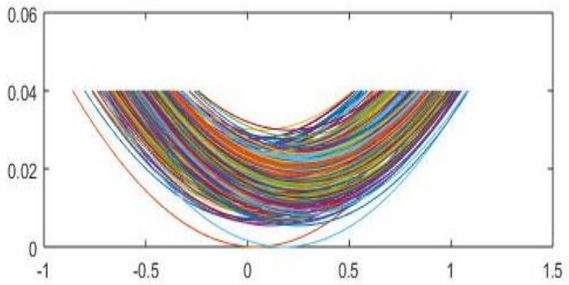



图25：表面粗糙度计算的轨迹

打印轨迹后，会忽略属于被动磨粒的轨迹。研究主动磨粒的轨迹。识别连续主动磨粒的交点以及工件表面的交点。找到共轭轨迹，该轨迹在工件上有接触。这条轨迹指示了最后印在工件表面上的表面，如图26所示。

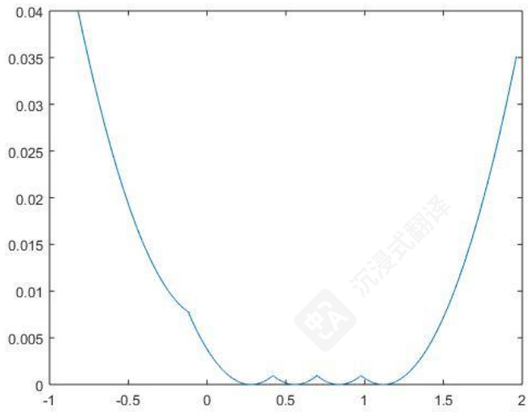



图26：最后表面

工件上最后一个表面内含切削方向的表面粗糙度信息，并且可以轻松找到该轨迹上坐标点的平均值。然而，这些信息对表面粗糙度的估计严重偏低，

surface roughness as it only represents the feed marks on the surface due to difference in consecutive active grit heights. The effect of vibration is included which has the dominant effect on the surface roughness in cutting direction which will be covered in next chapter.

The surface roughness in the perpendicular to cutting direction can be found when last surfaces of the grits reside on the rest of the elements are printed such as in Figure 27.

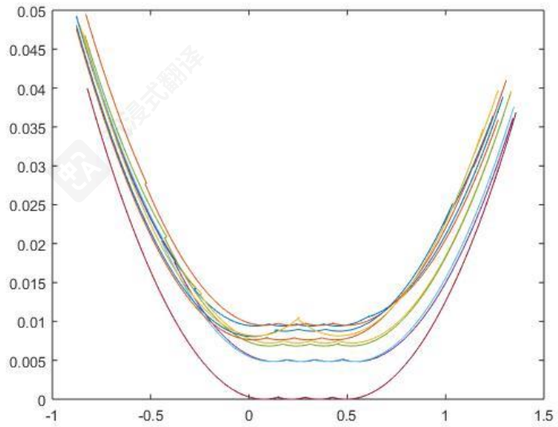



Figure 27: Last surfaces of all the elements

When all the last surfaces are created respective to each element, a basic 3Dprojection is applied to simulate the surface topography of the workpiece which is accurate in perpendicular to cutting direction. The surface is illustrated on Figure 28.

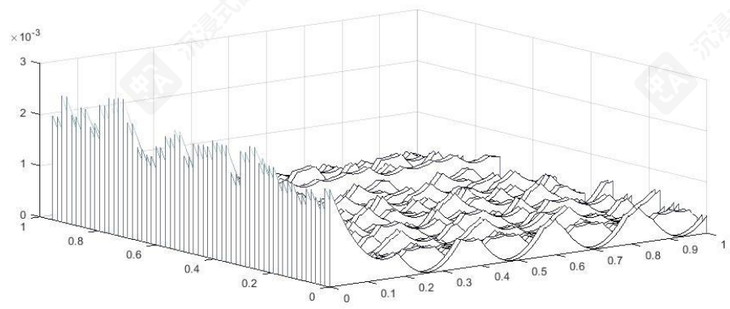



Figure 28: 3D projection of last surfaces



因为它仅代表由于连续工作砂粒高度差异而在表面留下的进给痕迹。振动的影响被包含在内，这对切削方向的表面粗糙度具有主导作用，将在下一章中详细讨论。

垂直于切削方向的表面粗糙度可以在磨粒最后表面的其余元素上打印时找到，例如图27所示。

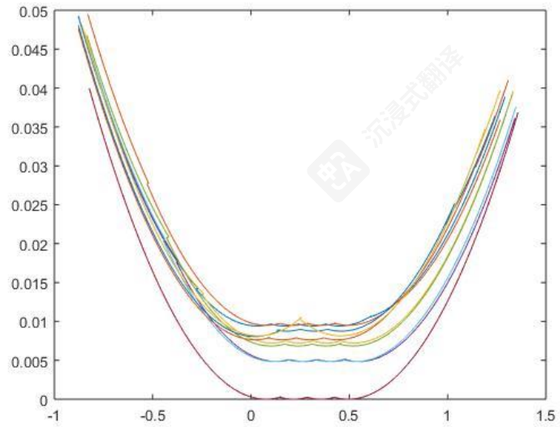



图27：所有元素的最后表面

当所有最后表面相对于每个元素创建完成后，会应用一个基本3D投影来模拟工件表面的形貌，该投影在垂直于切削方向上是准确的。该表面如图28所示。

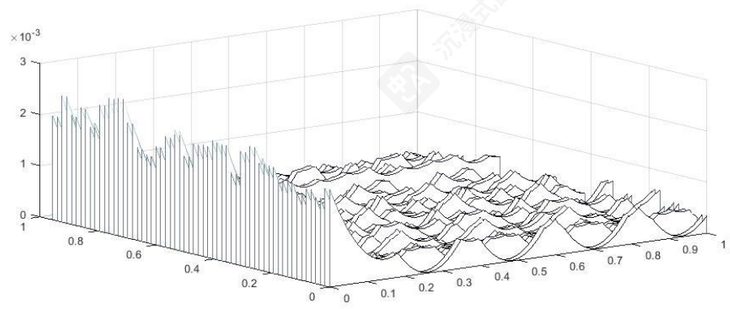



图28：最后表面的3D投影

The Ra value therefore can be calculated as the average height values of the last surface coordinates along the axial direction divided by the element number.

因此，Ra值可以计算为沿轴向方向最后表面坐标的平均高度值除以元素数量。

### 3.3 Grinding Force

### 3.3 磨削力

Since the milling analogy is used to simulate the wheel topography, the grinding force can be calculated after the resultant chip thickness is identified. The chip thickness in grinding is generally very low (in the magnitude of microns, sometimes lower) and there are a number of ways to predict it [1]. Apart from the empirical ways to calculate the chip thickness, the geometric-kinematic model can also be used to determine the resultant chip thickness during the grinding operation.

由于采用铣削类比来模拟轮廓形貌，因此可在确定合成切屑厚度后计算磨削力。磨削中的切屑厚度通常非常低（以微米计，有时更低），并且有多种方法可以预测它 [1]。除了通过经验方法计算切屑厚度外，几何运动学模型也可用于确定磨削过程中的合成切屑厚度。

#### 3.3.1 Undeformed Chip Thickness

#### 3.3.1 未变形切屑厚度

Just like the surface roughness the trajectories of the consecutive active grits and the intersections have the inherent data for the chip thickness calculation. If the milling analogy is applied again, having active grits which have different height distribution means the model is now analogous to milling cutter with run-out problem. The milling cutter with runout can be treated as it has different height values for each cutting flute. The milling cutter with run-out is simulated in Figure 29.

就像表面粗糙度一样，连续有效磨粒的轨迹和交点具有芯片厚度计算的固有数据。如果再次应用铣削类比，具有不同高度分布的有效磨粒意味着该模型现在类似于存在振刀问题的铣刀。存在振刀的铣刀可以视为每个切削刃具有不同高度值。存在振刀的铣刀在图29中进行了模拟。

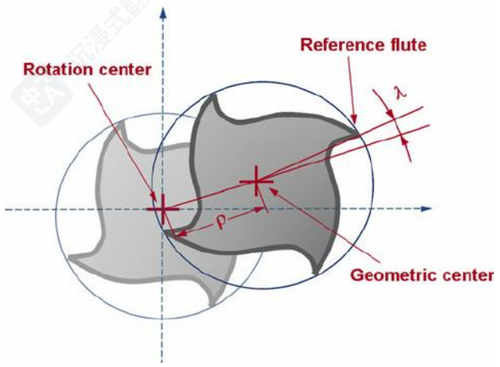



Figure 29: Milling cutter with run-out

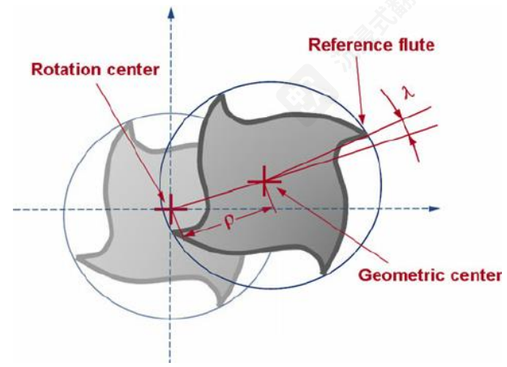



图29：带跑合的铣刀

The instantaneous chip thickness can be calculated as the geometric-kinematic simulation is being carried out. The chip thickness is the difference between the radial differences of the consecutive active grits in addition to the feed per active grit. The formula for the calculation is given in equations (4-5):

$$h_{i,j} = R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) \quad 0 < \theta_{i,j} < \theta_{exit,i,j} \tag{4}$$

$$h_{max} = R_n - R_m + (n-m).f_t \sin(\theta_{exit,i,j}) \tag{5}$$

where h is the undeformed chip thickness per grit, with the indices *i* and *j* represent the *i*'th active grit on the *j*'th element. Indices n and m represent the consecutive positions of the active grits along with the circumference of the wheel. Angle θ represents the immersion angle of the current grit. Finally, $f_t$ indicates the feed rate. The calculation of the grinding force is a straightforward multiplication of the calculated undeformed chip area with the cutting force coefficients.

### 3.3.2 Mechanistic Force Model

Since the milling analogy is employed, the methods to predict the force can be categorized in two ways. The force can be computed using mechanistic or analytical approaches. The mechanistic force model requires applications of a number of tests in order to calibrate the force coefficients.

Assuming grinding wheel acts similar to a milling tool where each grain removes material from the workpiece, the milling analogy can be applied for grinding process. Axial force can be neglected here because the random distribution of the oblique angles (can change between positive and negative values) results in a very small force on the dynamometer [12]. Force components can be projected in x (cutting direction) and y (normal direction) directions as given in eq.1 where $F_t$ and $F_n$ are tangential and normal forces and φ is instantaneous immersion angles:(Minus/plus signs are for up/down grinding, respectively):

$$F_x = -F_t \cos(\varphi) \mp F_n \sin(\varphi) \tag{6}$$

$$F_y = F_t \sin(\varphi) \mp F_n \cos(\varphi) \tag{7}$$



瞬时芯片厚度可以在几何-运动学模拟进行时计算。芯片厚度是连续有效磨粒的径向差之和，再加上每个有效磨粒的进给量。计算公式在公式(4-5)中给出：

$$h_{i,j} = R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) \quad 0 < \theta_{i,j} < \theta_{exit,i,j} \tag{4}$$

$$h_{max} = R_n - R_m + (n-m).f_t \sin(\theta_{exit,i,j}) \tag{5}$$

其中 h 表示每颗磨粒的未变形切屑厚度，索引 i 和 j 分别表示第 j 个元件上的第 i 颗有效磨粒。索引 n 和 m 表示有效磨粒沿轮缘的连续位置。角度 θ 代表当前磨粒的浸入角。最后，$f_t$ 表示进给速度。磨削力的计算是计算出的未变形切屑面积与切削力系数的简单乘积。

### 3.3.2 机理力模型

由于采用了铣削类比，预测力的方法可以分为两种。力可以使用机理或解析方法进行计算。机理力模型需要应用多次试验来校准力系数。

假设砂轮类似于铣刀，每个颗粒从工件上去除材料，则可将铣削类比应用于磨削过程。此处可忽略轴向力，因为倾斜角的随机分布（可正可负）导致在测力计[12]上的力非常小。力分量可按公式1投影到x（切削方向）和y（法向）方向，其中$F_t$ 和$F_n$ 分别为切向力和法向力， φ 为瞬时浸入角：（正负号分别对应向上/向下磨削）：

$$F_x = -F_t \cos(\varphi) \mp F_n \sin(\varphi) \tag{6}$$

$$F_y = F_t \sin(\varphi) \mp F_n \cos(\varphi) \tag{7}$$

Analytically derived average cutting forces can be obtained as it can be seen in formulae (8-9) (detailed calculation can be found in [58]):

$$\bar{F}_x = \left\{\frac{Nfa}{8\pi}[K_{tc}\cos(2\varphi) - K_{nc}(2\varphi - \sin(2\varphi))]\right\}_{\varphi_{st}}^{\varphi_{ex}} \quad (8)$$

$$\bar{F}_y = \left\{\frac{Nfa}{8\pi}[K_{tc}(2\varphi - \sin(2\varphi)) + K_{nc}\cos(2\varphi)]\right\}_{\varphi_{st}}^{\varphi_{ex}} \quad (9)$$

$K_{tc}$ and $K_{nc}$ are the cutting force coefficients in tangential and normal directions, respectively, $a$ is axial depth of cut, $N$ is number of active grits which are defined around any line on the wheel periphery which participate in chip formation process which are the output of the geometric-kinematic model. $f$ is feed per revolution per active grit.

In mechanistic approach in order to obtain force coefficients, instead of calculation of cutting variables such as shear angle, shear stress etc., a set of grinding tests are conducted at same wheel speed, axial and radial depth of cut but different feed rates. Results are used to identify force coefficients from Eq. 2. So, the coefficients are obtained experimentally [58]. Once the force coefficients are known, it is possible to predict grinding forces at any arbitrary condition in the range of experimental data.

### 3.3.3 Thermomechanical Force Model

In addition to mechanistic approach, force coefficients can be calculated analytically using mechanics of oblique cutting [58]. Analytical expressions for force coefficients are given below (10-11).

$$K_{tc} = \frac{\tau_s}{\sin\varphi_n}\frac{\cos(\beta_n-\alpha_n)+\tan i\tan\eta\sin\beta_n}{\sqrt{cos^2(\varphi_n+\beta_n-\alpha_n)+tan^2\eta\, sin^2\beta_n}} \quad (10)$$

$$K_{nc} = \frac{\tau_s}{\sin\varphi_n\cos i}\frac{\sin(\beta_n-\alpha_n)}{\sqrt{cos^2(\varphi_n+\beta_n-\alpha_n)+tan^2\eta\, sin^2\beta_n}} \quad (11)$$

where $\tau_s$ is shear stress, $\varphi_n$ is shear angle, $\beta_n$ is friction angle, $\alpha_n$ is rake angle, $\eta$ is chip flow angle and $i$ is oblique angle. In order to obtain force coefficients, three important variables i.e. shear stress ($\tau_s$), shear angle ($\varphi_n$) and friction angle ($\beta_n$) should be known. Chip flow angle ($\eta$) can be taken as equal to oblique angle ($i$) based on Stabler's empirical rule. Oblique and rake angles are obtained by investigation of grit



可通过公式（8-9）解析推导出平均切削力，如公式所示（详细计算可参见[58]）：

$$\bar{F}_x = \left\{\frac{Nfa}{8\pi}[K_{tc}\cos(2\varphi) - K_{nc}(2\varphi - \sin(2\varphi))]\right\}_{\varphi_{st}}^{\varphi_{ex}} \quad (8)$$

$$\bar{F}_y = \{\frac{Nfa}{8\pi}[K_{tc}(2\varphi - \sin(2\varphi)) + K_{nc}\cos(2\varphi)]\}_{\varphi_{st}}^{\varphi_{ex}} \quad (9)$$

$K_{tc}$ 和 $K_{nc}$ 分别是切向和法向的切削力系数，a是轴向切深，N是参与切削过程的有效磨粒数，这些磨粒被定义为在砂轮周边的任何线上参与切屑形成过程，它们是几何-运动学模型的输出。f是每转每有效磨粒的进给量。

在机理方法中，为了获得力系数，而不是计算剪切角、剪切应力等切削变量，而是在相同的砂轮速度、轴向和径向切深但不同的进给速率下进行一组磨削试验。结果用于根据公式2识别力系数。因此，这些系数是实验测得的[58]。一旦知道力系数，就可以在实验数据范围内预测任何任意条件下的磨削力。

3.3.3 热力力学力模型

除了机械方法外，还可以使用斜切力学 [58] 分析计算力系数。力系数的分析表达式如下（10-11）。

$$K_{tc} = \frac{\tau_s}{\sin\varphi_n}\frac{\cos(\beta_n-\alpha_n)+\tan i\tan\eta\sin\beta_n}{\sqrt{cos^2(\varphi_n+\beta_n-\alpha_n)+tan^2\eta\, sin^2\beta_n}} \quad (10)$$

$$K_{nc} = \frac{\tau_s}{\sin\varphi_n\cos i}\frac{\sin(\beta_n-\alpha_n)}{\sqrt{cos^2(\varphi_n+\beta_n-\alpha_n)+tan^2\eta\, sin^2\beta_n}} \quad (11)$$

在 $\tau_s$ 为剪切应力，$\varphi_n$ 为剪切角，$\beta_n$ 为摩擦角，$\alpha_n$ 为挖沟角，$\eta$ 为切屑流动角和 $i$ 为斜角。为了获得力系数，三个重要变量，即剪切应力 ($\tau_s$)，剪切角 ($\varphi_n$) 和摩擦角 ($\beta_n$) 应该是已知的。根据 Stabler 经验法则，切屑流动角 ($\eta$) 可以取等于斜角 ($i$)。斜角和挖沟角是通过研究沙粒获得的

properties which will be described in the next section. In this study, shear stress was obtained by using Johnson-Cook model [40]:

$$\tau = \frac{1}{\sqrt{3}}\left[A + B\left(\frac{\gamma}{\sqrt{3}}\right)^{n}\right]\left[1 + C.\ln\left(\frac{\dot{\gamma}}{\dot{\gamma}_0}\right)\right]\left[1 - \left(\frac{T - T_r}{T_m - T_r}\right)^{m}\right] \quad (12)$$

The Johnson-Cook equation constants is given in [59].

| A (Mpa) | B | C | m | n |
|---|---|---|---|---|
| 1485 | 904 | 0.015 | 1.689 | 0.77 |

Table 4: Constants of Johnson-Cook equation for Inconel 718.

The shear strain and the strain rate are given [58]:

$$\gamma = \frac{\cos(\alpha_n)}{\sin(\varphi_n)\cos(\varphi_n - \alpha_n)} \quad (13)$$

$$\dot{\gamma} = \frac{V\cos(\alpha_n)}{\mathrm{d}\cos(\varphi_n - \alpha_n)} \quad (14)$$

where $V$ is cutting speed and $d$ is the shear zone thickness which can be approximated to 0.15 of shear plane length ($l$) where [58]

$$\mathrm{l} = \frac{h}{\sin(\varphi_n)} \quad (15)$$

where h is the uncut chip thickness that is calculated through the geometric-kinematic model. $d$ was approximated as 1 micron in this study. During the analysis, it has been observed that thickness of shear plane did not affect the shear stress significantly. This fact was also reported in [46].

Friction angle and the shear angle (using the minimum energy principle) can be approximated as [58]

$$\beta_n = \tan^{-1}\frac{F_y}{F_x} + \alpha_n \quad (16)$$

$$\varphi_n = \frac{\pi}{4} - \frac{(\beta_n - \alpha_n)}{2} \quad (17)$$

A few grinding tests were done at different feed rates to identify the friction angle using equation (13) after subtracting the ploughing forces. It varied from 32.5 to 35.3 degree an average value of 33.9 degree was used here. Details can be found in [40].



下一节将描述的属性。在本研究中，剪切应力是通过使用 Johnson-Cook 模型 [40] 获得的：

$$\tau = \frac{1}{\sqrt{3}}\left[A + B\left(\frac{\gamma}{\sqrt{3}}\right)^{n}\right]\left[1 + C.\ln\left(\frac{\dot{\gamma}}{\dot{\gamma}_0}\right)\right]\left[1 - \left(\frac{T - T_r}{T_m - T_r}\right)^{m}\right] \quad (12)$$

Johnson-Cook方程的常数在[59]中给出。

| A (MPa) | B | C | m | n |
|---|---|---|---|---|
| 1485 | 904 | 0.015 | 1.689 | 0.77 |

表4：Inconel 718的Johnson-Cook方程常数。

剪切应变和应变率在[58]中给出：

$$\gamma = \frac{\cos(\alpha_n)}{\sin(\varphi_n)\cos(\varphi_n - \alpha_n)} \quad (13)$$

$$\dot{\gamma} = \frac{V\cos(\alpha_n)}{\mathrm{d}\cos(\varphi_n - \alpha_n)} \quad (14)$$

其中V是切削速度，d是剪切区厚度，可近似为剪切平面长度（l）的0.15倍，在[58]

$$\mathrm{l} = \frac{h}{\sin(\varphi_n)} \quad (15)$$

其中h是未切削芯片厚度，通过几何运动学模型计算。本研究中，d被近似为1微米。在分析过程中观察到，剪切平面厚度对剪切应力没有显著影响。这一事实也在[46]中有所报道。

摩擦角和剪切角（使用最小能量原理）可以近似为 [58]

$$\beta_n = \tan^{-1}\frac{F_y}{F_x} + \alpha_n \quad (16)$$

$$\varphi_n = \frac{\pi}{4} - \frac{(\beta_n - \alpha_n)}{2} \quad (17)$$

在多种进给速率下进行了若干磨削试验，通过减去犁削力后使用公式(13)来确定摩擦角。该角在32.5至35.3度之间变化，此处采用平均值为33.9度。详细信息可参见[40]。

The temperature in the shear zone can be calculated iteratively as the force is calculated. The detailed calculation can be found in [1].

A straightforward approach to identify the shear stress, friction and shear angles is using orthogonal database in which those values are defined experimentally for each pair of workpiece-tool at corresponding rake angle. Due to varying and high negative rake angle of the grain and very small undeformed chip thickness in grinding process, it is difficult and time consuming to conduct orthogonal cutting tests. Accordingly, in this work equations (12) to (17) was used instead. The total average force is obtained by summing up the force for all active grits.

Even though the mechanistic model can predict the grinding force accurately, it requires a set of experiments to be conducted for identification of the grinding force coefficients and calibrations are needed for various feed, speed, depth and grit parameters. The thermomechanical model on the other hand, can predict the grinding force with slightly poorer discrepancy while being applicable to wider range as no calibration is needed for varying process and wheel parameters. Direct analytical relationships can be developed between the inputs and outputs of the grinding process.

The flowchart for grinding force calculation is presented in Figure 30. As it can be seen, the first step for predicting the grinding force is optical measurements done on the grinding tools. Where the geometric properties of the abrasive grits can be identified for a substantial number of samples. Then the normal distribution is applied to the geometric properties to create a broader sample space in order to mimic the wheel topography. After the wheel is modeled, the kinematic trajectories of the grits are printed using the geometric-kinematic model. The geometric-kinematic model allows the prediction of;

i) The number of active grits
ii) Surface roughness in perpendicular to cutting direction
iii) Instantaneous uncut chip thickness

By using the newly found uncut chip thickness the simulation can predict the force either directly by mechanistically identified grinding force coefficients or through the material model using the thermo-mechanical force calculation.



剪切区的温度可随力的计算进行迭代计算。详细计算方法可参见[1]。

一种直接识别剪切应力、摩擦力和剪切角的方法是使用正交数据库，其中这些值通过实验为每个工件-刀具对的相应前角定义。由于晶粒具有变化且为高度负前角，并且在磨削过程中变形切屑厚度非常小，因此进行正交切削试验既困难又耗时。因此，在本工作中，使用了公式(12)至(17)。总平均力是通过将所有活动砂粒的力相加获得的。

尽管机械模型可以准确预测磨削力，但它需要一组实验来识别磨削力系数，并且需要对各种进给、速度、深度和砂粒参数进行校准。另一方面，热力学模型在预测磨削力时存在略微较差的差异，但适用范围更广，因为它不需要对变化的过程和轮子参数进行校准。可以直接开发磨削过程输入和输出之间的解析关系。

磨削力计算的流程图展示在图30中。可以看出，预测磨削力的第一步是在磨削工具上进行的的光学测量。其中，可以通过对大量样品进行测量来识别磨料颗粒的几何特性。然后，将正态分布应用于几何特性，以创建更广泛的样本空间，以便模拟砂轮形貌。在砂轮建模后，使用几何-运动学模型打印出磨料颗粒的运动学轨迹。几何-运动学模型允许预测；

i) 有效磨料颗粒的数量
ii) 垂直于切削方向的表面粗糙度
iii) 瞬时未切削切屑厚度

通过使用新发现的未切割芯片厚度，模拟可以直接通过机械识别的磨削力系数来预测力，或通过材料模型使用热力学力计算。

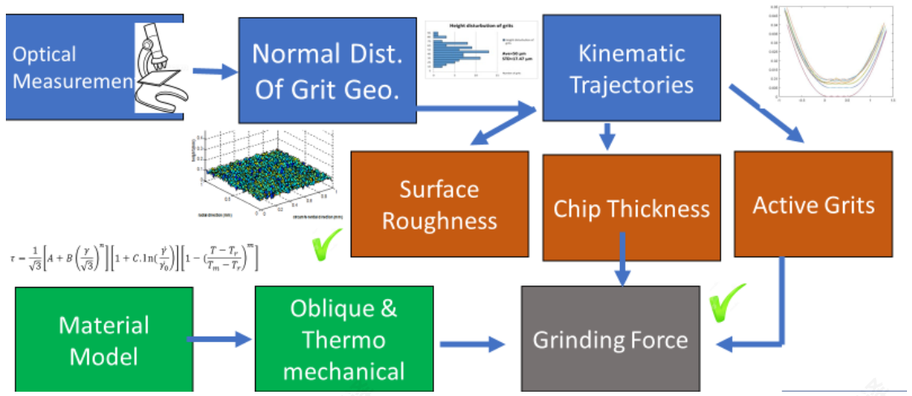



Figure 30: Flowchart for force calculation

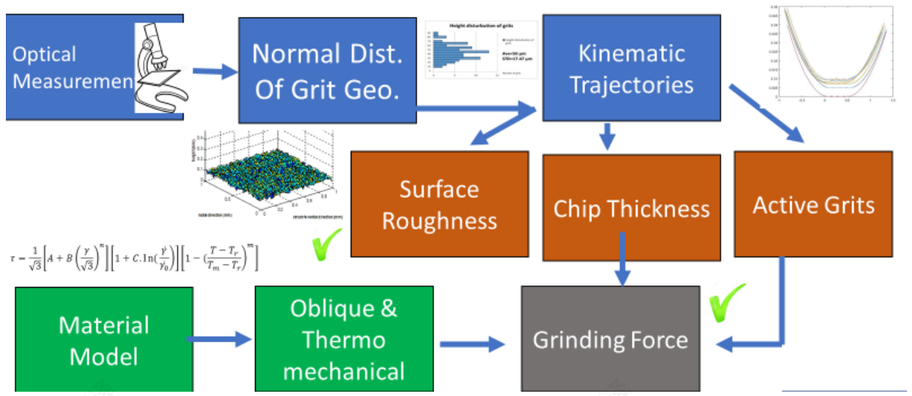



图30：力计算流程图

### 3.4 Results

In this section of the chapter, the results obtained for the active grits, surface roughness and grinding force are presented. For calculation of active grits, the experimental verification can be a challenging task which will be covered later, however the experimental verification for the workpiece surface roughness and grinding force is compared with the outputs from the simulations and the models are verified.

### 3.4 结果

在本章节的这一部分，展示了针对有效磨粒、表面粗糙度和磨削力的结果。对于有效磨粒的计算，实验验证可能是一项具有挑战性的任务，将在后面涵盖，但是工件表面粗糙度和磨削力的实验验证与模拟输出进行比较，模型得到验证。

#### 3.4.1 Active grits

First of all, large number of simulations are done in order to have a database of active grit numbers for different cutting conditions. This is the most crucial as it will be the key parameter for assigning the tooth number (milling analogy) and determining an average value for feed per tooth.

The algorithm for determining the number of active grits is a randomized problem. In the beginning of the simulation/computation a large amount of sample space is created from a smaller amount determined by the optical measurements. In this method, Gaussian distribution model is used (mean, std. Deviation values determine the larger sample space). [1,11,12]

#### 3.4.1 活齿

首先，进行了大量模拟，以建立不同切削条件下的活齿数量数据库。这是最关键的，因为它将是分配齿数（铣削类比）和确定每齿进给平均值的参数。

确定活动砂粒数量的算法是一个随机问题。在模拟/计算的开始阶段，会从光学测量确定的小量样本空间中创建大量样本空间。在此方法中，使用高斯分布模型（均值、标准差值决定了更大的样本空间）。[1,11,12]

This means that when the simulation is run, a new larger sample space is created from scratch based on the smaller sample space input. Although this will result in a different grinding wheel model in each run, the idea in this model is that if one includes sufficient number of elements for the wheel input, all program outputs such as grinding force, active grit number …etc. will converge around an average value [1], [11], [12]. (The reason being in the Gaussian distribution, the values closer to the average will have higher probabilities). Having enough amount of sample space is generally possible since in the scope of this thesis, axial depth of cuts between 7-10mm are used. The simulation results are given for 40k rpm in Figure 31.




Figure 31: Active grit number table

The active grit simulation results shown in Figure 31 belong to the wheel presented in Figure 16.

It is observed that the active grit number increase with the table speed is linear, and it increases with the radial depth as well. However, through a larger set of simulations it is realized that the number of active grits directly depend on feed per revolution and the effect of radial depth of cut is small compared to the effect of the feed per revolution. A number of simulations were also carried out for the wheels given in Table 5. A first glance shows that the radial depth of cut increase in the process does not increase the active grits very much, not much as the effect of the feed per revolution at least. This situation is shown in Figure 32 for the depth of cuts between 35$\mu$m and 200$\mu$m (this region stands in the experiments range for the scope of this thesis). It can be seen that



这意味着当运行模拟时，会根据较小的样本空间输入从零开始创建一个新的更大的样本空间。虽然这会导致每次运行时磨削轮模型不同，但该模型的思想是：如果为轮输入包含足够数量的元素，所有程序输出（如磨削力、活动砂粒数量等）都将趋近于一个平均值 [1], [11], [12]。（原因在于高斯分布中，更接近平均值的数值具有更高的概率）。由于在此论文的范围内，切深在7-10mm之间，因此通常有足够的样本空间。图31给出了40k rpm的模拟结果。




图31：有效磨料数量表

图31中展示的有效磨料模拟结果属于图16中所示的车轮。

观察到，随着工作台速度的增加，有效磨粒数呈线性增长，并且随着径向切深也增加。然而，通过更大规模的模拟可以发现，有效磨粒数直接取决于每转进给量，而径向切深的影响相比之下较小。表5中给出的砂轮也进行了多次模拟。初步观察显示，在加工过程中径向切深增加并不会显著增加有效磨粒数，至少其影响不如每转进给量的影响大。这种情况在图32中对于35$\mu$m至200$\mu$m范围内的切深（该区域在本论文的实验范围内）得到了体现。可以看出，

Wheel 1's number of active grits doesn't have a major change when the radial depth of cut is increased. It is worth to mention that the numbers of active grits are not integer as average values are calculated from larger simulations.

| Wheel no | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type | Mounted point | Mounted point | Mounted point | Mounted point | Mounted point |
| Grit Material | CBN | CBN | CBN | CBN | CBN |
| Diameter (mm) | 10 | 5 | 30 | 10 | 10 |
| Grit size | B126 | B64 | B151 | B126 | B126 |
| Bond Type | Electoplated | Electoplated | Electoplated | Electoplated | Electoplated |
| Condition | Fresh | Fresh | Fresh | Dressed | Worn |
| C number | 33 | 97 | 23 | 33 | 31 |
| Grit width (μm) | 65 | 41 | 73 | 67 | 39 |
| Edge radius (μm) | 1.15 | 1 | 1.07 | 1 | 1.37 |
| Mean and std. dev. Of grit height (μm) | 46 & 18 | 30 & 11 | 88 & 17 | 37 & 8 | 25 & 0.2 |
| Mean and std. dev. Of width of cut (μm) | 18 & 6 | 9 & 4 | 23 & 11 | 17 & 7 | 16 & 7 |
| Average rake angle (degrees) | -52 | -51 | -54 | -53 | -57 |
| Average oblique angle (degrees) | 24 | 21 | 25 | 31 | 30 |

Table 5: Wheel specifications for the active grit calculations



当径向切深增加时，轮1的有效磨粒数没有发生明显变化。值得一提的是，有效磨粒数并非整数，因为平均值为从更大规模的模拟中计算得出。

| 轮号 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type | 安装点 | 安装点 | 安装点 | 挂载点 | 挂载点 |
| 磨料材料 | CBN | CBN | CBN | CBN | CBN |
| 直径 (mm) | 10 | 5 | 30 | 10 | 10 |
| 磨料尺寸 | B126 | B64 | B151 | B126 | B126 |
| 粘结类型 | 电镀 | 电镀 | 电镀 | 电镀 | 电镀 |
| 条件 | 新鲜 | 新鲜 | 新鲜 | 已处理 | Worn |
| C数量 | 33 | 97 | 23 | 33 | 31 |
| 磨粒宽度(μm) | 65 | 41 | 73 | 67 | 39 |
| 边缘半径(μm) | 1.15 | 1 | 1.07 | 1 | 1.37 |
| 平均值和标准差磨粒的偏差高度(μm) | 46 & 18 | 30 & 11 | 88 & 17 | 37 & 8 | 25 & 0.2 |
| 均值和标准差 dev. 宽度 切割 (μm) | 18 & 6 | 9 & 4 | 23 & 11 | 17 & 7 | 16 & 7 |
| 平均抽水率角度（度） | -52 | -51 | -54 | -53 | -57 |
| 平均斜角角度（度） | 24 | 21 | 25 | 31 | 30 |

表5：主动研磨计算用轮规格

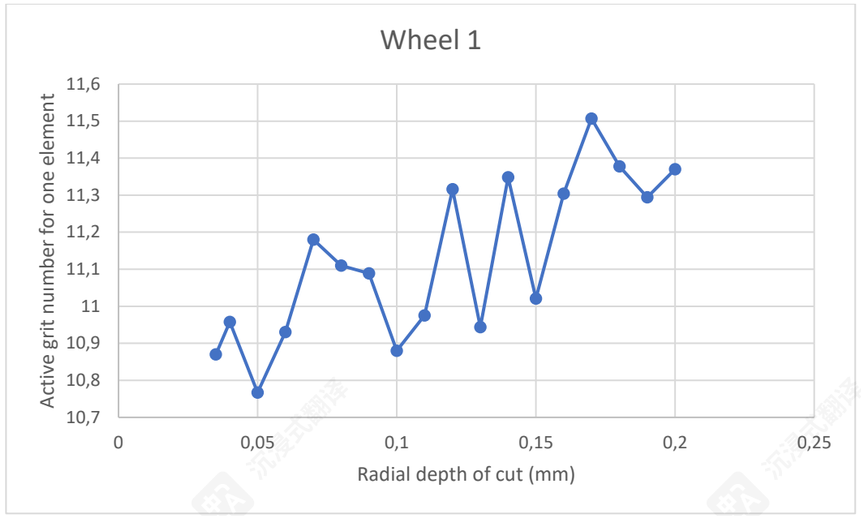



Figure 32: Active grit number for wheel 1 for varying depth of cut

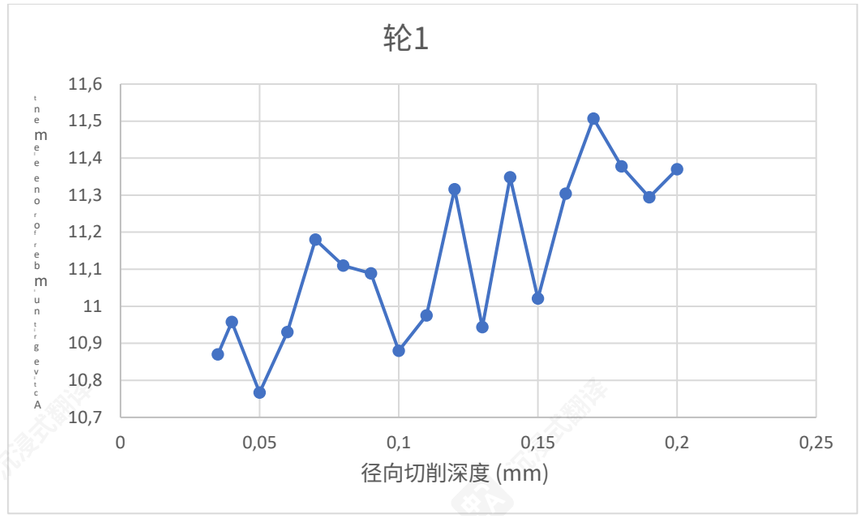



图32：轮1在不同切削深度下的有效磨料数量

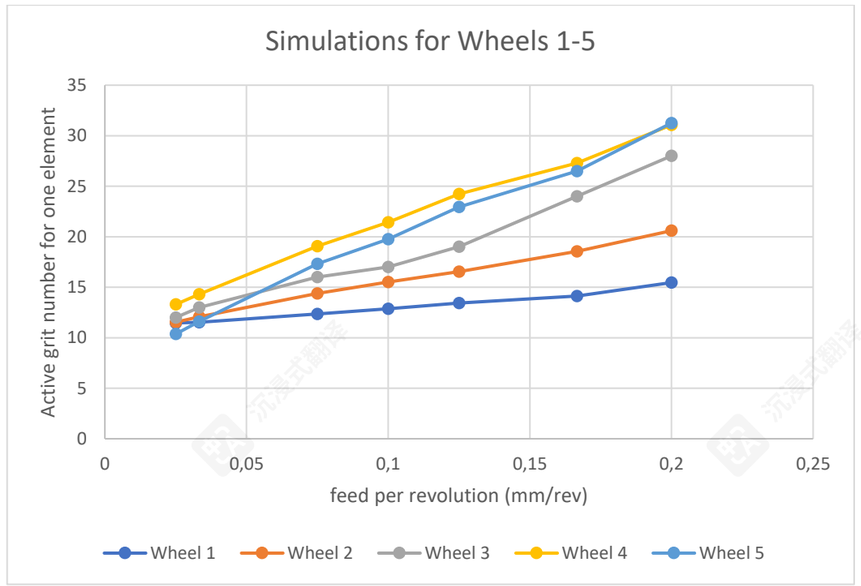



Figure 33: Active grit number for varying feed rate and 5 wheels

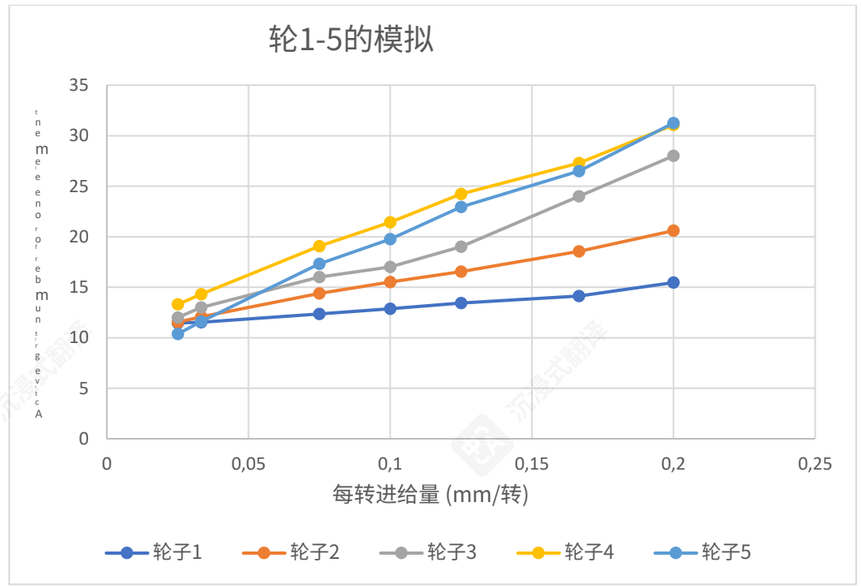



图33：在不同进给速度和5个轮子下的有效磨料数量

The active grit simulations for all of the 5 wheels are given in Figure 33. As it can be seen, the effects of wheel condition, diameter and grit size is presented. In terms of wheel condition, the Wheels 1,4 and 5 are fresh, dressed and worn wheels in a consecutive order. The active grit number is increased as the wheel is dressed or worn because of the decrease in the standard deviation of the height values of the grits. The active grits on Wheel 2 is larger than the ones on Wheel 1 because the grit size is lower on Wheel 2 as the grit size is B64. When the overall grit size is decreased, the number of active and passive grits in $1mm^2$ is increased as more grits may now fit in the unit square. Furthermore, Wheel 3 has a diameter of 30mm and even though the grits size is big and the grit per 1 $mm^2$ is low, the surface area of the wheel much higher which results in a higher number of active grits for Wheel 3 comparing to Wheel 1. The most active grits then will be on dressed and worn wheels as their standard deviation is significantly low comparing to other wheels.

所有 5 个轮子的主动砂粒模拟结果如图 33 所示。可以看出，轮子状态、直径和砂粒尺寸的影响都呈现出来了。就轮子状态而言，轮子 1、4 和 5 依次是新鲜、修整和磨损的轮子。随着轮子被修整或磨损，主动砂粒数量会增加，因为砂粒的高度值的标准差减小了。轮子 2 上的主动砂粒比轮子 1 上的更大，因为轮子 2 的砂粒尺寸较低，为 B64。当整体砂粒尺寸减小时，$1mm^2$ 内的主动和被动砂粒数量会增加，因为现在有更多砂粒可以适应单位正方形。此外，轮子 3 的直径为 30mm，尽管砂粒尺寸较大，每 $1mm^2$ 的砂粒数量较低，但轮子的表面积更大，导致轮子 3 的主动砂粒数量比轮子 1 更多。那么，主动砂粒最多的将是修整和磨损的轮子，因为它们的标准差与其他轮子相比显著较低。

### 3.4.2 Grinding Force and surface roughness

### 3.4.2 磨削力和表面粗糙度

The verification experiments of force and surface roughness were done with the 20mm Electroplated CBN tool grit size: B126 which was shown in Figure 16, together with a conventional wheel of the same grit size and diameter. The experiments were done for the following process parameters:

力和表面粗糙度的验证实验使用的是图16中所示的20mm电镀立方氮化硼砂轮，砂轮粒度为B126，同时配用相同粒度和直径的普通砂轮。实验在以下工艺参数下进行：

Feed rates ($\mu$m/rev): 2.5, 5, 7.5, 10.

进给速度（$\mu$m/rev）：2.5，5，7.5，10。

Radial depths of cut ($\mu$m): 20, 40, 60.

径向切深（$\mu$m）：20，40，60。

Axial depth of cut: 9mm

轴向切深：9mm

Grinding speed: 40m/s.

磨削速度：40m/s。

Wheels: CBN B126 20mm and Conventional B120 20mm.

轮子：CBN B126 20mm 和传统 B120 20mm。

The comparison of the model outputs and the experiments are given in the following figures (Figure 34 and Figure 35). As it can be seen, the forces tend to increase linearly while the feed rate is increased. The prediction accuracy of the mechanistic force model is better compared to that of the analytical model but since it requires more time and

模型输出与实验结果的对比展示在以下图表中（图34和图35）。可以看出，当进给率增加时，力趋于线性增加。与解析模型相比，力学力模型的预测精度更好，但由于它需要更多的时间和能量，因此热力学力模型更优越，因为它的力预测也足够充分。

energy, the thermo-mechanical force model is more favorable with the fact that it is also sufficient for force prediction.

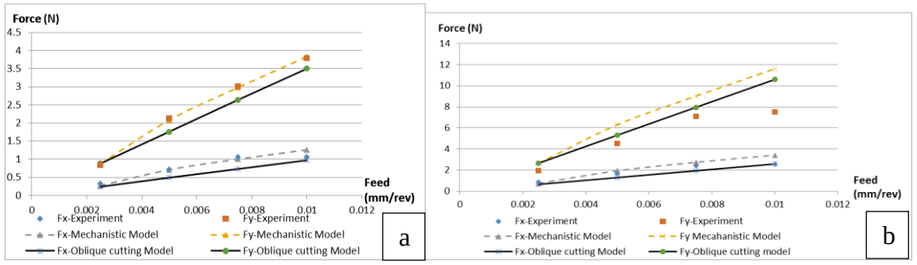



Figure 34: Verification of force model: Depth of cut: 20$\mu$m. a) CBN wheel b) conventional wheel

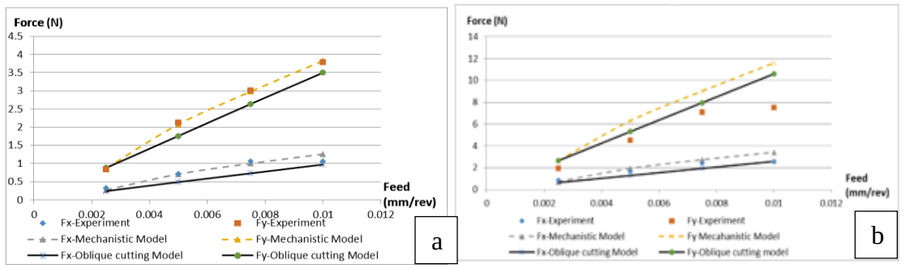



Figure 35: Verification of the force model. Depth of cut: a)40 $\mu$m b) 60 $\mu$m. (CBN wheel)

Another point is that the forces with the conventional wheel is high comparing to the CBN wheel. This is the reason that the conventional wheels are dressed before use making the height distribution with smaller deviation which increases the number of active grits. The increased number of active grits means more contact and increased edge forces. The ploughing forces are also increased with the conventional wheel due to the fact that the edge radii of the grits of the conventional wheel are approximately 5 times higher than that of the CBN wheel, thus creating more contact.

The phenomena of increased force with the number of active grits is also verified in Figure 36. As it can be seen, when the grit number per mm$^2$ increases in a wheel which has lower standard height deviation, the forces tend to increase. The reason is that with



热力学力模型更优越，因为它的力预测也足够充分。

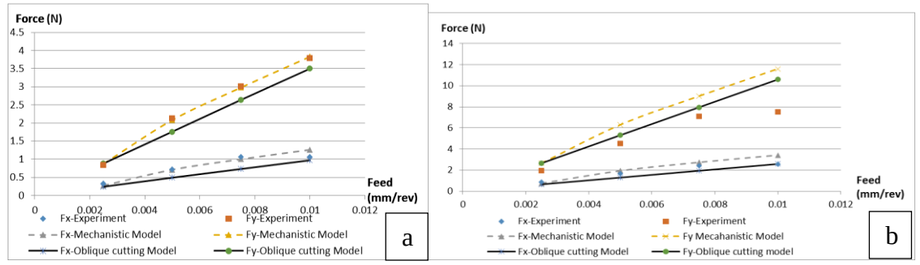



图34：验证力模型：切削深度：20$\mu$米。a) CBN砂轮 b) 普通砂轮

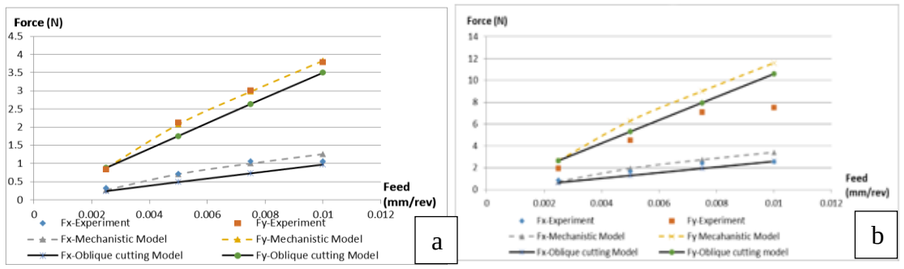



图35：验证力模型。切削深度：a)40 $\mu$米 b) 60 $\mu$米。(CBN砂轮)

另一个点是，与CBN砂轮相比，普通砂轮的力较高。这是普通砂轮在使用前需要修整的原因，修整后高度分布的偏差更小，从而增加了有效磨粒的数量。有效磨粒数量的增加意味着更多的接触和更大的边缘力。由于普通砂轮的磨粒边缘半径大约是CBN砂轮的5倍，因此接触更多，普通砂轮的耕犁力也相应增加。

图36也验证了随着主动砂粒数量的增加，力会增强的现象。如图所示，当每毫米$^2$ 砂粒数量在一个标准高度偏差较低的车轮中增加时，力会倾向于增大。原因是，在标准高度偏差较低的情况下，主动砂粒数量会增加，从而导致与工件接触更多。砂粒标准高度偏差较高的车轮不会出现这种效果，因为随着静态砂粒数量的增加，主动砂粒数量不会发生显著变化。

low standard height deviation the active number of grits will increase and will result in more contact with the workpiece. The wheel with higher standard height deviation of grits will not experience this effect since with the increased static number of grits the active grit number will not change significantly.

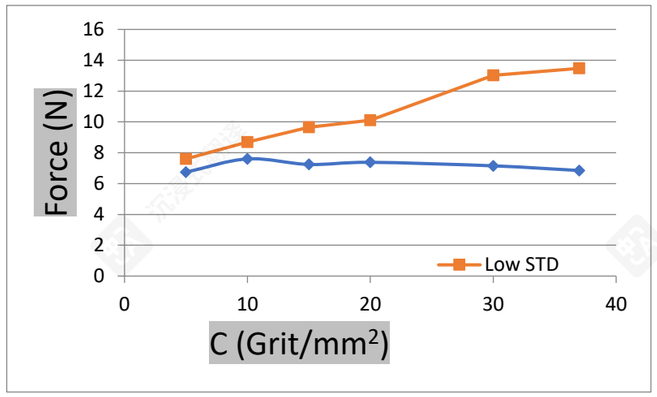



Figure 36: The effect of grit number on the forces

砂粒标准高度偏差较低时，主动砂粒数量会增加，从而导致与工件接触更多。砂粒标准高度偏差较高的车轮不会出现这种效果，因为随着静态砂粒数量的增加，主动砂粒数量不会发生显著变化。

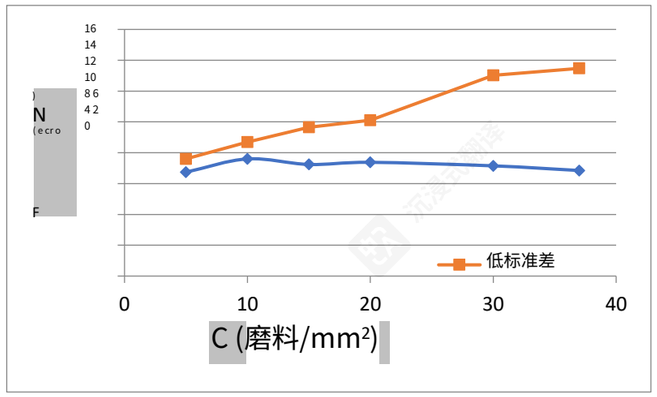



图36：磨料数量对力的影响

## 3.5 Summary

As one of the most complicated subtractive processes in the industry in terms of modelling and prediction, grinding process has stochastic nature and the first step of modelling grinding starts from coping with the effect of randomness of the process. The initial approach is to determine the wheel parameters such as the geometric data and the distribution of the grits on the surface of the grinding wheel. In this thesis a method for virtual wheel generation is developed by utilizing optical measurement methods to identify the geometric properties of the individual grits [1] and then applying random distribution to create a sample space of grits for the whole wheel [11]. The grinding wheel then is assumed to be discretized along the axial direction into thin disks which are named as the elements. Recognizing that some of the abrasive grits do not participate in the chip formation action, the active grits are identified with the geometric-kinematic model by analyzing the trajectories of the individual grits as the grinding wheel is turned over the workpiece virtually. The geometric-kinematic model



## 3.5 总结

作为行业内建模和预测方面最为复杂的减法过程之一，磨削过程具有随机性，建模的第一步始于应对过程随机性的影响。初始方法是确定砂轮参数，例如几何数据和磨粒在砂轮表面的分布。在本论文中，通过利用光学测量方法识别单个磨粒的几何特性[1]，然后应用随机分布来创建整个砂轮的磨粒样本空间[11]，从而开发了一种虚拟砂轮生成方法。砂轮然后被假设沿轴向离散成薄盘，这些薄盘被称为元素。认识到有些磨粒不参与切屑形成过程，通过分析虚拟状态下砂轮转过工件时单个磨粒的轨迹，利用几何运动学模型识别了有效磨粒。几何运动学模型

allows calculation of the active number of grits, instantaneous chip thickness and the surface roughness of the workpiece. The chip thickness calculation allows the prediction of the grinding force and two methods namely mechanistic and thermo-mechanical force model are applied in order to verify the models and to compare the performance of both. The thermo-mechanical force model's performance surpasses the other model by being applicable to wider range and not relying calibration methods for different grinding conditions. The results for the number of the active grits conclude that the number depends on the specifications of the wheel and the process parameters such as the radial depth of cut and the feed rate. The surface roughness of the workpiece surface along the perpendicular to cutting direction is observed that it is not affected by the process parameters as the wheel prints the topography and of its surface onto the surface of the workpiece.



允许计算砂轮的主动砂粒数量、瞬时切屑厚度和工件表面粗糙度。切屑厚度计算允许预测磨削力，并应用了机械力和热力学力模型这两种方法，以验证模型并比较两者的性能。热力学力模型的性能通过适用范围更广且不依赖于不同磨削条件下的校准方法而优于其他模型。关于主动砂粒数量的结果表明，该数量取决于砂轮规格和工艺参数，如切深和进给率。观察到工件表面垂直于切削方向的表面粗糙度不受工艺参数的影响，因为砂轮将其顶形及其表面印在了工件表面。

## 4 DYNAMIC FORCE MODEL FOR ABRASIVE PROCESSES

The dynamics of grinding operations have always been an obstacle for abrasive operations since the main goal of the abrasive processes is to finalize the workpiece geometry and the surface roughness given the fact that the instability will hinder the part quality in a considerable way. One of the first stability models for grinding was developed by Inasaki [60] where the problem was defined as a one-degree-of-freedom model and the whole grinding geometry was constructed as a single point representation. This representation resembles a turning operation as the dynamics are constructed as one-degree-of-freedom and the varying chip thickness is calculated along the radial depth of cut of the system. One may say that considering this representation turns grinding chatter into macro level phenomenon. As described in the literature survey of this thesis, the following literature consists of either the analytical solutions to the problem for the mentioned simplifications, or numerical solutions such as time domain simulations where the problem is enhanced in terms of complexity. On the other hand, by utilizing the advantages of the geometric-kinematic model, the micro-level representation of the grinding process is possible. In this thesis, a dynamic model for surface grinding is developed. Applying the milling analogy and the geometrical random distribution of the grits on the grinding wheel, the simulation can predict the dynamic forces and vibrations. A geometric-kinematic model is developed in order to predict the grit trajectories during the grinding process. The active grits are calculated and the chip thickness per active grit is computed with the trajectory simulation of the wheel. The dynamic model is created with the inclusion of the regenerative effect of the workpiece surface. The aim of this model is to model the dynamic behavior of the surface grinding and predict the region of grinding parameters which cause instability in the closed loop time domain system.

### 4.1 Dynamic Chip thickness and Dynamic Forces

As a reminder, the geometric and kinematic simulation, the model inherently obtains the undeformed chip thickness by the trajectories of the grits and their intersections. Once the active grit number is calculated the passive grits are disregarded and the



## 4 动态力模型用于磨削过程

磨削操作的动力学一直是对磨削过程的主要障碍，因为磨削过程的主要目标是在工件几何形状和表面粗糙度方面达到最终效果，而事实上，不稳定性会以相当大的方式影响零件质量。Inasaki [60] 开发了磨削的第一个稳定性模型，其中将问题定义为一个自由度模型，并将整个磨削几何形状构建为单点表示。这种表示类似于车削操作，因为动力学被构建为一个自由度，并且沿系统径向切深计算的切削厚度是变化的。可以说，考虑到这种表示，磨削颤振变成了宏观现象。正如本论文的文献综述所述，以下文献包括针对所提及简化的问题的解析解，或数值解，如时域仿真，其中问题在复杂性方面得到了增强。另一方面，通过利用几何运动学模型的优势，可以实现磨削过程的微观级表示。在本论文中，开发了一个用于表面磨削的动态模型。应用铣削类比和磨头上砂粒的几何随机分布，仿真可以预测动态力和振动。开发了一个几何运动学模型，以预测磨削过程中的砂粒轨迹。计算了活动砂粒，并使用砂轮的轨迹仿真计算了每个活动砂粒的切削厚度。动态模型在工件表面再生效应的包含下创建。该模型的目的是模拟表面磨削的动态行为，并预测在闭环时域系统中导致不稳定的磨削参数区域。

### 4.1 动态芯片厚度和动态力

提请注意的是，几何和运动学仿真中，该模型通过砂粒轨迹及其交点来获得未变形的切屑厚度。一旦计算出有效砂粒数量，被动砂粒就会被忽略，并且

problem transforms into a milling cutter analogy around the discretized element of the wheel surface. The active grits have different heights which make them have different radial distance from the center of the wheel. This similar to a milling cutter with runout. Thus, the chip thickness is calculated as the chip thickness in a milling cutter with a runout with some modifications [61]. The equation is given:

$$h_{i,j} = R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) \quad 0 < \theta_{i,j} < \theta_{exit,i,j} \tag{18}$$

where h is the undeformed chip thickness per grit, with the indices *i* and j mean the *i*'th active grit on the *j*'th element. Indices n and m represent the consecutive positions of the active grits along with the circumference of the wheel. Angle $\theta$ represents immersion angle of the current grit. Also, $f_t$ indicates the feedrate. In the dynamic model, the calculation of the grinding force is a straightforward multiplication of the calculated undeformed chip area with the cutting force coefficients.

$$Fn_{i,j} = K_{nc}.w.h_{i,j} \tag{19}$$

$$Ft_{i,j} = K_{tc}.w.h_{i,j} \tag{20}$$

where $F_n$ and $F_t$ represent the forces in normal and tangential directions respectively. $K_{nc}$ and $K_{tc}$ are the experimentally defined cutting force coefficients in the respective directions. The edge forces are also experimentally identified with linear regression method and added to the overall force values after the cutting forces are calculated. *w* is the average width for a grit. The forces are then projected into the machine coordinates. The summation of individual grit and the elemental forces yield to the total forces:

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \sum_{i,j} \left( \begin{bmatrix} -cos\theta_{i,j} & -sin\theta_{i,j} \\ sin\theta_{i,j} & -cos\theta_{i,j} \end{bmatrix} \times \begin{bmatrix} Ft_{i,j} \\ Fn_{i,j} \end{bmatrix} \right) \tag{21}$$

Calculating the dynamic undeformed chip thickness is the crucial part when simulating the grinding dynamics in the time domain. Applying milling cutter analogy in grinding allows the use of the dynamic milling equations presented in [3,4]. After the calculation of active grit number per one element of the wheel, the active grits can be assumed to be distributed with their respective pitch angles along the circumference of the wheel. This way the wheel is discretized into thin disk-shaped n number of milling tools each having different number of teeth. The problem here is that the constructed structure does not have any predefined from. The grit distributions are totally random, and this will create random delays within the elements. For this reason, the regeneration of the



问题就转化为一个围绕轮面离散单元的铣刀类比。有效砂粒具有不同高度，这使得它们到轮子中心的径向距离不同。这类似于一个有振刀的铣刀。因此，切屑厚度被计算为有振刀铣刀的切屑厚度，并进行了一些修改 [61]。方程如下：

$$h_{i,j} = R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) \quad 0 < \theta_{i,j} < \theta_{exit,i,j} \tag{18}$$

其中 h 表示每颗磨粒的未变形切屑厚度，索引 i 和 j 分别表示第 j 个元件上的第 i 颗有效磨粒。索引 n 和 m 表示有效磨粒沿轮周连续的位置。角度 $\theta$ 代表当前磨粒的浸入角。此外，$f_t$ 表示进给速度。在动态模型中，磨削力的计算是将计算出的未变形切屑面积与切削力系数直接相乘的结果。

$$Fn_{i,j} = K_{nc}。w。h_{i,j} \tag{19}$$

$$Ft_{i,j} = K_{tc}.w.h_{i,j} \tag{20}$$

其中 $F_n$ 和 $F_t$ 分别表示法向和切向方向的力。$K_{nc}$ 和 $K_{tc}$ 是实验确定的相应方向的切削力系数。边缘力也通过线性回归方法实验识别，并在切削力计算后加到总力值中。w 是磨粒的平均宽度。然后将力投影到机床坐标系中。单个磨粒和元件力的总和得到总力：

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \sum_{i,j} \left( \begin{bmatrix} -cos\theta_{i,j} & -sin\theta_{i,j} \\ sin\theta_{i,j} & -cos\theta_{i,j} \end{bmatrix} \times \begin{bmatrix} Ft_{i,j} \\ Fn_{i,j} \end{bmatrix} \right) \tag{21}$$

在时域内模拟磨削动力学时，计算动态未变形芯片厚度是关键部分。将铣刀类比应用于磨削，允许使用[3,4]中提出的动态铣削方程。在计算每个砂轮元素中的有效砂粒数量后，可以假设这些有效砂粒沿砂轮圆周分布其各自的节距角。这样，砂轮被离散化为n个薄盘形铣削工具，每个工具具有不同的齿数。这里的问题是，构建的结构没有预定义的起始点。砂粒分布完全随机，这将导致元素内出现随机延迟。因此，芯片厚度的再生研究是逐个元素进行的，因为所有元素都被视为独立的盘形铣削工具。轴向离散化也在[3,4]中进行了公式化；然而，对于标准铣削工具，切削刃数量和节距角是固定的。螺旋角的引入也可以通过工具轴上离散化元素之间的解析关系来处理。但在磨削情况下，节距角不会是一个常数，因为切削刃被放置在砂轮圆周上的任意位置。每个元素的节距角和砂粒数量将不同。

chip thickness is studied element by element as all the elements are treated as a separate disk shape milling tool. The axial discretization is also formulated in [58]; however in the case of a standard milling tool the number of cutting edges and the pitch angle are fixed. The introduction of helix angle can also be coped with analytical relations between the discretized elements along the axis of the tool. But in the case of grinding, the pitch angle will not be a constant number as the cutting edges are placed on any arbitrary positions on the circumference of the wheel. The pitch angles and the number of grits will be different for each element.

In turning operations, the chip thickness is a constant value over the surface. In milling analogy, the chip thickness is an immersion angle dependent value under the circular tooth path assumption. If the tool is subjected to the vibrations in x and y coordinates as a result of the grinding forces, the equation of chip thickness should be modified.

The initial step of these equations to transform the vibrations into machine coordinates to the surface normal (associated with the *i*'th number of grit in the *j*'th number of the element) of the current grits according to the instantaneous immersion angle.

$$n_{i,j} = -xsin(\theta_{i,j}) - ycos(\theta_{i,j}) \quad (22)$$

where x and y are the global vibrations of the wheel in the respective directions which affect each element. The instantaneous chip thickness H for tooth i in element j becomes:

$$H_{i,j} = (h_{i,j} + n_{i-1,j} - n_{i,j}) \times g(\theta_{i,j}) \quad (23)$$

where n_(i-1,j) indicates the vibration of the previous active grit on the same element. The change in the surface normal creates a dynamic chip thickness, thus creating a dynamic force in normal and tangential directions. The projected dynamic forces create the vibrations due to the frequency response functions in which the modal parameters are obtained by hammer tests. The switching function $g(\theta_{i,j})$ is equal to 1 if the current active grit in the current element is in between its respective start and exit angles:

$$g(\theta_{i,j}) = \begin{cases} 1, & when \quad \theta s_{i,j} \le \theta_{i,j} \le \theta e_{i,j} \\ 0, & when \quad \theta_{i,j} < \theta s_{i,j}\ ,\ \theta_{i,j} > \theta e_{i,j} \end{cases} \quad (24)$$

where $\theta s_{i,j}$ and $\theta e_{i,j}$ are the start and exit angles of the current active grit of the current element, respectively. The vibration on the surface normal is included in the chip



芯片厚度的再生研究是逐个元素进行的，因为所有元素都被视为独立的盘形铣削工具。轴向离散化也在[58]中进行了公式化；然而，对于标准铣削工具，切削刃数量和节距角是固定的。螺旋角的引入也可以通过工具轴上离散化元素之间的解析关系来处理。但在磨削情况下，节距角不会是一个常数，因为切削刃被放置在砂轮圆周上的任意位置。每个元素的节距角和砂粒数量将不同。

在车削加工中，切屑厚度在工件表面上是恒定值。在铣削类比中，切屑厚度是在圆形齿迹假设下，随浸入角变化的值。如果刀具因磨削力在x轴和y轴方向上受到振动，则切屑厚度方程应进行修正。

这些方程的初始步骤是将振动转换为机器坐标系下的表面法向（与当前砂粒的i'个数的j'个元素相关联）的瞬时浸入角。

$$n_{i,j} = -xsin(\theta_{i,j}) - ycos(\theta_{i,j}) \quad (22)$$

其中x和y是砂轮在相应方向上的全局振动，这些振动会影响每个元素。对于齿i和元素j的瞬时切屑厚度H为：

$$H_{i,j} = (h_{i,j} + n_{i-1,j} - n_{i,j}) \times g(\theta_{i,j}) \quad (23)$$

其中n_(i-1,j)表示同一元素上先前活动的砂粒的振动。表面法向的变化会产生动态切屑厚度，从而在法向和切向方向上产生动态力。投影的动态力会因频率响应函数产生振动，其中模态参数通过锤击试验获得。切换函数$g(\theta_{i,j})$ 在当前元素中的当前活动砂粒位于其相应的起始角和退出角之间时等于1:

$$g(\theta_{i,j}) = \begin{cases} 1, & when \quad \theta s_{i,j} \le \theta_{i,j} \le \theta e_{i,j} \\ 0, & when \quad \theta_{i,j} < \theta s_{i,j}\ ,\ \theta_{i,j} > \theta e_{i,j} \end{cases} \quad (24)$$

$\theta s_{i\,j}$ 和$\theta e_{i\,j}$ 分别是当前元素当前活动砂轮的起始角和退出角。表面法向的振动包含在芯片中

thickness formula. The chip thickness formula in terms of x and y directions can be calculated as shown in equation (25). The static chip thickness may be removed if the equation is going to be used for calculating the stability only. For the calculation of the dynamic forces and the real values of the vibration amplitudes, the static chip thickness should also be added:

$$H_{i,j} = [\, R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) - x_{j-1}sin(\theta_{i,j}) - y_{j-1}cos(\theta_{i,j}) - \left(x_j sin(\theta_{i,j}) - y_j cos(\theta_{i,j})\right)] \times g(\theta_{i,j}) \quad (25)$$

Introducing the conventions;

$$x_j - x_{j-1} = dx \quad \& \quad y_j - y_{j-1} = dy \quad (26)$$

Equation (25) can be rewritten as:

$$H_{i,j} = [\, R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) + dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \quad (27)$$

And if the stability will be investigated in equation (27) the static chip thickness can be removed:

$$H_{i,j} = [dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \quad (28)$$

By using the equations (19) and (20) the dynamic forces in the x and y directions can be written as shown in equation (29).

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \sum_{i,j} \left( \begin{bmatrix} -cos\theta_{i,j} & -sin\theta_{i,j} \\ sin\theta_{i,j} & -cos\theta_{i,j} \end{bmatrix} \times \begin{bmatrix} K_{tc}.w.[dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \\ K_{nc}.w.[dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \end{bmatrix} \right) \quad (29)$$

The total force acting on the wheel disturbs the system though the equations of motions in 2 directions.

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \begin{bmatrix} m_x\ddot{x} + c_x\dot{x} + k_x x \\ m_y\ddot{y} + c_y\dot{y} + k_y y \end{bmatrix} \quad (30)$$

where $m_x$ and $m_y$ are the modal mass, $c_x$ and $c_y$ are the modal damping and $k_x$ and $k_y$ are the modal stiffness of the grinding system in the respective directions of x and y.



厚度公式。芯片厚度公式在x和y方向上可以按方程(25)计算。如果方程仅用于计算稳定性，则可以移除静态芯片厚度。对于动态力和振动幅值的实际值计算，静态芯片厚度也应添加：

$$H_{i,j} = [\, R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) - x_{j-1}sin(\theta_{i,j}) - y_{j-1}cos(\theta_{i,j}) - \left(x_j sin(\theta_{i,j}) - y_j cos(\theta_{i,j})\right)] \times g(\theta_{i,j}) \quad (25)$$

介绍规范；

$$x_j - x_{j-1} = dx \quad \& \quad y_j - y_{j-1} = dy \quad (26)$$

方程式(25)可以重写为：

$$H_{i,j} = [\, R_n - R_m + (n-m).f_t \sin(\theta_{i,j}) + dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \quad (27)$$

如果要在方程式(27)中研究稳定性，则可以移除静态芯片厚度：

$$H_{i,j} = [dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \quad (28)$$

通过使用方程式(19)和(20)，x和y方向的动态力可以写成方程式(29)中所示的形式。

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \sum_{i,j} \left( \begin{bmatrix} -cos\theta_{i,j} & -sin\theta_{i,j} \\ sin\theta_{i,j} & -cos\theta_{i,j} \end{bmatrix} \times \begin{bmatrix} K_{tc}.w.[dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \\ K_{nc}.w.[dxsin(\theta_{i,j}) + dycos(\theta_{i,j})] \times g(\theta_{i,j}) \end{bmatrix} \right) \quad (29)$$

作用在车轮上的总力通过2个方向的运动方程扰动系统。

$$\begin{bmatrix} Fx \\ Fy \end{bmatrix} = \begin{bmatrix} m_x\ddot{x} + c_x\dot{x} + k_x x \\ m_y\ddot{y} + c_y\dot{y} + k_y y \end{bmatrix} \quad (30)$$

其中，$m_x$ 和$m_y$ 分别为砂轮系统在x和y方向上的模态质量，$c_x$ 和$c_y$ 分别为模态阻尼，$k_x$ 和$k_y$ 分别为模态刚度。

## 4.2 Vibration and surface roughness in the cutting direction

The surface roughness of the workpiece holds quite an importance in grinding. The surface roughness in perpendicular to the cutting direction was shown in the previous chapter to be the imprint of the topography of the grinding wheel onto the workpiece. The surface roughness in perpendicular to the cutting direction can be calculated and predicted with a sufficient accuracy by using the geometric-kinematic model. However, the geometric-kinematic model is unsatisfactory for predicting the surface roughness in the cutting direction as the model predict only the feed marks of the active grits on the workpiece surface. The real surface roughness in the cutting direction is shown to be larger and it is the result of the dynamic behavior of the process. The roughness that is created by the vibrations of the grinding wheel is more dominant compared to the other process effects.

In the dynamic model that is proposed in this thesis the vibrations of the grinding wheel in both directions, namely x and y, are considered. The calculated vibrations are stored in an array and then the grit path profiles are printed with the geometric kinematic model, such as shown in Figure 37. The path profiles are updated in each step.

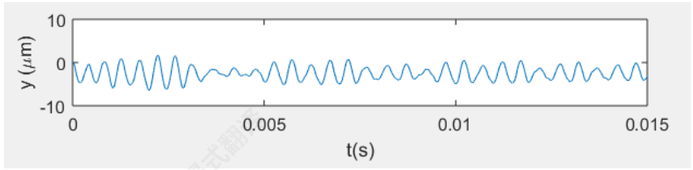



Figure 37: Y path profile

The surface roughness of the workpiece is calculated using the stored position values in the model. The average roughness value Ra is calculated as shown in equation (31)

$$R_a = \frac{1}{n}\sum_{i=1}^{n}|y_i| \quad (31)$$

where n is the number of samples and y is the displacement values of the tool from the surface of the workpiece.



## 4.2 切削方向上的振动与表面粗糙度

工件表面粗糙度在磨削过程中至关重要。前一章已表明，垂直于切削方向的表面粗糙度是砂轮形貌在工件上的印记。通过几何-运动学模型，垂直于切削方向的表面粗糙度可以足够精确地计算和预测。然而，几何-运动学模型不适用于预测切削方向的表面粗糙度，因为该模型仅预测了工件表面上有效磨粒的进给痕迹。实际切削方向的表面粗糙度被证明更大，这是加工动态行为的结果。由砂轮振动产生的粗糙度比其他工艺效应更为显著。

在本论文提出的动态模型中，考虑了砂轮在x和y两个方向上的振动。计算出的振动数据存储在数组中，然后使用几何运动学模型打印砂粒路径轮廓，如图37所示。路径轮廓在每一步都会更新。

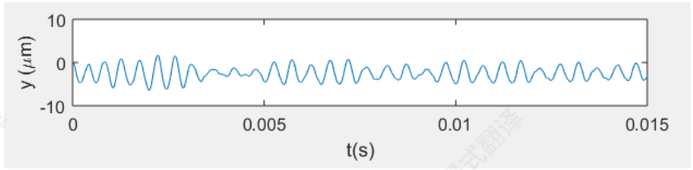



图37：Y路径轮廓

工件表面粗糙度是根据模型中存储的位置值计算的。平均粗糙度值Ra的计算方法如公式(31)所示。

$$R_a = \frac{1}{n}\sum_{i=1}^{n}|y_i| \quad (31)$$

其中 n 是样本数量，y 是刀具相对于工件表面的位移值。

### 4.3 Time domain Simulation

In the time domain simulation of the grinding process, the time is discretized into a small number of steps determined with the consideration of the interested frequencies. The time step should yield enough frequency contend for the natural frequency of the tool. Each step number k represents an angular step dθ and a time step dt. The angular position of the wheel can be found as follows,

$$\theta_w = k \times d\theta \tag{32}$$

The instantaneous angular immersion angles of the active grits per each element can be found by angular incrementation of each grit considering the angular position of the wheel and the pitch angle corresponding to the current element. The algorithm consists of three loops in sequence k, j and i namely the loops for angular increment, elemental increment, and the grit number increment respectively.

In the grit loop, the kinematic-geometric model is applied. All grits along the current element's circumference are spanned. With the trajectory and the intersection calculation, the active grit number is identified, and thus the pitch angles are found. The forces are calculated with the beforementioned formulae (considering the undeformed chip thickness and the dynamic chip thickness) at that angular position of the wheel if the corresponding grit is between the corresponding contact angles with the workpiece. The forces are accumulated at the end of this loop. The element loop provides that the calculations are done for each element of the wheel. At the end of the elemental loop, the accumulated forces are summed and transformed into the x and y direction vibrations for the specific angular position of the wheel. The outermost loop provides every calculation is repeated for each angular position of the loop.

It is worth to mention that this work's difference from the milling cutter problem is that each element acts as a disk-shaped milling tool with a different number of teeth as a result of the stochastic nature of the grinding process. The elements are lumped and the relative vibration between them is neglected. The total forces from each grit from each element are accumulated and then the vibrations are created, affecting each element of the wheel. This allows a continuous and simultaneous calculation of the dynamic grinding.



4.3 时域仿真

在磨削过程的时域仿真中，时间被离散化为少数几个步骤，这些步骤是根据感兴趣的频率确定的。时间步长应产生足够多的频率内容以覆盖刀具的固有频率。每个步骤编号 k 代表一个角步 dθ 和一个时间步长 dt。轮子的角位置可以按如下方式计算，

$$\theta_w = k \times d\theta \tag{32}$$

每个元素中，每个主动砂粒的瞬时角浸入角度可以通过考虑轮子的角位置和当前元素对应的节距角，对每个砂粒进行角增量计算得到。该算法依次包含三个循环，分别为角增量循环、元素增量循环和砂粒数量增量循环。

在砂粒循环中，应用运动学-几何模型。沿当前元素圆周的所有砂粒都被扫描。通过轨迹和交点计算，识别出主动砂粒数量，从而找到节距角。如果对应的砂粒位于与工件对应的接触角范围内，则在轮子该角位置使用前述公式（考虑未变形芯片厚度和动态芯片厚度）计算力。该循环结束时累积这些力。元素循环确保对轮子的每个元素进行计算。元素循环结束时，累积的力被汇总并转换为轮子特定角位置的x和y方向振动。最外层循环确保对循环的每个角位置重复每次计算。

值得提及的是，这项工作与铣刀问题的区别在于，每个元素都作为一个具有不同齿数的盘形铣削工具，这是由于磨削过程的随机性所致。元素被集总，并且忽略了它们之间的相对振动。每个元素中每个砂粒的总力被累积，然后产生振动，影响轮子的每个元素。这使得动态磨削的连续和同时计算成为可能。

The overall flowchart for the operation is shown in Figure 38.

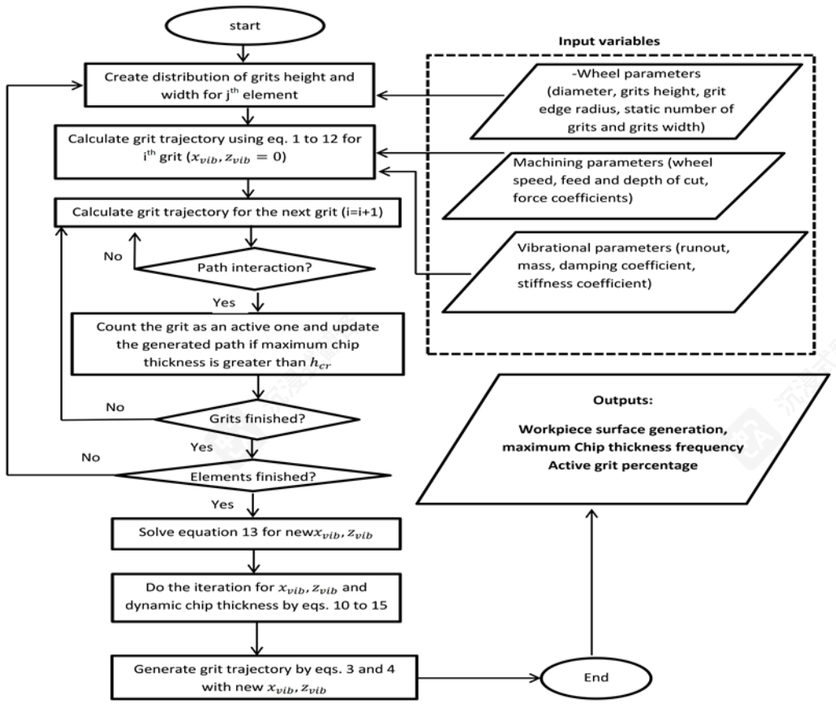



Figure 38: The dynamic grinding flowchart

The governing delayed differential equation of the system is given in block diagram in Figure 39 where x and x(t-$\tau$) are the dynamic and delayed displacements of the tool tip in the x direction respectively. Similarly, y and y(t-$\tau$) are the dynamic and delayed displacements of the tool tip in the z direction respectively. h and delay amount are the instantaneous chip thickness and delay term respectively that is calculated from the kinematic model. *Kt* and *Kn* are the cutting force coefficients in the tangential and normal directions, respectively. In “thickness to force” block equations 10, 11 and 12 are applied where the forces in x and z are obtained. The dynamic model can be used to calculate the stability of the system and it is also used to predict the accurate dynamic force and displacement values in order to predict the surface roughness in the cutting direction. The delayed differential equation is solved through Runge – Kutta method ODE solver provided by Simulink application. The delayed feedback system is developed after the calculated force values are transferred to displacements in x and z



操作的整体流程图如图38所示。

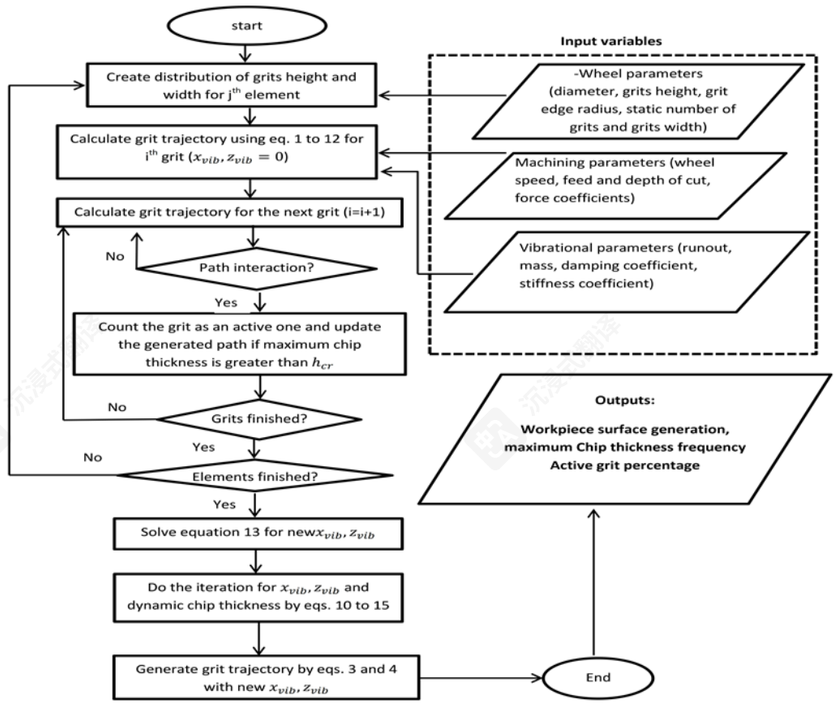



图38：动态磨削流程图

系统的控制延迟微分方程在图39中以框图形式给出，其中x和x(t-$\tau$)分别是工具尖端在x方向上的动态和延迟位移。类似地，y和y(t-$\tau$)分别是工具尖端在z方向上的动态和延迟位移。h和延迟量分别是瞬时切屑厚度和延迟项，该值根据运动学模型计算。Kt和Kn分别是切向和法向方向上的切削力系数。在“厚度到力”块方程10、11和12中应用了这些方程，其中获得了x和z方向上的力。动态模型可用于计算系统的稳定性，并且也用于预测精确的动态力和位移值，以预测切削方向上的表面粗糙度。延迟微分方程通过Simulink应用程序提供的Runge-Kutta方法ODE求解器进行求解。在计算出的力值转移到x和z方向上的位移后，延迟反馈系统被开发，其中延迟位移被反馈到进行力计算的块中。

directions and the delayed displacements are fed back to the block where the force calculations are done.

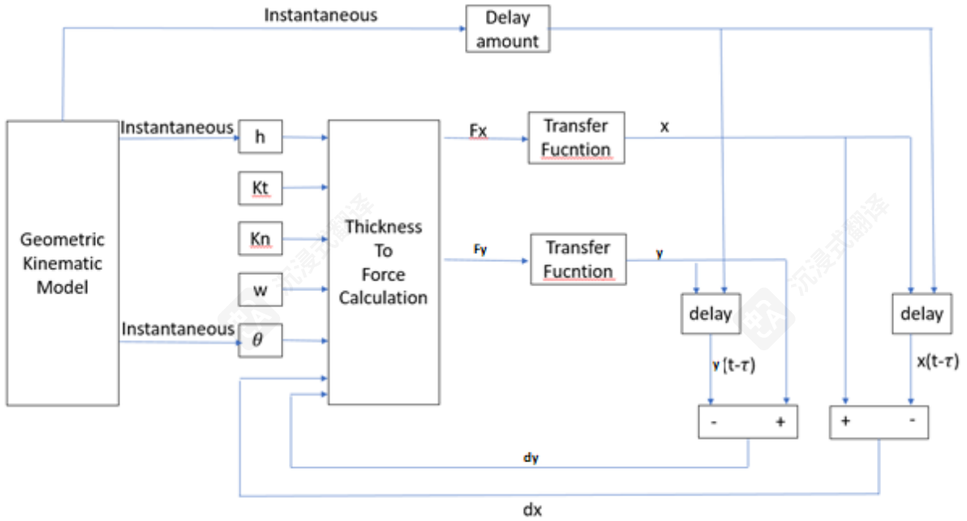



Figure 39: The block diagram

It is necessary to note that within this model two algorithms are running simultaneously. This is a requirement of the stochastic nature of the grinding process as it contains a number of non-linearities that are not in common sense when studying milling and turning problems. The main problem in grinding is that if the geometric-kinematic model is applied and the grinding wheel is discretized into elements, all active grits will be placed on the periphery of the elements in a random fashion. There will be no pre-defined rules to create the wheel after the grits have been measured. This results in random delays for the delayed-differential system.

The way that the random delays are handled is a straightforward method. Remember that initially the wheel element is created with all grits (active or passive) before the active grit calculation as shown in Figure 40. Let us assume that there are n number of grits on the periphery of this element and they all have index values. These elements



方向，并且延迟位移被反馈到进行力计算的块中。

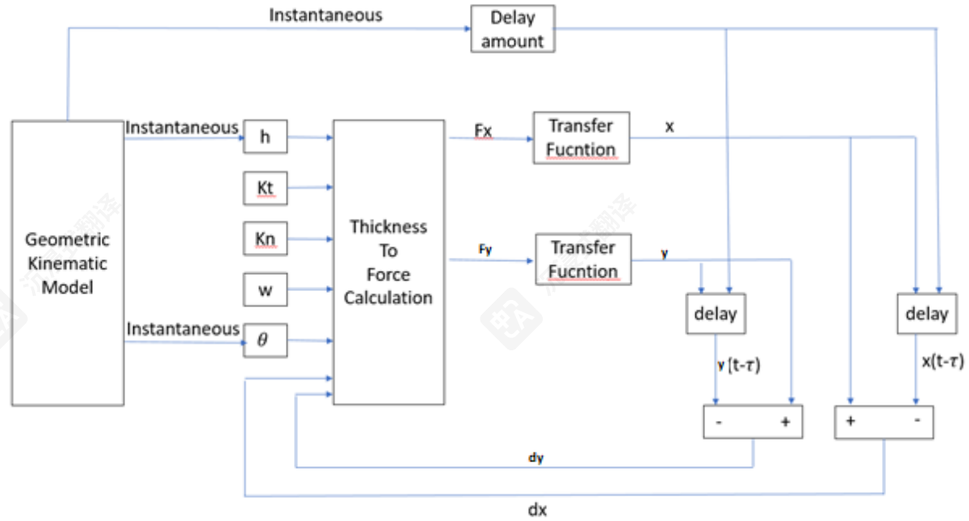



图39：框图

需要注意的是，在此模型中，有两个算法在同时运行。这是由磨削过程的随机性所决定的，因为磨削过程包含许多非线性因素，而这些因素在研究铣削和车削问题时通常不被认为是常识性的。磨削的主要问题在于，如果应用几何-运动学模型并将砂轮离散化为单元，所有激活的磨粒将随机地分布在单元的周边。在磨粒被测量后，将没有预定义的规则来创建砂轮。这会导致延迟微分系统出现随机延迟。

随机延迟的处理方式是一种直接的方法。请记住，在执行主动颗粒计算之前，轮元素会先创建所有颗粒（主动或被动），如图40所示。假设该元素边缘有n个颗粒，并且它们都具有索引值。这些元素

will be created with equal spacing and thus will produce a fixed amount of delay if all of the grits are active.

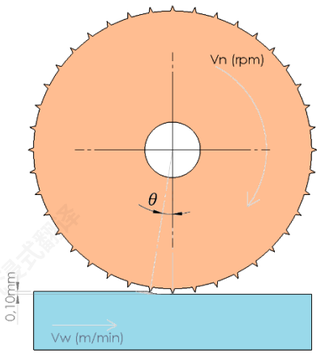



Figure 40: The discretized wheel element

Of course, this is not the case, thus some of the grits will be active as shown in Figure 41. The active grits are shown as bold particles.

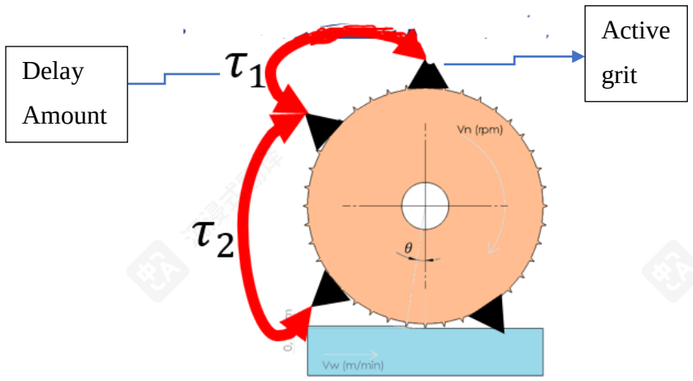



Figure 41: The active grits and delays

The active grits will have their respective index values. For example, let's say that out of n number of grits on the elements, the 1st, 50th and the 10$^{0th}$ grits are active. Than the



将以等间距创建，因此如果所有颗粒都处于活动状态，就会产生固定的延迟量。

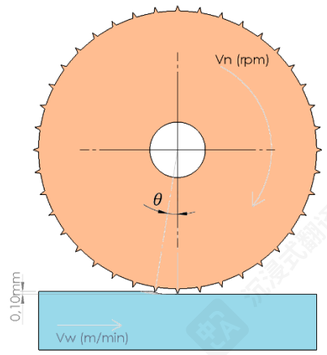



图40：离散化的车轮单元

当然，情况并非如此，因此一些粗粒将被激活，如图41所示。被激活的粗粒显示为粗体颗粒。

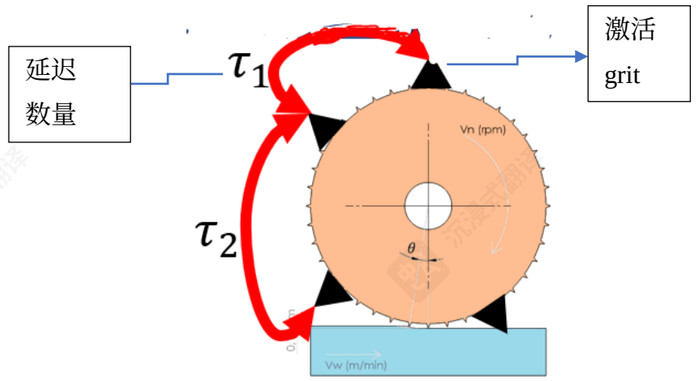



图41：被激活的粗粒和延迟

活跃的粗粒将拥有各自的索引值。例如，假设在元素上的 n 个粗粒中，1st、 50th 和 10$^{0th}$ 粗粒是活跃的。那么

delay between the consecutive active grits will have the delay amount calculated for every passive grit, times the number of passive grits that are missed until the next active grit. In our example, the delay between the first and the second active grit will be 49 times the delay amount per all passive grits.

The other non-linearity that is covered in the model is the change of the number of the active grits during the vibration of the grinding tool. In grinding operations, the depths of cut are selected in the range of microns and some cutting edges can easily go out of the cut due the tool vibrations. This phenomenon is handled through a modification in the model. In the dynamics part of the model after a single step is executed, the geometric-kinematic model is run through a shadow (or a copy) of the element. In the shadow element a full rotation is simulated, and the active grits are calculated. This means a full rotation of a shadow element is always executed after each step is calculated. This is shown in Figure 42.

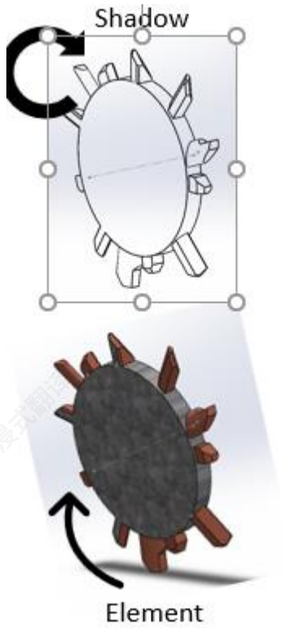



Figure 42: The shadow-element method visualization



连续活跃粗粒之间的延迟将是每个非活跃粗粒的延迟量乘以直到下一个活跃粗粒之前错过的非活跃粗粒数量。在我们的例子中，第一个和第二个活跃粗粒之间的延迟将是所有非活跃粗粒的每个延迟量的 49 倍。

模型中涵盖的另一种非线性是磨削工具振动时有效砂粒数量的变化。在磨削操作中，切削深度通常选择在微米范围内，由于工具振动，一些切削刃可能会轻易偏离切削区域。该现象通过模型修改进行处理。在模型动力学部分执行单步后，几何运动学模型会通过元素的阴影（或副本）运行。在阴影元素中模拟完整旋转，并计算有效砂粒。这意味着每次计算步骤后，阴影元素都会执行完整旋转。这如图42所示。

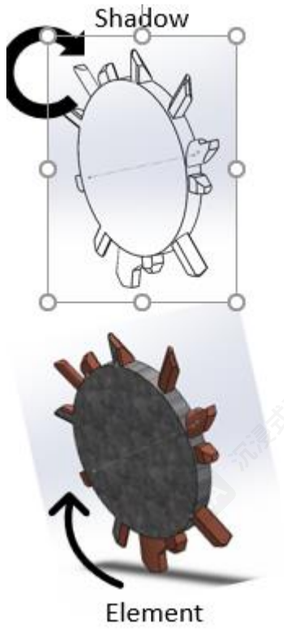



图42：阴影元素法可视化

## 4.4 Experimental Procedure

For the proposed dynamic model, the modal parameters of the grinding tools are measured by the hammer tests. The hammer used in the tests is PCB model number 086C01 and the accelerometer is Kistler model number 8728A500. The data is collected through National Instruments USB Carrier model number NI USB-9162. The FRF of the tool and the modal parameters are calculated with the software CutPro. The measurement setup is given in Figure 43. This is Wheel 1 from Figure 16 which is a 10mm EP CBN B126 fresh wheel as shown in Figure 44.

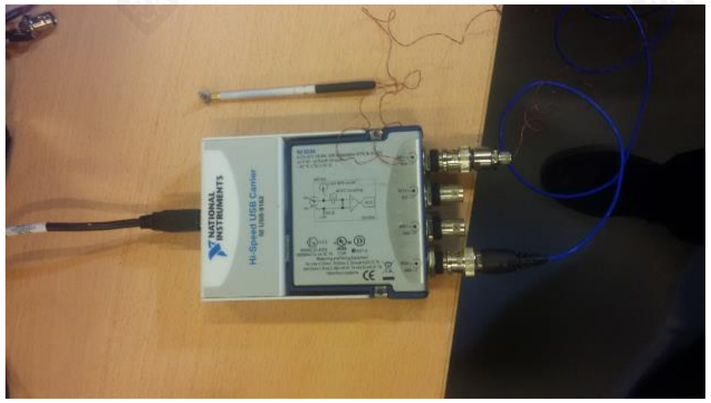



Figure 43: The hammer test setup

The modal parameters for the x and y directions obtained from the program analyzing the information transmitted to the data acquisition board by the accelerometer are given in Table 6.

| Direction | Frequency (Hz) | Damping Ratio (%) | Modal Stiffness (N/mm) |
|---|---|---|---|
| X | 2100 | 3,9 | 3124287 |
| Y | 2023 | 2,3 | 1326700 |

Table 6: Modal parameters of the wheel



## 4.4 实验步骤

对于所提出的动态模型，磨削工具的模态参数通过锤击测试进行测量。测试中使用的锤子是PCB型号086C01，加速度计是Kistler型号8728A500。数据通过国家仪器USB载体型号NI USB-9162收集。工具的频响函数和模态参数使用CutPro软件计算。测量设置如图43所示。这是图16中的轮子1，如图44所示，是一个10mm EP CBN B126的新轮。

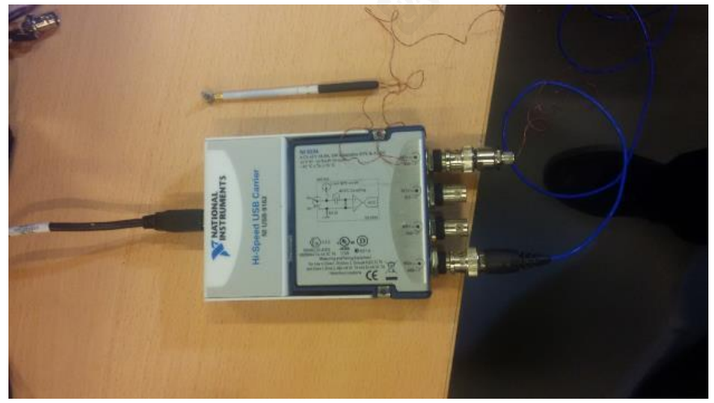



图43：锤击测试装置

程序分析加速度计向数据采集板传输的信息所获得的 x 和 y 方向的模态参数见表6。

| 方向 | 频率 (Hz) | 阻尼比 (%) | 模态刚度 (N/mm) |
|---|---|---|---|
| X | 2100 | 3,9 | 3124287 |
| Y | 2023 | 2,3 | 1326700 |

表6：车轮模态参数

The main aim of this work is the dynamic force investigation of the CBN wheels on hardened alloys. In this manner, experiments are carried out on the Chevalier Smart H/B 818III CNC surface grinding machine with the combination of Pferd BZY-N 20x10/6 B126 CBN wheel and 7mmx90mm 50HRC hardened AISI1050 carbon steel. Force measurements are gathered via Kistler 9129AA table type 3-axis dynamometer with 5kHz external signal amplifier due to the magnetic chuck limitations. Keyence LK-G5000 series laser displacement sensor has been installed which has 50kHz sampling rate onto the wheel shaft to detect vibration of the tool. A simple microphone is used to measure sound data to detect if there is instability during the grinding process. All devices are connected to Labview program via NI USB-6251 BNC with 50kHz sampling rate for each channel. The experimental setup is illustrated in Figure 44 with the described devices.

In the test matrix, workpiece velocity is scanned from 100mm/min to 20m/min with the consecutive radial depth of cut from 10 to 50µm for constant spindle speed of 40000 rpm. The axial depth is selected as 7mm. In each data set, spark-out operations have been performed to supply a consistency of actual depth of cut. Stability is investigated through different feed rate and depth of cut since higher spindle speed resulted in higher vibration amplitude of the wheel. The grinding wheel is selected as a 20mm diameter electroplated CBN with concentration B126.

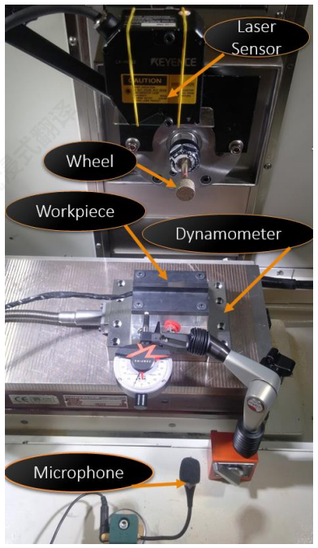



Figure 44: The experiment setup



这项工作的主要目的是对 CBN 轮在硬化合金上的动态力进行探究。为此，在 Chevalier Smart H/B 818III 数控表面磨床上进行实验，结合使用 Pferd BZY-N 20x10/6 B126 CBN 轮和 7mmx90mm 50HRC 硬化 AISI1050 碳钢。由于磁力夹的限制，通过 Kistler 9129AA 台式三轴测力计收集力测量数据，并使用 5kHz 外部信号放大器。安装了 Keyence LK-G5000 系列激光位移传感器，其采样率为 50kHz，用于检测工具的振动，并将其安装在轮轴上。使用一个简单的麦克风测量声音数据，以检测磨削过程中是否存在不稳定性。所有设备都通过 NI USB-6251 BNC 连接到 Labview 程序，每个通道的采样率为 50kHz。实验装置如图 44 所示，包括所描述的设备。

在测试矩阵中，工件速度从 100mm/min 扫描到 20m/min，连续的径向切削深度从 10 到 50µm，主轴转速恒定为 40000 rpm。轴向切削深度选择为 7mm。在每个数据集中，都进行了清根操作以确保实际切削深度的稳定性。通过不同的进给率和切削深度来研究稳定性，因为较高的主轴转速会导致砂轮振动幅度增大。选择的砂轮是直径 20mm 的电镀立方氮化硼，浓度为 B126。

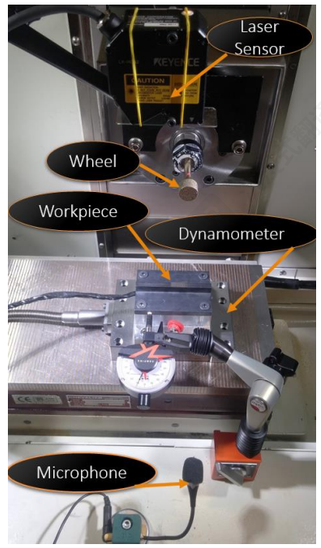



图44：实验装置




Figure 45: The measured forces of 40µm radial depth for varying feed rate

The measurement of dynamic forces is not feasible with this setup. The dynamic forces detection using the dynamometer may be insufficient as the force measurements will lack the sufficient frequency. The dynamometer used in this research has the natural frequency of 2.3kHz; however, when workpiece is clamped on it due to its mass its bandwidth reduces. The wheel natural frequency is measured to be 2.1kHz.

Another significant note here is that the laser sensor was initially intended to measure the vibrations of the grinding wheel. The laser sensor (Keyence) act as a photoreceptor and measures the distance as of the reflected light from the target. The reflected light's brightness determines the distance to the sensor. For this setup, the orientation of the laser sensor was a significant problem as the wheel arbor is thin and cylindrical which will cause reflection errors to the sensor. Another problem is about the fixture of the laser sensor. A fixture was designed and manufactured for this laser sensor to measure the wheel while the machine is in operation. The operation conditions would easily disturb the sensor orientation just enough to lose the reflection orientation. Hence, the sensor was used only for the detection of the true rpm of the machine.



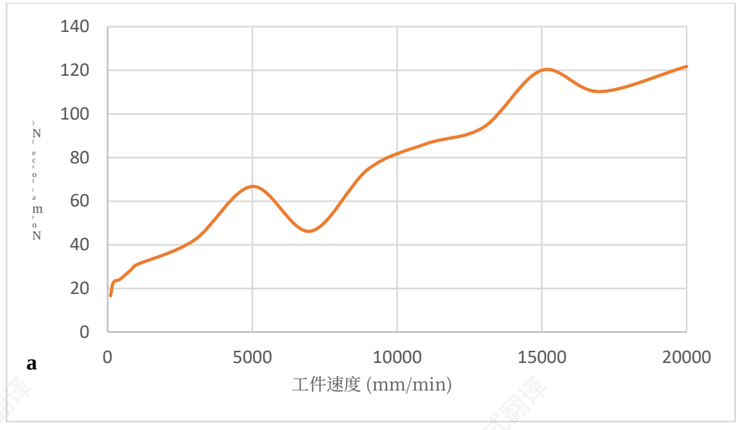



图45：不同进给速度下的40µm径向深度测量力

使用该装置无法测量动态力。使用测力计进行动态力检测可能不足，因为力测量的频率不够。本研究中使用的测力计固有频率为 2.3kHz；然而，当工件因质量被夹持在上面时，其带宽会降低。砂轮的固有频率测量为 2.1kHz。

此处另一点值得注意的是，激光传感器最初旨在测量磨轮的振动。激光传感器（Keyence）充当光电接收器，并测量目标反射光线的距离。反射光的亮度决定了与传感器的距离。对于此设置，激光传感器的方向是一个重要问题，因为轮轴是薄而圆柱形的，这会导致反射误差到传感器。另一个问题是关于激光传感器的固定。为此设计并制造了一个固定装置，用于在机器运行时测量轮子。运行条件会轻易干扰传感器方向，足以失去反射方向。因此，该传感器仅用于检测机器的真实转速。

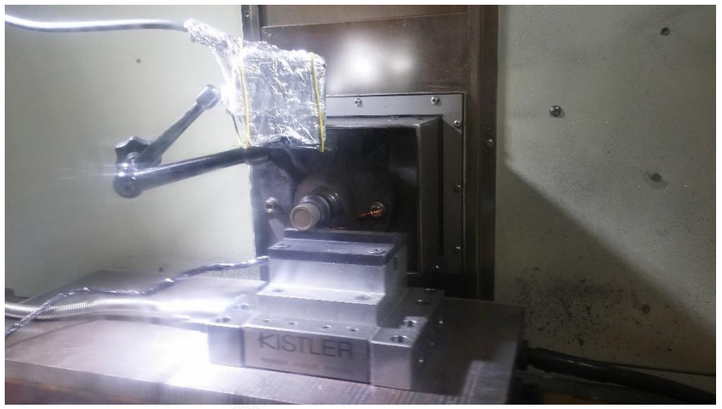



Figure 46: The laser sensor mounting

As these measurement methods could not be utilized, an indirect method to verify the dynamic force model is established.

The measured force in the test is given in Figure 45. The increase in the force in the normal direction along with the increasing feed rate almost follows a linear pattern. It is observed that trend is broken at the feed rates of 5000 and 15000mm/min. The measurements were repeated several times and the same profile is measured. This force profile is a result of the stochastic nature of the grinding process. It is known from the geometric-kinematic model that, as the feed rate increases so does the number of active grits. The increased number of active grits together with the decrease of the cutting force coefficients may have an effect on the dynamics of the system. This phenomenon is further investigated in the results section of this chapter.

## 4.5 Dynamic Force Results

It is assumed that there are no wheel vibrations in the geometric-kinematic model mentioned in the previous chapter. This is not very accurate because, unlike other cutting operations, low radial depths are used for surface grinding and the effect of the vibrations on the process variables will be significant. For example, a 10-micron deflection in a conventional milling operation may not significantly affect the process



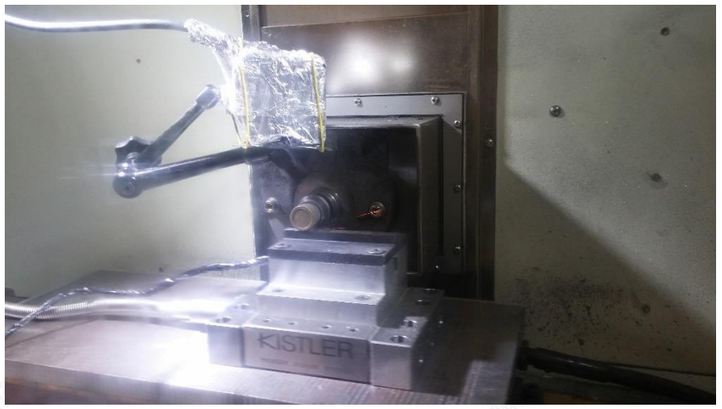



图46：激光传感器安装

由于这些测量方法无法使用，建立了一种间接方法来验证动态力模型。

测试中的测量力如图45所示。法向力随进给速度的增加几乎呈线性变化。观察到在进给速度为5000和15000mm/min时，该趋势被打破。多次重复测量，得到相同的轮廓。这种力轮廓是磨削过程随机性的结果。根据几何运动学模型可知，随着进给速度的增加，激活的磨粒数量也随之增加。激活磨粒数量的增加以及切削力系数的减小可能对系统的动力学产生影响。这种现象将在本章结果部分进一步研究。

## 4.5 动态力结果

假设前一章提到的几何运动学模型中没有轮子振动。这并不太准确，因为与其他切削操作不同，表面磨削使用的是低径向深度，而振动对工艺变量的影响将会显著。例如，在传统的铣削操作中，10微米的偏移可能不会显著影响工艺结果，但在磨削过程中，它可能会使砂轮脱离工件接触。

result, but in an abrasive process it can make the wheel come out of the contact with the workpiece.

As described in the previous chapters, the dynamic grinding formulation was developed using the milling analogy. The dynamic chip thickness is calculated by using particle orientations, and thus the dynamic forces are obtained. The modal parameters found for the small CBN wheel (B126 20mm) are used in the time domain simulations.

An output of the dynamic model is presented in Figure 47.

In the dynamic force model, the mechanistic force model is used, and the force coefficients are identified using the maximum force output obtained from the dynamometers.

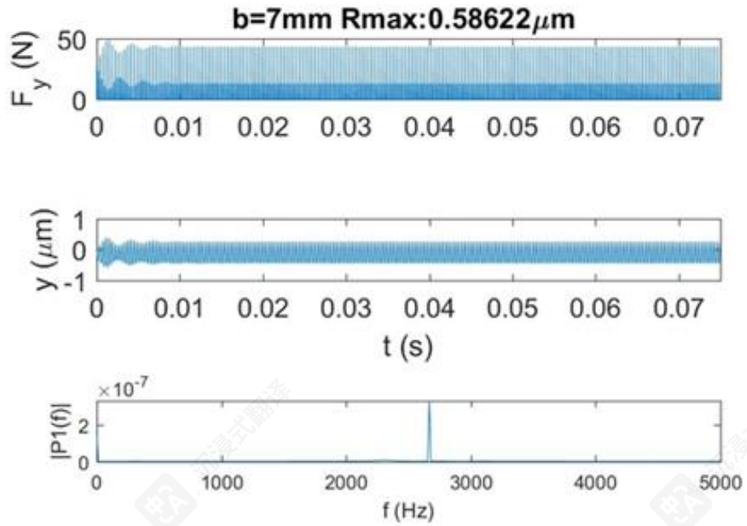



Figure 47: Dynamic forces and vibrations, 40000rpm, 0.125 mm/rev, 40μm radial depth (a) forces (b) vibration (c) vibration frequency

The process parameters given in Figure 47 are:

Rpm: 40000 / min

Feed: 0.125 mm / rev



例如，在传统的铣削操作中，10微米的偏移可能不会显著影响工艺结果，但在磨削过程中，它可能会使砂轮脱离工件接触。

如前几章所述，动态磨削公式是利用铣削类比开发的。通过使用颗粒方向计算动态切屑厚度，从而得到动态力。为时域仿真所用的小尺寸CBN砂轮（B126 20mm）的模态参数已被确定。

动态模型的输出如图47所示。

在动态力模型中，采用机理力模型，并通过从测力计获得的最大力输出识别力系数。

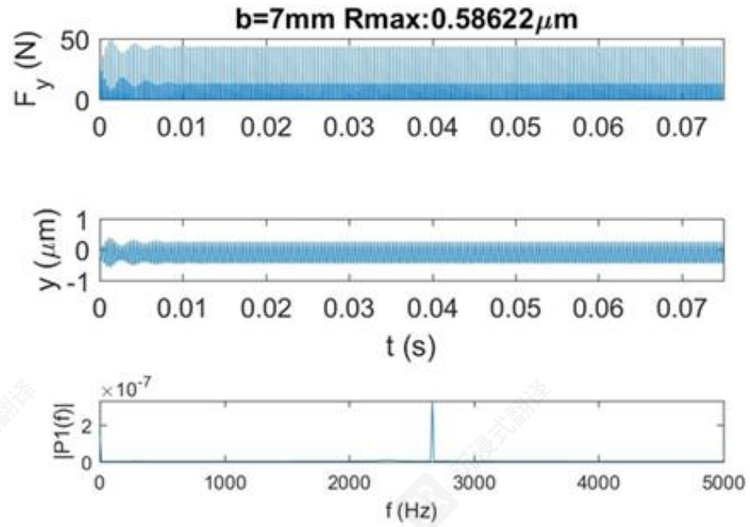



图47：动态力与振动，40000rpm，0.125 mm/rev，40μm径向深度 (a) 力 (b) 振动 (c) 振动频率

图47中给出的工艺参数为：

转速：40000 / min

进给：0.125 mm / 转

Radial depth: 40 µm

Axial depth: 7mm

In the figure, dynamic cutting forces in the y direction and Fourier transformation spectrum of the y direction vibration are given, respectively.

The forces are obtained after the cutting coefficients are determined. When the active grit detection algorithm is used, the number of particles falling on an element is 8, depending on the process parameters given in Figure 47. Therefore, the feed per tooth is approximately 0.015 mm. Please consider that the tooth number 8 is only for one element. Considering the diameter of the tool (20mm) and the axial depth of cut (7mm) the element number is nearly 150. So, on the contact surface, the number of active grits are nearly 1200. A closer look to the force predictions are given in Figure 48.




Figure 48: Comparison of the experimental force data. Wheel speed 40m/s, radial depth of cut 40µm, table speed: 5000mm/min



径向深度：40 µm

轴向深度：7mm

在图中，分别给出了 y 方向的动态切削力和 y 方向振动的傅里叶变换频谱。

在确定切割系数后，才能获得这些力。当使用主动磨粒检测算法时，落在每个元素上的颗粒数量为8，具体取决于图47中给出的工艺参数。因此，每齿进给量约为0.015毫米。请注意，8齿仅针对一个元素。考虑到刀具直径（20毫米）和轴向切深（7毫米），元素数量接近150。因此，在接触面上，主动磨粒的数量接近1200。图48给出了更详细的力预测结果。

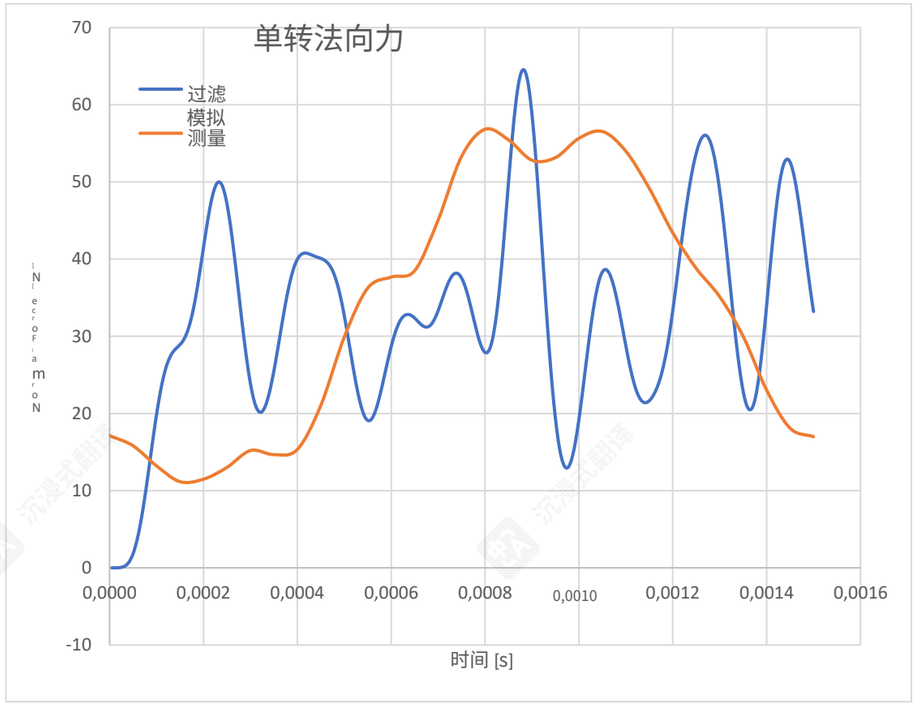



图48：实验力数据的对比。轮速40m/s，切深径向40µm，工作台速度：5000mm/min

The maximum and minimum values of both simulated and measured forces are in a good order. There is no significant change in the average values of the simulated force. So, the max., min. and the average force values can be predicted. The differences at the peak points is due to the insufficiency of the dynamometer for catching the high frequency output.

The observation about the discrepancy of the linear increasing trend during the experiments was studied with the dynamic model. The comparisons of the measured and predicted force values for both models (geometric-kinematic and dynamic) are given in Figure 49 and Figure 50. The geometric kinematic model behaves as predicted such as in the previous chapter but has lower performance predicting the force at feedrates of 5000 and 15000 mm/min. The model cannot predict the unusual increase in the force at those feedrates. However, the dynamic model is able to predict the forces at feedrates of 5000 and 15000 mm/min with higher accuracy as can be seen in Figure 49 and Figure 50.

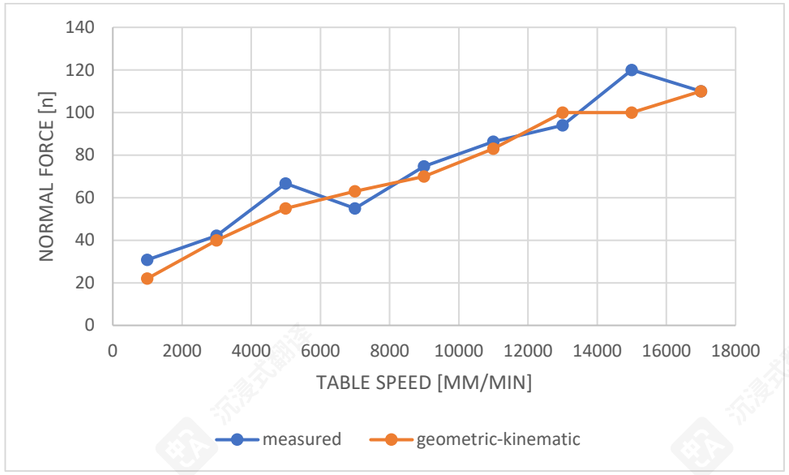



Figure 49: Comparison of measured and predicted force (Geometric-kinematic model) data for varying feed rate, wheel speed 40m/s, radial depth of cut 40µm



模拟和实测力的最大值和最小值都在合理范围内。模拟力的平均值没有明显变化。因此，可以预测最大值、最小值和平均值。峰值点处的差异是由于测力仪无法捕捉高频输出的不足。

实验中线性增长趋势的差异观察是通过动态模型进行研究的。两种模型（几何运动学和动态模型）的测量值与预测值比较如图49和图50所示。几何运动学模型表现符合预期，如前一章所述，但在进给速度为5000和15000 mm/min时预测力的性能较低。该模型无法预测这些进给速度下力的异常增加。然而，动态模型能够以更高的精度预测5000和15000 mm/min时的力，如图49和图50所示。

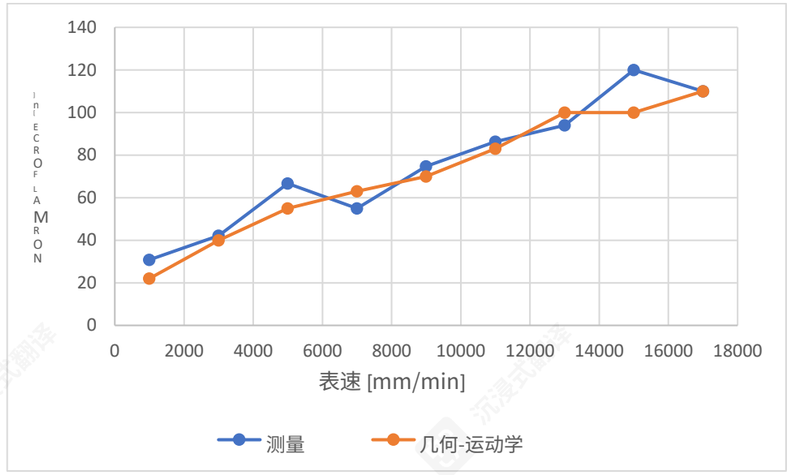



图49：不同进给速度下，轮速40m/s，切深40µm的测量与预测力（几何-运动学模型）数据对比

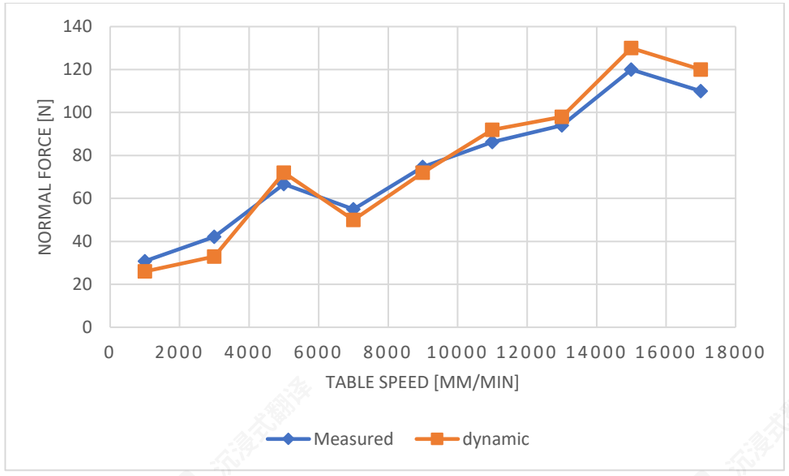



Figure 50: Comparison of measured and predicted force (Dynamic model) data for varying feed rate, wheel speed 40m/s, radial depth of cut 40µm

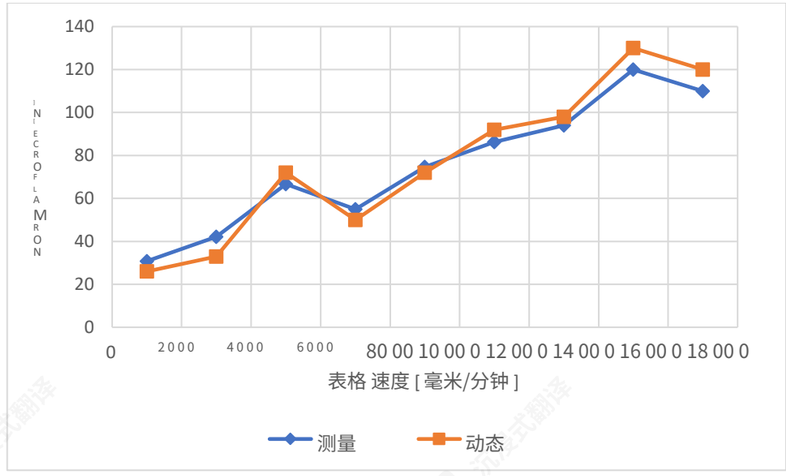



图50：不同进给速度、轮速40m/s、径向切削深度40µm的测量与预测力（动态模型）数据对比

Three different test cases, simulated above, are considered. The active particle counts are as follows.

Test 1: Feed 3000 mm / min, number of active particles: 6, Kn: 5.4 GPa

Test 2: Feed 5000 mm / min, number of active particles: 8, Kn: 3.5 GPa

Test 3: Feed 7000mm / min, number of active particles: 9, Kn: 3.1 GPa

where Kn is the force coefficient in the normal direction which was determined experimentally.

The frequency of the process forces coming into the system depends on the number of active grits. In grinding, the increase in feed rate affects the number of active grits and the cutting force coefficient itself. I will be seen clearly in the next chapter from the stability charts that the condition of Test 2 is much closer to the stability limit than Test 1 and Test 3. This is the reason for the unusual increased forces at the feed rates of 5000mm/min and 15000mm/min. The dynamic forces are closely investigated at the feed rates of 5000 mm/min and 7000 mm/min and shown in Figure 51 and Figure 52. It is shown that in the case of 5000 mm/min feed rate, the vibration amplitudes are higher and the system is less stable. When the feed rate is increased to 7000mm/min the forces



考虑了上述模拟的三个不同测试案例。活动粒子数量如下。

测试1：进给3000 mm/min，活跃粒子数：6，Kn：5.4 GPa

测试2：进给5000 mm/min，活跃粒子数：8，Kn：3.5 GPa

测试3：进给7000mm/min，活跃粒子数：9，Kn：3.1 GPa

其中Kn是法向力系数，通过实验测定得到。

该工艺进入系统的频率取决于活动砂粒的数量。在磨削过程中，进给率的增加会影响活动砂粒的数量以及切削力系数本身。从稳定性图表中可以清楚地看到，第2次试验的条件比第1次试验和第3次试验更接近稳定性极限。这就是在进给率为5000mm/min和15000mm/min时出现异常增加力的原因。动态力在进给率为5000 mm/min和7000 mm/min时被密切研究，并显示在图51和图52中。结果表明，在进给率为5000 mm/min的情况下，振动幅度较高，系统稳定性较差。当进给率增加到7000mm/min时，力

become more stable. The detailed discussion with stability will be given in the next chapter.

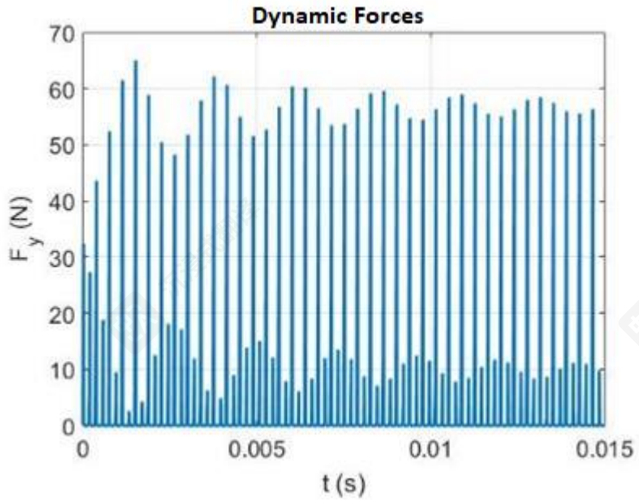



Figure 51: Dynamic forces when feedrate is 5000mm/min

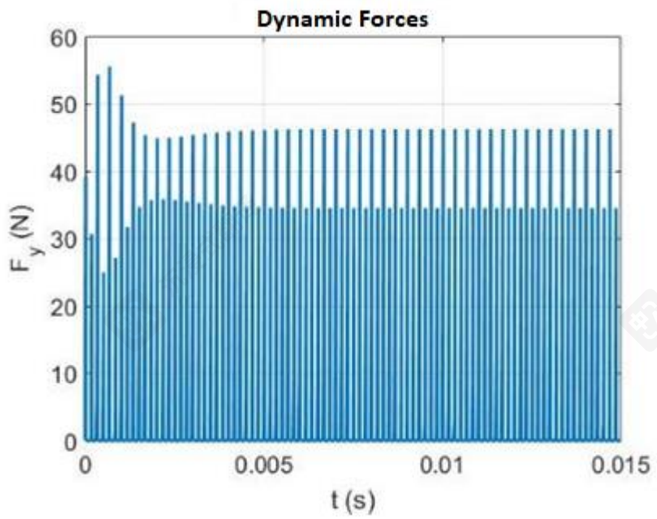



Figure 52: Dynamic forces when feedrate is 7000mm/min



变得更加稳定。与稳定性的详细讨论将在下一章给出。

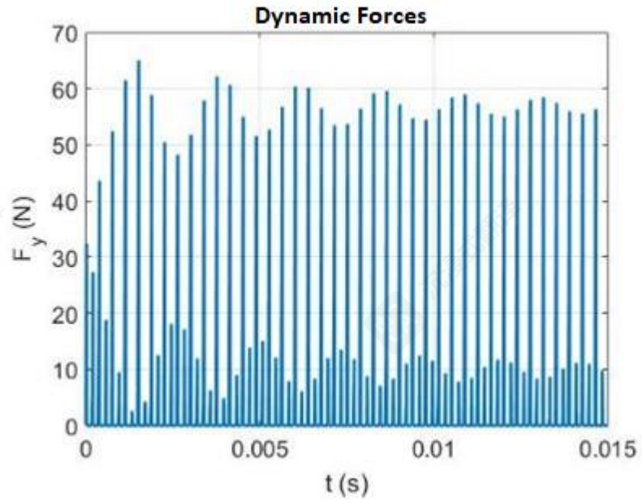



图51：进给速度为5000mm/min时的动态力

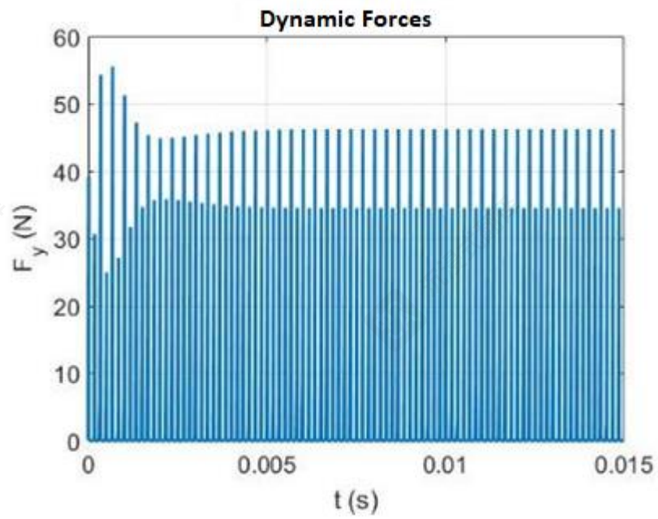



图52：进给速度为7000mm/min时的动态力

## 4.6 Surface Roughness Results

A series of experiments were conducted to verify the surface roughness prediction with the dynamic model. Thin-walled Inconel 718 (2 mm thick) test piece was selected to minimize the force and wheel vibration. The experiments were planned to eliminate the elastic deflection of the wheel to assure that certain radial depth of cut is removed. Hence, the wheels were mounted with a minimum over-hang to minimize the static deflection. The experimental setup is shown in Figure 53. The surface roughness of the workpiece is measured with the Talysurf surface tracer as well as the 3D $\mu$surf Nanofocus microscope. Two wheels were tested: Wheel 1 and Wheel 2 as presented in Figure 8, which are the 10mm EP CBN B126 and B64, respectively. Experiments were carried out at different feed rates: 0.01 mm/rev, 0.03 mm/rev, 0.05 mm/rev and 0.1 mm/rev. The radial depth of cut was 40 μm and the wheel speed was 40m/s. The surface roughness results are shown in Figure 54 Figure 54 and Figure 55.

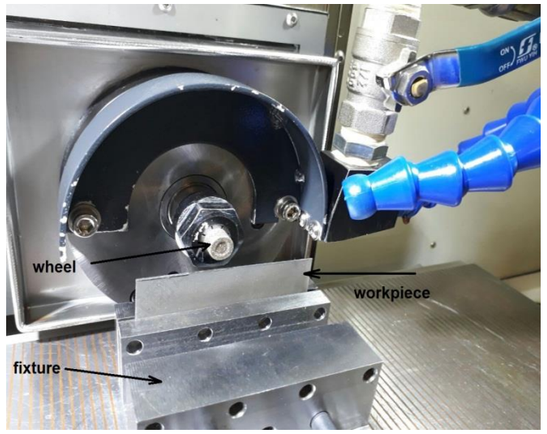



Figure 53: The experimental setup



## 4.6 表面粗糙度结果

进行了一系列实验，以验证动态模型下的表面粗糙度预测。选用2毫米厚的薄壁Inconel 718测试件，以最小化力和轮子振动。实验计划消除轮子的弹性变形，以确保去除一定的径向切深。因此，轮子以最小悬臂安装，以最小化静态变形。实验装置如图53所示。工件表面粗糙度使用Talysurf表面跟踪仪和3D $\mu$surf纳米聚焦显微镜测量。测试了两个轮子：图8中所示的轮子1和轮子2，分别为10mm EP CBN B126和B64。实验在不同进给率下进行：0.01 mm/rev、0.03 mm/rev、0.05 mm/rev和0.1 mm/rev。径向切深为 40 μm，轮速为40m/s。表面粗糙度结果如图54、图54和图55所示。

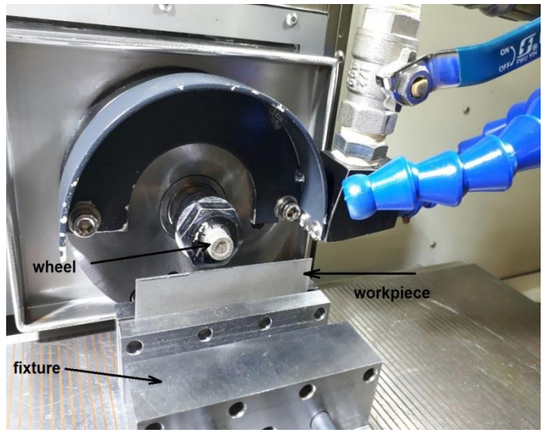



图53：实验装置

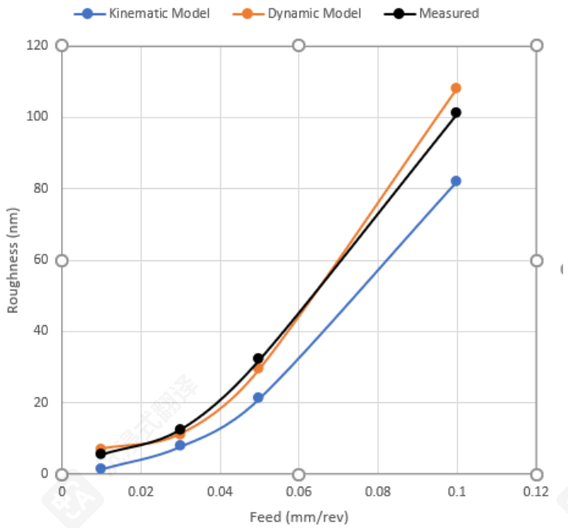



Figure 54: Model and experimental results of the surface roughness for wheel 1 at different feed rates.

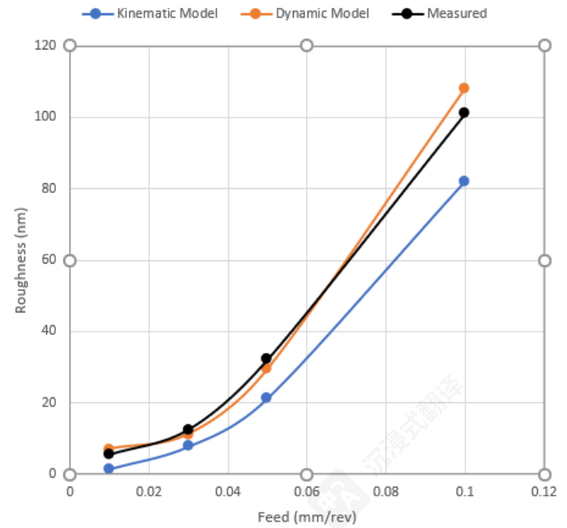



图 54：轮 1 在不同进给速度下的表面粗糙度模型和实验结果。

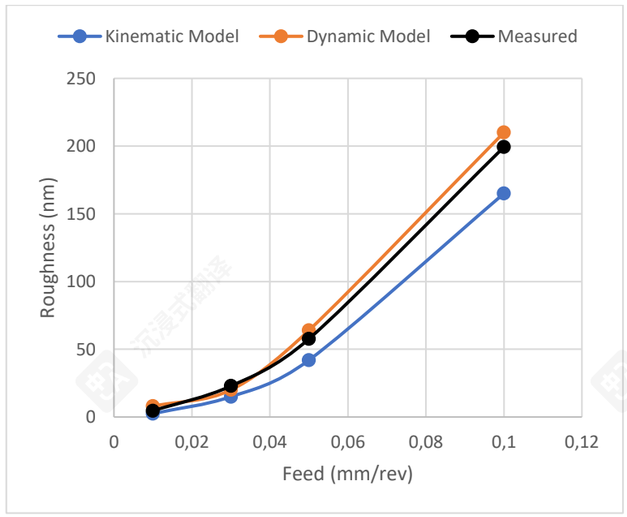



Figure 55: Model and experimental results of the surface roughness for wheel 2 at different feed rates.

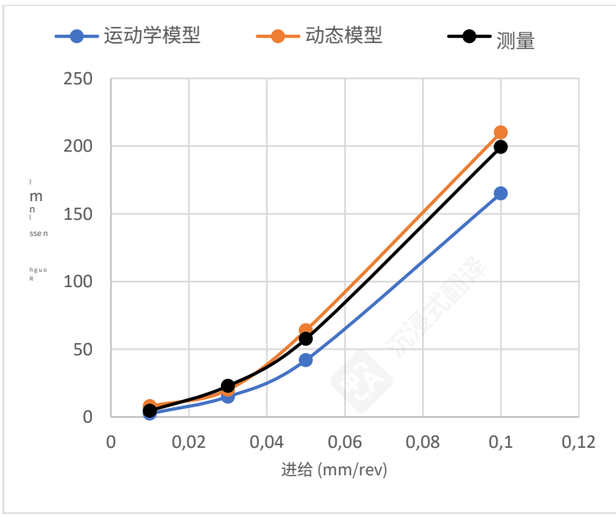



图 55：轮 2 在不同进给速度下的表面粗糙度模型和实验结果。

As it can be seen from the figures the performance of the geometric – kinematic model on predicting the surface roughness in cutting direction is lower than the proposed dynamic model. The reason for this is the fact that the geometric- kinematic model does not account for the dynamics of the grinding process and represents only the feed marks on the surface of the workpiece whereas the dynamic model is able to predict the vibrations of the grinding wheel and their imprints on the surface. The surface roughness of the workpiece along the feed direction is higher when ground with Wheel 2. There are two reasons for that. The first one is the active grit number on Wheel 2. As shown in Figure 33, the active number of grits is higher on Wheel 2 which will experience higher grinding forces due to ploughing. The resultant dynamic forces may result in higher amount of vibrations which will produce lower surface quality. The other reason is that the modal stiffness of Wheel 2 is lower as compared to Wheel 1. The lower modal stiffness also leads to higher amplitude of vibrations of the grinding wheel.

从图中可以看出，几何-运动学模型在预测切削方向表面粗糙度方面的性能低于所提出的动态模型。这是因为几何-运动学模型没有考虑磨削过程的动力学特性，仅表示工件表面的进给痕迹，而动态模型能够预测砂轮的振动及其在表面上的印记。使用砂轮2磨削时，工件沿进给方向的表面粗糙度更高。这有两个原因。第一个是砂轮2上的有效磨粒数量。如图33所示，砂轮2上的有效磨粒数量更多，这将由于切削作用导致更高的磨削力。产生的动态力可能导致更高的振动量，从而产生更低的表面质量。另一个原因是砂轮2的模态刚度低于砂轮1。模态刚度较低也会导致砂轮振动的振幅更大。

## 4.7 Summary

## 4.7 摘要

In this chapter a dynamic force model is developed for surface grinding operations. The dynamic model was needed in order to predict the grinding forces in some cases also to predict the surface roughness in the cutting direction. The geometric-kinematic model had shortcomings to describe the process in terms of the grinding force and the surface roughness in the cutting direction.

本章针对表面磨削操作开发了一种动态力模型。需要该动态模型是为了预测某些情况下的磨削力，有时也需要预测切削方向上的表面粗糙度。几何运动学模型在描述磨削力和切削方向上的表面粗糙度方面存在不足。

Applying the milling analogy and the geometrical random distribution of the grits on the grinding wheel, the simulation can predict the dynamic forces and vibrations. In order to simulate the geometry of the grinding process, the geometric-kinematic model is used and integrated with the dynamic model. The active grits are calculated and the chip thickness per active grit is computed with the trajectory simulation of the wheel. The dynamic model is created with the inclusion of the regenerative effect of the workpiece surface. An algorithm is developed to simultaneously use the dynamic and the geometric-kinematic model in order to represent some of the non-linearities of the grinding process such as the change of active grit number during the wheel vibration and the random delays of the active grits. The dynamic model is used with the time

应用铣削类比和磨粒在砂轮上的几何随机分布，仿真可以预测动态力和振动。为了模拟磨削过程的几何形状，采用了几何运动学模型，并将其与动态模型集成。计算了活动磨粒，并通过砂轮的轨迹仿真计算了每个活动磨粒的切屑厚度。动态模型在考虑工件表面再生效应的情况下创建。开发了一种算法，以同时使用动态模型和几何运动学模型，以表示磨削过程的一些非线性特性，例如砂轮振动时活动磨粒数量的变化以及活动磨粒的随机延迟。动态模型用于时间域仿真，以预测力和切向方向的表面粗糙度。模型和实验结果显示出良好的相关性，且动态模型比几何运动学模型性能更优。

domain simulations in order to predict forces and the surface roughness is cutting direction. The model and the experiment results show a good correlation and the dynamic model has better performance than that of the geometric-kinematic model.

仿真结果和实验结果显示出良好的相关性，且动态模型比几何运动学模型性能更优。



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## 5 CHATTER STABILITY OF ABRASIVE PROCESSES

Generating the final surface and geometry is often critical in grinding operations, as they are after other machining operations. If conventional material removal methods are used, hard to machine materials may cause a higher cost. The low speed cutting will decrease the producibility while the high-speed milling will decrease the tool life dramatically when machining hard-to-cut materials. The super abrasives however, can be used in high-speed machining applications that will result in increased productivity. When high material removal rate is set as a goal, the cutting depths must be increased for high productivity. Elevated cutting depths will play an important role on the stability of the machining process. The stability is very important issue as the surface roughness of the workpiece is crucial on the finishing operations and the vibrations caused by chatter would diminish the surface. When the aim is to increase the productivity, the stability regions must be taken into account.

In literature, the stability of the abrasive processes has mostly been investigated for cylindrical grinding which most likely to face chatter. In cylindrical grinding, the workpiece is usually slender and long and susceptible to vibrations as the rigidity is low. Surface grinding has not been much of focus in the grinding chatter literature as in the surface grinding conditions the workpiece and the tool is usually rigid and high depth of cuts are not required. The most usual abrasive process conditions for the surface grinding is to have low depths of cut, relatively high feed rate and high diameter grinding tools (around 200 mm and higher). However, in the scope of this thesis, the grinding the stability model is developed for small tools (around 10-20 mm) as the measurement of the smaller tools are more useful with the microscope. When mounted on the grinding machine the small tools can be used in high-speed grinding conditions as a sufficient grinding speed is required in abrasive operations.

The time domain dynamic model proposed in the previous chapter is used to determine the stability regions of the abrasive processes which will also be studied with the frequency domain solution as well.



## 5 磨削过程的 chatter 稳定性

在磨削操作中，生成最终表面和几何形状通常至关重要，因为它们是在其他加工操作之后进行的。如果使用传统的材料去除方法，难加工材料可能会导致更高的成本。低速切削会降低生产率，而高速铣削在加工难加工材料时会显著降低刀具寿命。然而，超硬磨料可以在高速加工应用中使用，从而提高生产率。当将高材料去除率设定为目标时，必须增加切削深度以实现高生产率。提高切削深度将在加工过程稳定性中发挥重要作用。稳定性是一个非常关键的问题，因为工件表面粗糙度对精加工操作至关重要，而 chatter 造成的振动会降低表面质量。当目标是提高生产率时，必须考虑稳定性区域。

在文献中，磨削过程的稳定性主要针对圆柱磨削进行了研究，因为圆柱磨削最有可能出现 chatter。在圆柱磨削中，工件通常细长且易受振动影响，因为刚度较低。表面磨削在磨削 chatter 文献中关注较少，因为在表面磨削条件下，工件和刀具通常较刚性，且不需要高深度切削。表面磨削最常用的磨削工艺条件是低切削深度、相对较高的进给率和直径较大的磨削工具（约 200 毫米及以上）。然而，在本论文的范围内，磨削稳定性模型是针对小工具（约 10-20 毫米）开发的，因为使用显微镜测量小工具更有用。当安装在磨床上的小工具可以用于高速磨削条件时，因为磨削操作需要足够的磨削速度。

前一章提出的时域动态模型用于确定磨削过程的稳定性区域，该区域也将与频域解一起进行研究。

## 5.1 Frequency domain solution

During the studies [1,11,40], the grinding operation was modeled analogous to a micro-milling operation. According to this assumption, each grit on the grinding wheel behaves almost like a micro milling tooth. In this case, to model the dynamic of the grinding wheel, the force coefficients representing the relationship between the forces and the dynamic chip thickness must be determined. The first step is to calculate the dynamic chip thickness and cutting forces. Then an eigenvalue problem of multidimensional dynamic equations will be solved, and the stability limits will be calculated.

The frequency domain solution of the grinding operation is based on the model of multi – teeth milling problem presented in [51,52]. The grinding wheel is modeled as a milling tool with many teeth. In the analytical solution, the wheel is treated as a milling tool with an average active grit number that is distributed over the grinding wheel with average pitch angles. The averaging method is shown to be feasible in [11] as the normal distribution method leads to larger number of values around the mean value. The average pitch angles would also mean that an average delay is performed. The zero-order application may not provide very accurate results as the low radial depth of cut creates impulse-like force which will excite a larger spectrum of frequency and higher frequencies may be included to the solution.

The grinding wheel vibrates under dynamic forces leaves vibration marks on the surface. The abrasive grit that follows it also travels on the wavy surface left by previous grit. If, in any way, the process is evolving to self-oscillating waves, that is, if the amplitude of a current grit's vibration is more than or equal to the amplitude of the previous grit, then this process becomes unstable and chatter occurs This process is called regeneration. In the developed model [11], the workpiece is assumed to be a rigid and the dynamic characteristics are not considered. By using the zero-order approach, relationship between the spindle speed and the limiting depth of cut for the instability can be calculated using the analytical equations below [51,52].

$$a_{lim} = \frac{2\pi \times \Lambda_R}{N \times K_t}(1 + \kappa^2) \quad (33)$$

$$\kappa = \frac{\Lambda_{Im}}{\Lambda_{Re}} = \frac{T \times sin\omega_c}{1 - T \times cos\omega_c} \quad (34)$$



## 5.1 频域解法

在研究过程中，磨削操作被类似于微铣削操作进行建模。根据这一假设，磨轮上的每个磨粒几乎表现得像一个微铣削齿。在这种情况下，为了建模磨轮的动态特性，必须确定表示力与动态切屑厚度之间关系的力系数。第一步是计算动态切屑厚度和切削力。然后，将求解多维动态方程的特征值问题，并计算稳定性极限。

磨削操作的频域解法基于 [51,52] 中提出的多齿铣削问题模型。砂轮被建模为具有多个齿的铣削工具。在解析解中，砂轮被视为一个具有平均有效砂粒数的铣削工具，该砂粒数以平均节距角均匀分布在砂轮上。平均化方法在 [11] 中被证明是可行的，因为正态分布方法会导致更多值集中在均值附近。平均节距角也意味着执行了平均延迟。零阶应用可能不会提供非常准确的结果，因为低径向切削深度会产生脉冲状力，从而激发更宽的频率范围，并且高频可能会被包含在解中。

砂轮在动态力作用下振动，会在表面留下振动痕迹。随后跟上的磨粒也在前一个磨粒留下的波状表面上运动。如果，无论如何，该过程演变成自激振荡波，也就是说，当前磨粒的振动幅度大于或等于前一个磨粒的振动幅度，那么这个过程就会变得不稳定并发生颤振。这个过程称为再生。在开发的模型 [11], 中，工件被假定为刚性的，不考虑动态特性。通过使用零阶方法，可以使用下面的解析方程计算主轴转速与不稳定性极限切削深度之间的关系[51,52]。

$$a_{lim} = \frac{2\pi \times \Lambda_R}{N \times K_t}(1 + \kappa^2) \quad (33)$$

$$\kappa = \frac{\Lambda_{Im}}{\Lambda_{Re}} = \frac{T \times sin\omega_c}{1 - T \times cos\omega_c} \quad (34)$$

$$T = \frac{\epsilon + 2k\pi}{\omega_c} \quad (35)$$

$$n = \frac{60}{N \times T} \quad (36)$$

$$\epsilon = \pi - 2\psi \quad (37)$$

$$\psi = tan^{-1}\,\kappa \quad (38)$$

In these equations (33-38) *T* is the tooth passing period. *κ* is the ratio of the imaginary and real parts of the dynamic transfer function of the wheel end point. *N* is the number of grits, *K*t is the predetermined cutting force coefficient, and *k* is the stability lobe number. *n* is the spindle speed.

The algorithm that is used for the zero-order application is as follows [51,52]:

- The dominant mode of the system is determined.
- The algorithm starts around a chatter frequency that is close to the dominant mode.
- The limiting depth of cut and the corresponding spindle speed is calculated. This procedure is continued for the consecutive lobe numbers (k=1, 2, 3…)
- The frequency range of 1.01 times to 1.5 times of the natural frequency of the wheel is spanned and the stability lobes are created.

The frequency response model is adopted and presented along with the time domain solution. The algorithm is provided with this thesis but the theory and the detailed calculation of the zero-order application and the Fourier series approach to the frequency domain solution to the chatter problem is provided in [51,52].

## 5.2 Time domain solution

For the stability limit calculation, the time domain solution that is provided in the previous chapter is used. The chatter can be predicted by the observation of the growing vibrations of the tool from the vibration outputs and the frequency spectrum of those vibrations. If the vibrations are in a growing manner and the frequency of the vibrations are at the chatter frequency (slightly higher than the natural frequency), the process is



$$T = \frac{\epsilon + 2k\pi}{\omega_c} \quad (35)$$

$$n = \frac{60}{N \times T} \quad (36)$$

$$\epsilon = \pi - 2\psi \quad (37)$$

$$\psi = tan^{-1}\,\kappa \quad (38)$$

在这些方程(33-38)中，T是齿通过周期。*κ* 是砂轮端点动态传递函数的虚部和实部之比。N是磨粒数量，Kt是预定的切削力系数，k是稳定性波瓣数。n是主轴转速。

用于零阶应用的算法如下 [51,52]：

- 确定系统的主导模态。
- 算法从接近主导模态的颤振频率附近开始。
- 计算限制切削深度和对应的轴转速。该步骤对连续的波数（k=1、2、3…）重复执行。
- 跨越轮子固有频率的1.01倍到1.5倍的频率范围，并创建稳定性波。

采用并展示了频域响应模型和时域解。该算法在本论文中提供，但零阶应用的原理和详细计算以及用于颤振问题的频域解的傅里叶级数方法在 [51,52] 中提供。

## 5.2 时域解

对于稳定性极限计算，采用前一章提供的时域解。通过观察工具振动输出及其频谱中振动的增长情况，可以预测颤振。如果振动呈增长趋势且振动频率处于颤振频率（略高于固有频率），则该过程被

identified as chatter. This may be a tiresome approach, but the dynamic model proposed in this thesis can handle some nonlinearities that will enhance the prediction. The dynamic model is run for different cases and the stability regions are determined. The method is described with sample outputs.

Figure 56 shows an example of an output.

The modal parameters identified for the small CBN wheel (B126 20mm) are used in the algorithm (given in Table 6).

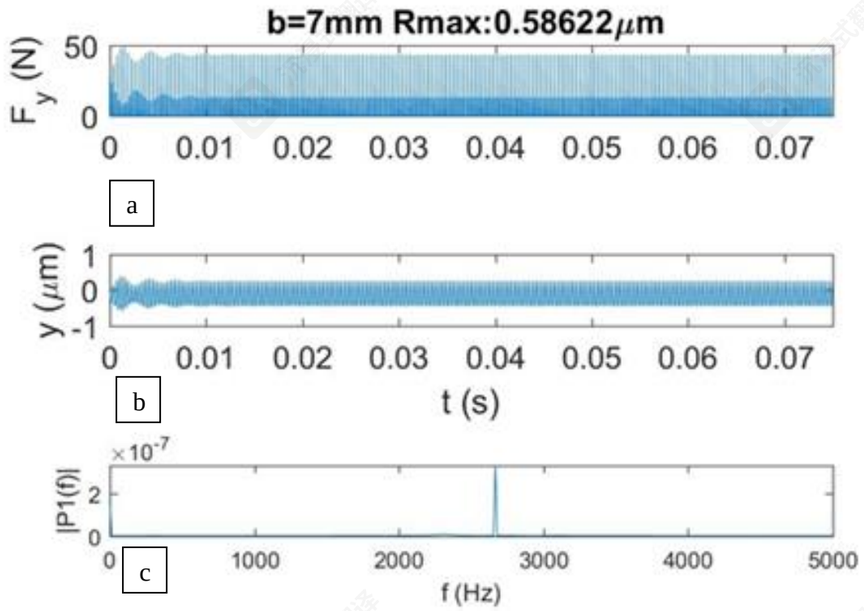



Figure 56: A sample dynamic grinding output (a) force (b) tool tip vibration (c) vibration spectrum

The process parameters given in Figure 56 are:

Rpm: 40000 / min

Feed: 0.125 mm / rev

Radial depth: 40 μm

Axial depth: 7mm



识别为颤振。这可能是一种繁琐的方法，但本论文提出的动态模型可以处理一些增强预测的非线性因素。动态模型针对不同案例进行运行，并确定了稳定性区域。该方法通过示例输出进行描述。

图56展示了一个输出示例。

为小CBN砂轮（B126 20mm）识别的模态参数用于该算法（见表6）。

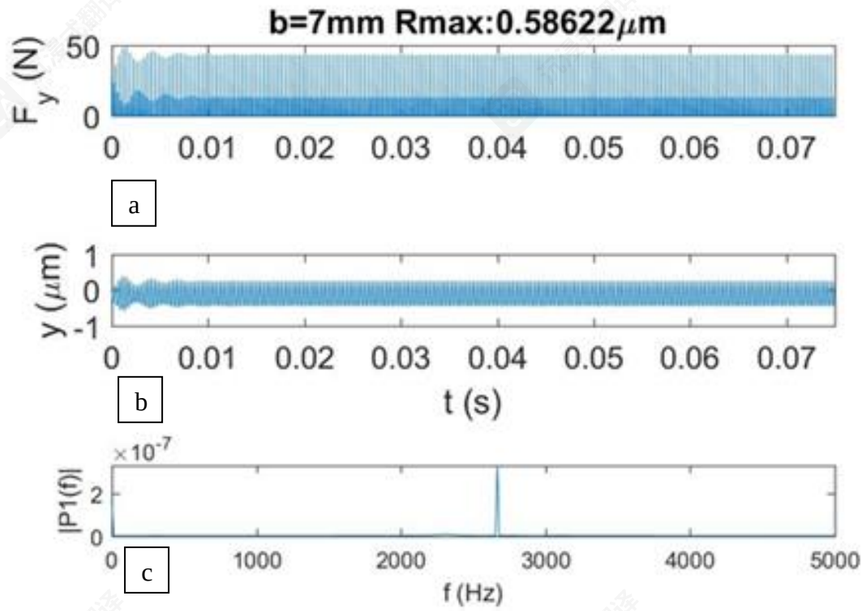



图56：一个动态磨削输出样本 (a) 力 (b) 刀具尖端振动 (c) 振动频谱

图56中给出的工艺参数为：

转速：40000 / min

进给：0.125 mm / 转

径向深度：40 μ米

轴向深度：7mm

In Figure 56, dynamic cutting forces in the y direction and Fourier transformation spectrum of the y direction vibration and tool tip vibration are given, R max shows the maximum roughness in order to indicate the vibration amplitude.

In time domain simulations, the axial depth is scanned in a range of 0 to 20 mm with 0.1 mm steps. Then for all spindle speeds between 30000 and 40000rpm, (with an increasing step of a thousand rpm) the different axial depths are scanned. When the chatter is observed the current values are captured and noted.

Figure 57 shows example of time simulation output for 40000 cycles of stable grinding at a speed of 5000 mm / min.

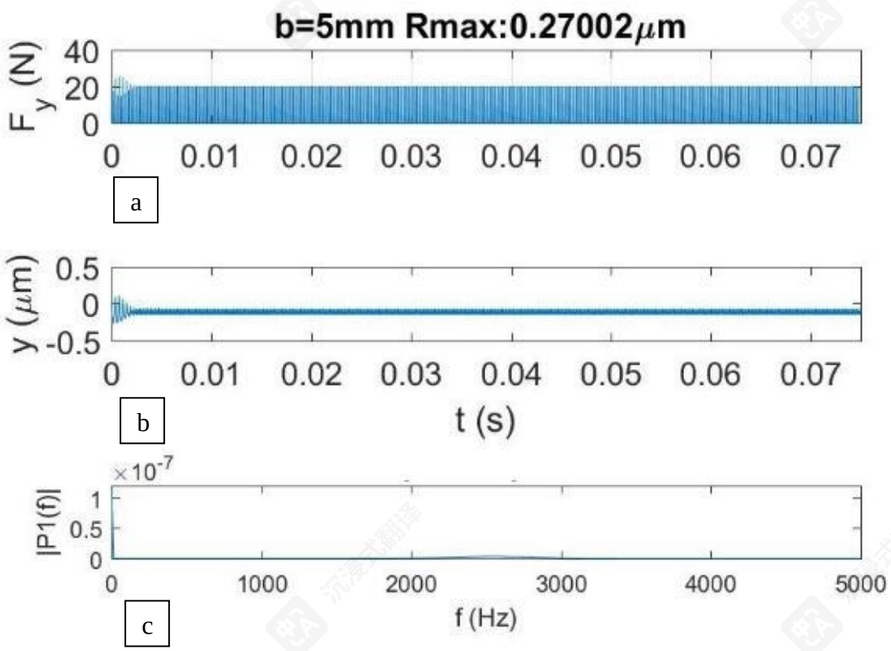



Figure 57: 40000 rpm, 5 mm depth, 5000 mm feed (a) force (b) tool tip vibration (c) vibration spectrum

Figure 57 shows a stable grinding example. For these conditions 5mm depth is stable. Vibrations and dynamic forces continue to diminish whereas there is no distinct vibration frequency in the spectrum.



在图56中，给出了y方向的动态切削力和y方向振动及刀具尖振动的傅里叶变换频谱，R max表示最大粗糙度，以指示振动幅度。

在时域仿真中，轴向深度以0.1毫米的步长扫描0到20毫米的范围。然后，对于所有30000到40000 rpm之间的主轴转速（以每千 rpm 的步长递增），扫描不同的轴向深度。当观察到颤振时，记录并保存当前值。

图57显示了在5000毫米/分钟速度下进行40000个稳定磨削循环的时域仿真输出示例。

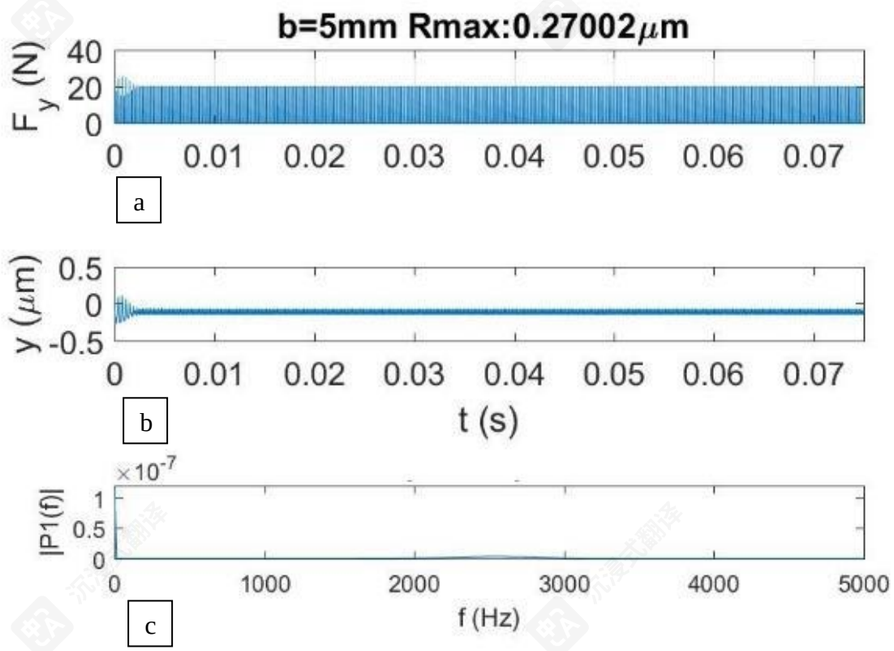



图57：40000 rpm，5毫米深度，5000毫米进给 (a) 力 (b) 刀具尖端振动 (c) 振动频谱

图57展示了一个稳定的磨削示例。在这些条件下，5毫米深度是稳定的。振动和动态力持续减小，而频谱中没有明显的振动频率。

No instability was seen when the depth was increased to 7 mm (Figure 58). When the vibrations were somewhat in the initial stage, they damped off later and settled on a stable profile. The frequency of the vibration is 2600 Hz. It is a harmonic of the spindle speed. (spindle speed 40000 rpm = 666 Hz).

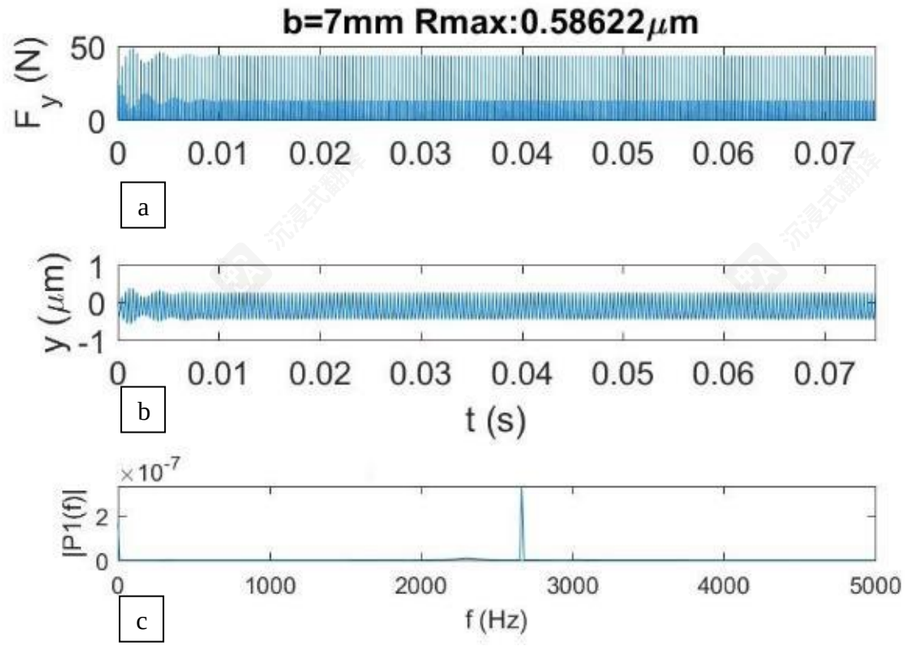



Figure 58: 40000 rpm, 7 mm depth, 5000 mm feed (a) force (b) tool tip vibration (c) vibration frequency

A marginally stable process is illustrated in Figure 59. The axial depth is increased to 7.7 mm. The frequency graph shows a small peak at 2300 Hz showing the sign of chatter. It is related to the natural frequency of the system (the measurements show there is one mode). The vibrations are settled a little bit later comparing to the previous case.

Figure 60 shows a time domain simulation of an unstable process where the axial depth is at 10mm. As can be seen, the dynamic forces and vibrations are in a state of great oscillation. The difference between the peak and the average forces and tool tip



当深度增加到7毫米时（图58），没有出现不稳定性。当振动处于初始阶段时，它们后来逐渐衰减并稳定在一个稳定轮廓上。振动频率为2600赫兹。它是主轴转速的谐波。（主轴转速40000 rpm = 666 Hz）。

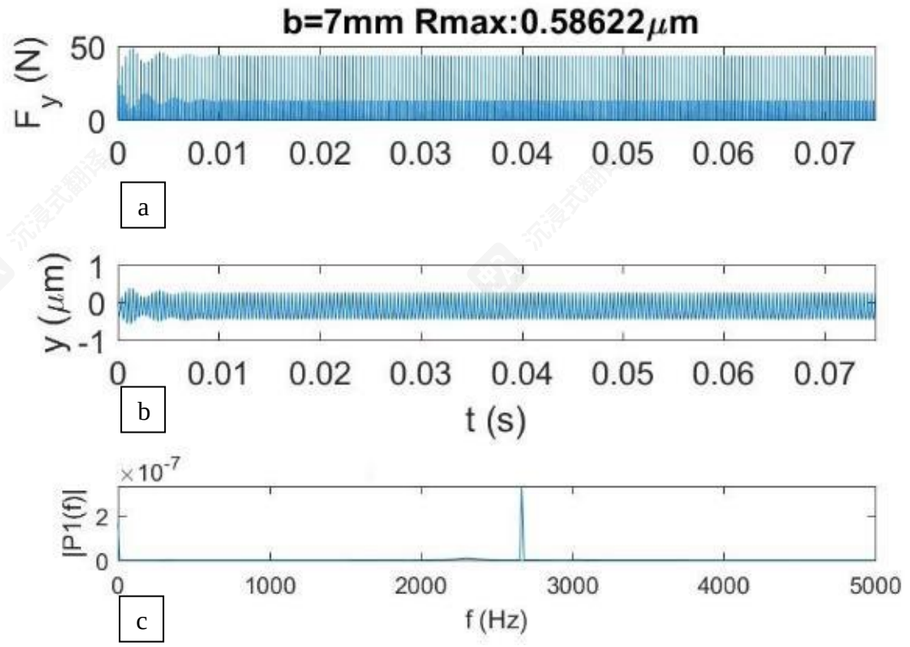



图58：40000 rpm，7 mm 深度，5000 mm 进给 (a) 力 (b) 刀尖振动 (c) 振动频率

图59展示了一个临界稳定过程。轴向深度增加到7.7 mm。频率图在2300 Hz处显示一个小峰值，表明存在颤振。这与系统的固有频率相关（测量显示存在一个模态）。与之前的情况相比，振动稍晚一些才稳定下来。

图60显示了轴向深度为10mm的不稳定过程的时间域仿真。如图所示，动态力和振动处于剧烈振荡状态。峰值力与平均力以及刀具尖端之间的差异

vibrations increased dramatically. The frequency of the vibration was found to be 2300 Hz, that is, the frequency of the chatter.

This procedure is repeated for the remaining spindle speeds and the stability limit corresponding to each spindle speed is found. By combining these limits, a stability diagram is obtained.

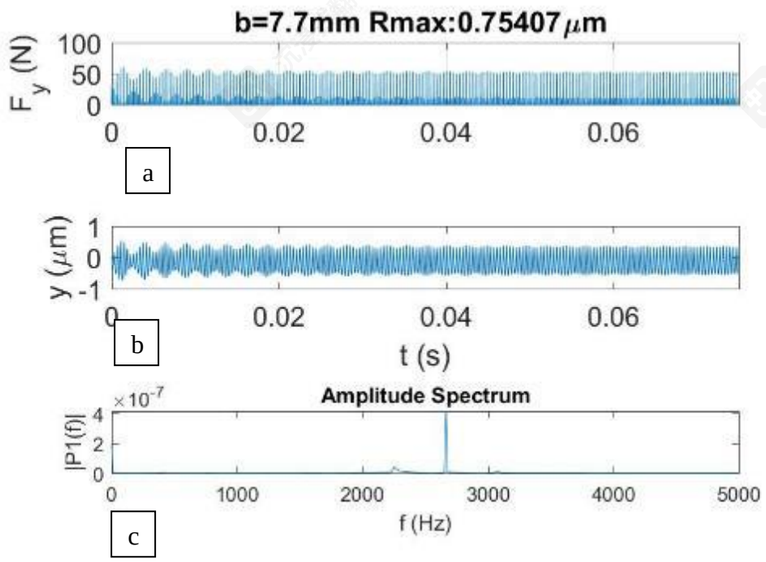



Figure 59: 40000 rpm, 7.7 mm depth, 5000 mm feed (a) force (b) tool tip vibration (c) vibration spectrum.



振动急剧增加。发现振动频率为2300赫兹，即颤振频率。

该步骤对剩余主轴转速重复执行，并找到与每个主轴转速对应的稳定性极限。通过结合这些极限，即可得到稳定性图。

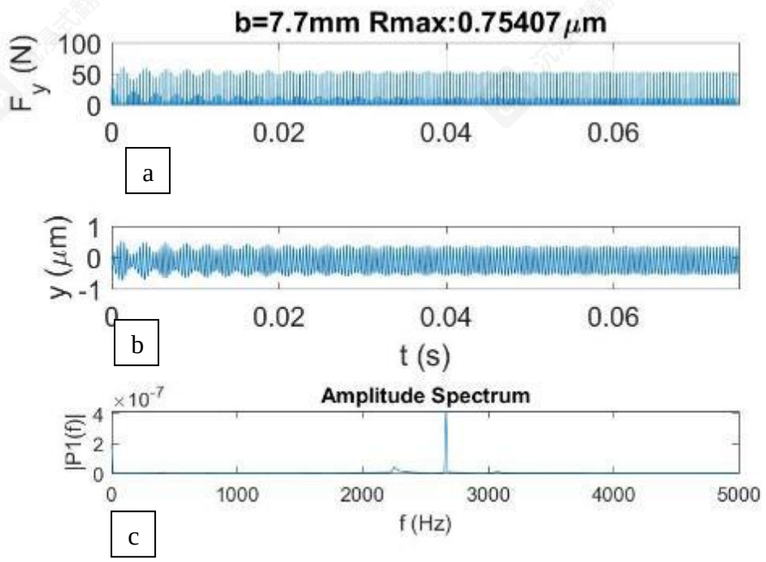



图59：40000 rpm，7.7 mm 深度，5000 mm 进给 (a) 力 (b) 刀具尖端振动 (c) 振动频谱。

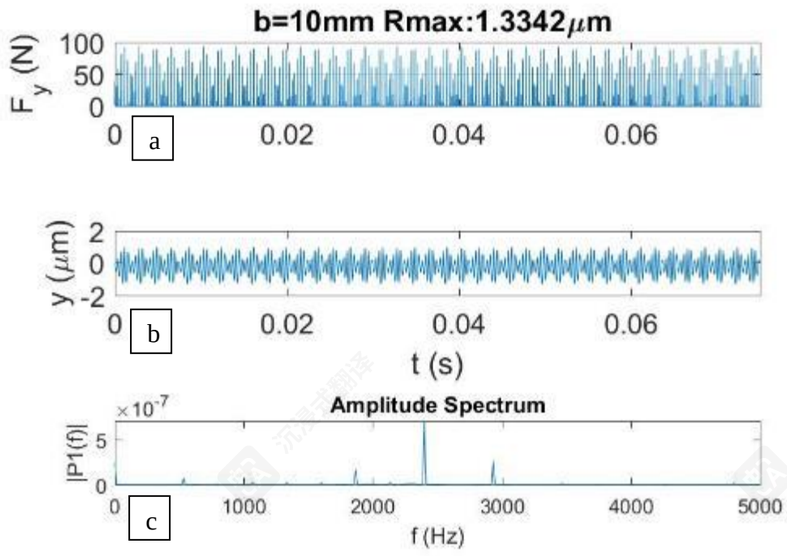



Figure 60: 40000 rpm, 10 mm depth, 5000 mm feed (a) force (b) tool tip vibration (c) vibration spectrum.

There are a few things to be careful about when determining if the process is unstable. To understand that a process is stable or unstable, the simulations should be taken at specific depths and the vibrations should be visually inspected. The distinction between stable and unstable vibrations must be detected with naked eye. The implications of the chatter reside in the repetition of the vibration “pulses” in the state of unstable case where the frequency spectrum also provides the clue. In this point an algorithm can be developed to detect the chatter automatically as the chatter occurs at most times in shapes of pulses but the pattern of the pulses may change from case to case. Therefore, a more “brute force” method is applied here. The simulations presented above cover the time including 50 rotational cycles of the grinding wheel. To be examined more closely, the following figures can be obtained in 5-cycle simulations. Figure 61 shows a stable grinding simulation at 40 000 rpm and axial depth of 5mm.



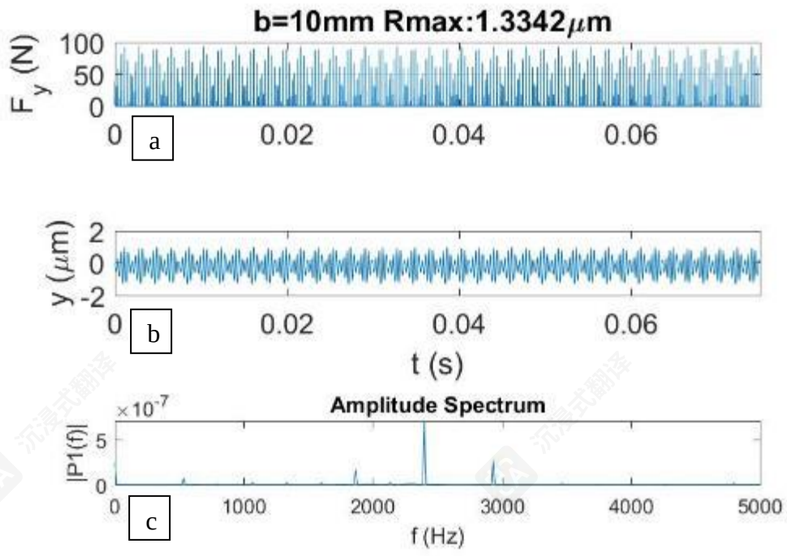



图60：40000 rpm，10 mm 深度，5000 mm 进给 (a) 力 (b) 刀具尖端振动 (c) 振动频谱。

在判断过程是否不稳定时，需要注意以下几点。要理解过程是否稳定，应在特定深度进行模拟，并直观检查振动情况。稳定与不稳定的振动区别必须用肉眼识别。颤振的影响体现在不稳定情况下振动“脉冲”的重复，而频谱图也提供了线索。在此点上，可以开发算法自动检测颤振，因为颤振通常以脉冲形式出现，但脉冲模式可能因情况而异。因此，这里采用更“蛮力”的方法。上述模拟覆盖了砂轮50个旋转周期的时间。为更仔细地检查，可在5周期模拟中获得以下图形。图61显示了40 000 rpm转速和轴向深度5mm时的稳定磨削模拟。

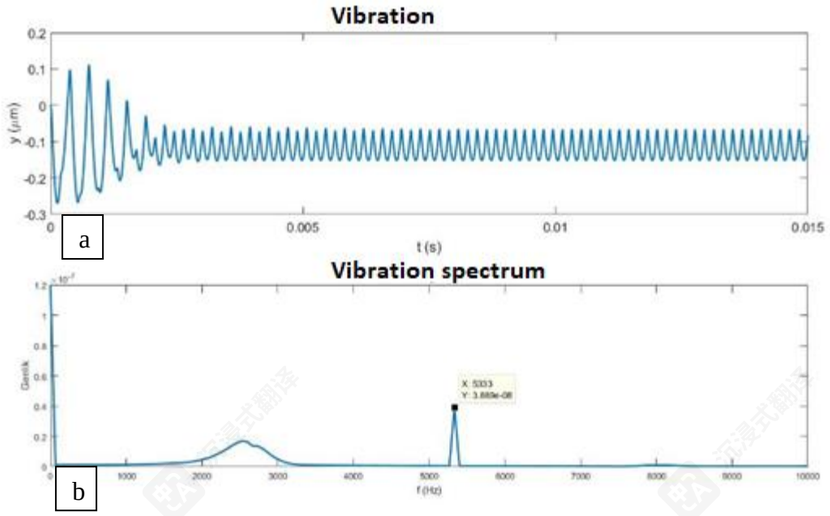



Figure 61: 40000 rpm, 5 mm depth, 5000 mm feed (a) tool tip vibration (b) vibration spectrum

Since the radial depths of cut for the grinding operations are comparably lower than other machining operations (40 microns in the example), two things come into importance. The first is the contact and the contact length of the workpiece is minimal. Therefore, the contact of the abrasive particles with the workpiece won't be a long lasting one, resulting in discrete forces rather than continuous ones. The dynamic forces are shown in Figure 62. The natural frequency of the wheel is also excited as the forces are in the form of impulses. Thus, when the grinding process is started, in the first few cycles, the wheel vibrates at higher amplitude, but then these vibrations are damped and remain under forced vibrations in a stable fashion. The frequency of the forced vibrations is the spindle speed frequency, which corresponds to 5333 Hz (666Hz spindle speed x 8 active grits).



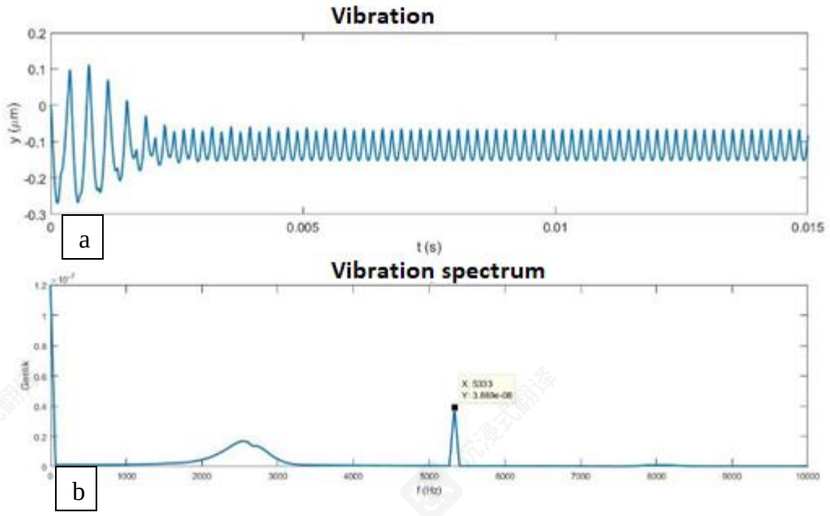



图61：40000 rpm，5 mm 深度，5000 mm 进给 (a) 刀尖振动 (b) 振动频谱

由于磨削加工的径向切深与其他加工工序相比相对较低（例如为40微米），因此有两点变得重要。第一点是接触，工件接触长度最小。因此，磨粒与工件的接触不会持续很长时间，导致产生离散力而非连续力。动态力如图62所示。由于力以脉冲形式存在，因此砂轮的固有频率也会被激发。因此，当磨削过程开始时，在最初的几个周期内，砂轮以较高振幅振动，但随后这些振动被阻尼，并以稳定的方式保持受迫振动。受迫振动的频率是主轴转速频率，对应于5333 Hz（666Hz主轴转速 x 8个有效磨粒）。

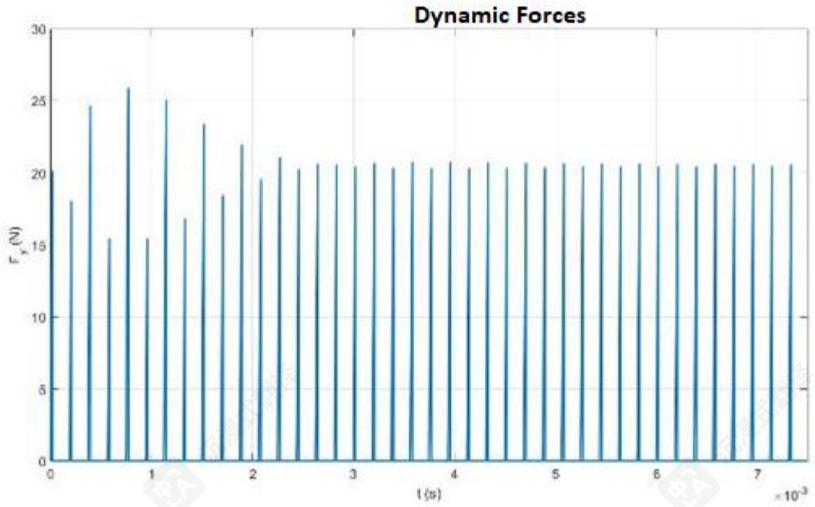



Figure 62: Stable dynamic forces, depth: 5mm, speed: 40000 rpm

The second problem caused by the low radial depth of cut in grinding is that the uncut chip thickness is very low (this can be observed by the forces that are shown in Figure 62). The process is sensitive to vibrations precisely because the difference between start angle of the grit and the exit angle of the grit is very small (can be approximated around 5 degrees). The slightest vibration can interrupt the contact of the grit with the workpiece due to the low depth, and momentarily the force is reduced to zero. Such non-linear properties are reflected in the solution of the system.

An unstable process vibration is given below (Figure 63 and Figure 64). If the spindle speed is set to 35000 rpm in the given condition, the system becomes unstable (Where the depth is 5mm and spindle speed is 35000rpm). First, a great increase in vibration amplitude is observed. This increase is much more than the average amplitude of the forces comparing to the stable case. The pulse-like pattern can be seen. The frequency of these oscillations coincides with the frequency of chatter (2333 Hz) which should locate around the natural frequency.



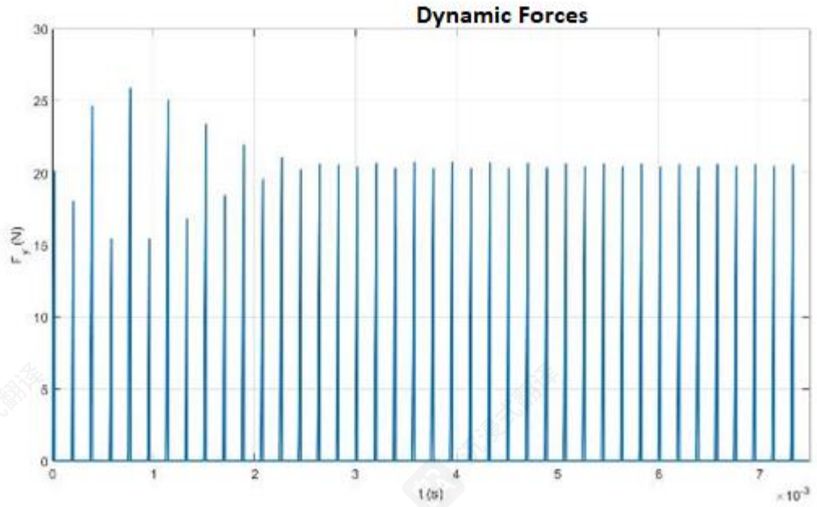



图62：稳定动态力，深度：5mm，速度：40000 rpm

磨削中低径向切深引起的第二个问题是未切除的切屑厚度非常低（这可以通过图62中所示的力观察到）。该过程对振动非常敏感，正是因为磨粒的起始角与磨粒的出口角之间的差值非常小（可以近似为5度）。由于切深较低，最轻微的振动都会由于低深度而中断磨粒与工件的接触，并且瞬间力会降至零。这种非线性特性体现在系统的解决方案中。

一个不稳定的振动过程如下（图63和图64）。在给定条件下，如果主轴转速设置为35000 rpm，系统变得不稳定（此时深度为5mm，主轴转速为35000 rpm）。首先观察到振动幅值大幅增加。这种增加远大于稳定情况下力的平均幅值。可以看到脉冲状模式。这些振荡的频率与颤振频率（2333 Hz）一致，该频率应位于固有频率附近。

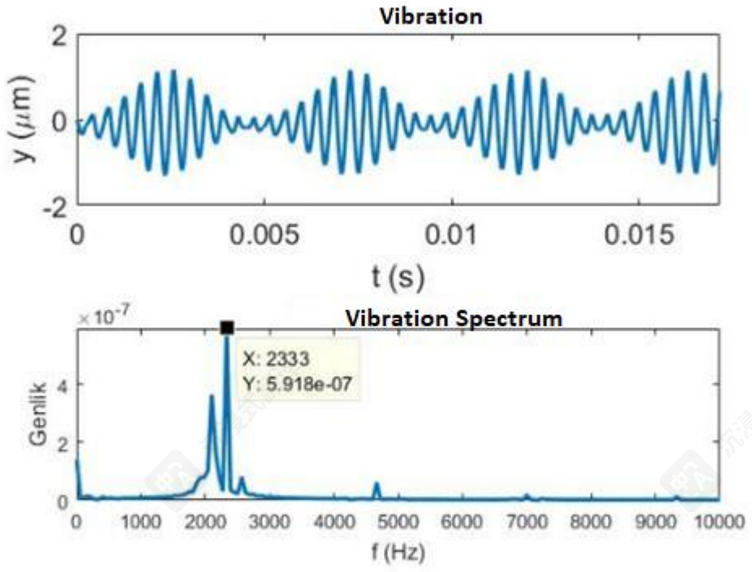



Figure 63: Unstable grinding. Depth: 5mm, Spindle speed: 35000rpm (a) tool tip vibration (b) vibration spectrum

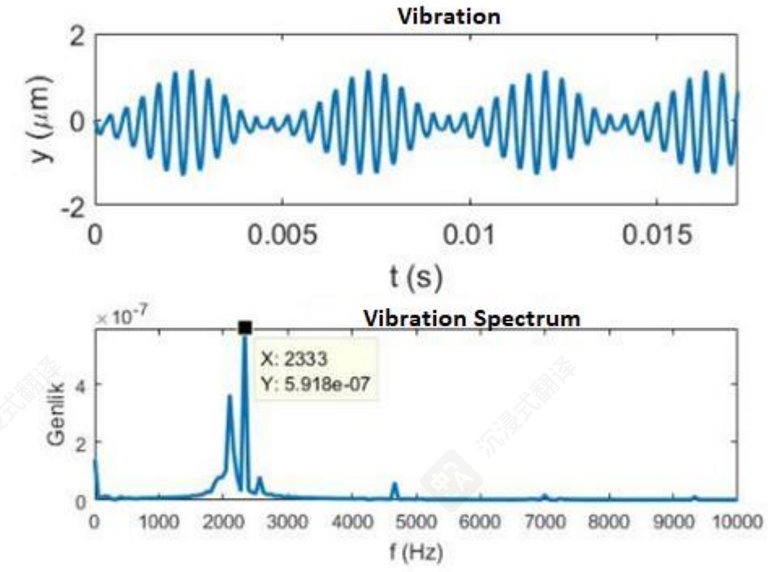



图63：不稳定磨削。深度：5mm，主轴转速：35000rpm (a) 刀具尖端振动 (b) 振动频谱

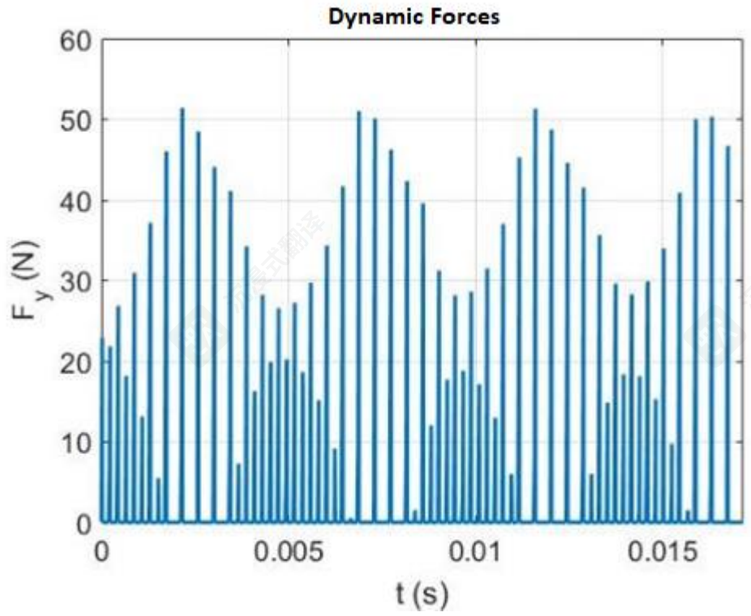



Figure 64: Unstable grinding forces. Depth: 5mm. Spindle Speed: 35000 rpm.

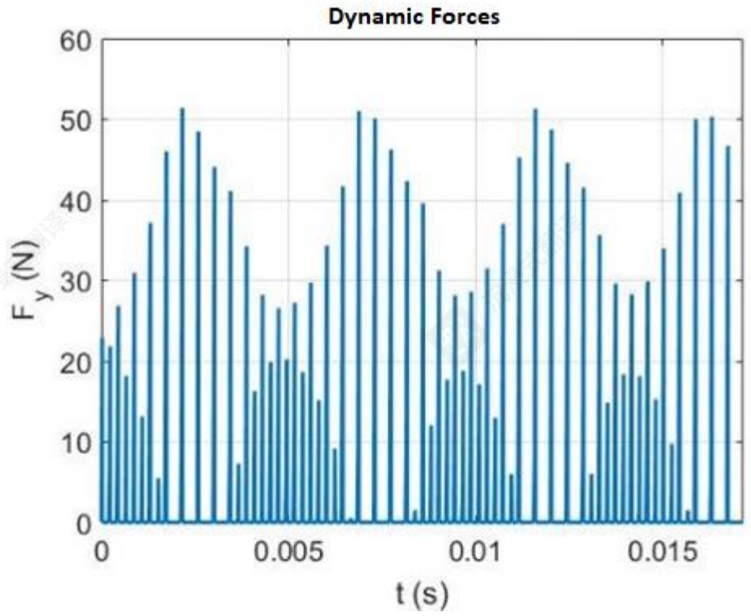



图64：不稳定磨削力。深度：5mm。主轴转速：35000 rpm。

## 5.3 Stability diagrams

The stability charts are obtained from the time domain model simulations. A series of simulations are conducted in order to identify the limits of chatter. A sample stability chart is given in Figure 65.




Figure 65: Stability diagram for 5000mm/min feed rate

In Figure 65, the given stability graph is for the small CBN wheel (B126 20mm), radial depth of 40 μm and feed rate of 5000 mm/min. The average active grit number residing in one element is 8 in this example. The area above the curve is the unstable region and the area under the curve is stable region. The stability graph resembles the ones for milling operations. The range for spindle speed is selected to be between 30000 and 40000 rpm as this range is the reasonable speed for grinding for this machine and tools setup in this research.

The stability diagrams do not usually change with feed rate for turning and milling operations. But in grinding operations, the stability diagrams alter because with a change in the feed rate, the active number of grits per element changes which affects cutting coefficients as well. The diagrams for different feed rates is given in Figure 66.



## 5.3 稳定性图

稳定性图是根据时域模型仿真获得的。进行了一系列仿真以确定颤振的极限。图65给出了一个稳定性图的示例。




图65：进给速度为5000mm/min的稳定性图

在图65中，给定的稳定性图适用于小尺寸CBN砂轮（B126 20mm）、径向深度为40 μm和进给速度为5000 mm/min的情况。在此示例中，每个元件中平均存在的有效磨粒数为8。曲线以上的区域是不稳定区域，曲线以下的区域是稳定区域。稳定性图与铣削操作的图相似。主轴转速范围选择在30000至40000 rpm之间，因为该范围是本研究中此机器和工具设置下合理的磨削速度。

稳定性图通常不会随着进给速度的变化而改变，适用于车削和铣削操作。但在磨削操作中，稳定性图会发生变化，因为进给速度的变化会导致每个元件的切削刃数量改变，从而影响切削系数。不同进给速度的图表如图66所示。

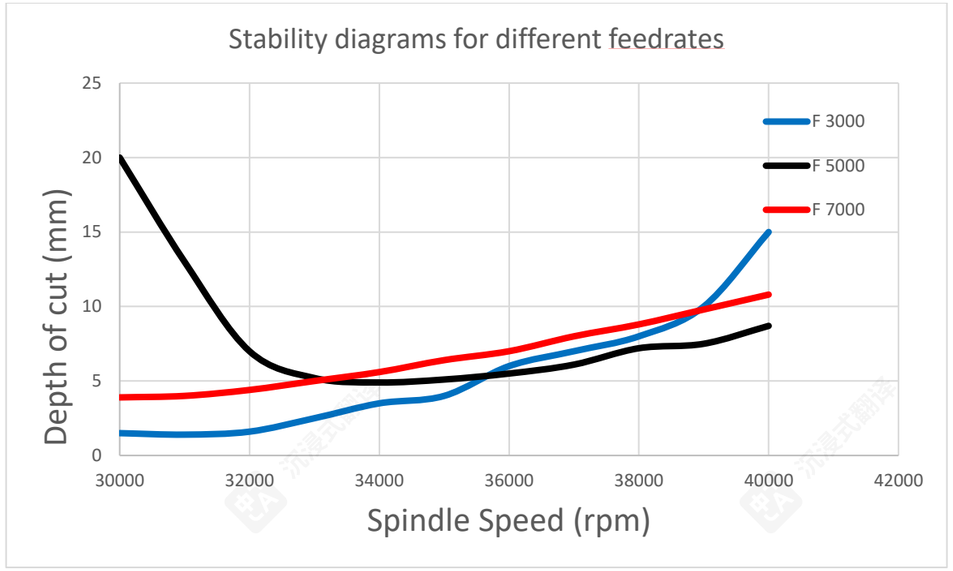



Figure 66: Stability diagrams for different feedrates

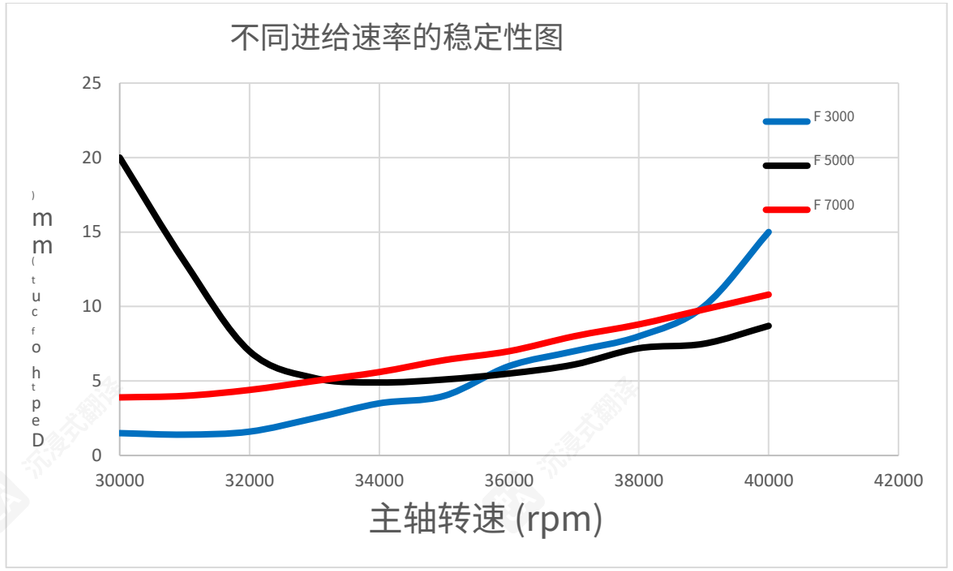



图66：不同进给速率的稳定性图

As seen from Figure 66, the stability diagrams with the CBN 20mm wheel and for the radial depth of 40 μm, could have a different profile for feed rates of 3000, 5000 and 7000 mm/ min. The respective cutting coefficients per grit and the active grit number per single element are given in Table 7. The feed rate has an indirect effect on the process stability as it can inherently change the cutting force coefficients and the number of active grits. The table shows that the increase in feed rate has an effect on increase in the active grit number. The force coefficients have respective decreases.

从图66可以看出，使用CBN 20mm砂轮和径向深度为40 μm时，进给速度为3000、5000和7000 mm/min的稳定性图可能会有不同的轮廓。相应的每颗切削刃的切削系数和每个元件的切削刃数量如表7所示。进给速度对工艺稳定性有间接影响，因为它会改变切削力系数和切削刃数量。表显示，进给速度的增加会导致切削刃数量的增加，而力系数则相应减少。

| Feed rate (mm/min) | Cutting Coefficient (GPa) | Active grit number per element |
|---|---|---|
| 3000 | 5.4 | 6 |
| 5000 | 3.5 | 8 |
| 7000 | 3.1 | 9 |

Table 7: Process parameters for different feed rate conditions

| 进给速率 (mm/min) | 切削系数 (GPa) | 每个元件的激活砂粒数 |
|---|---|---|
| 3000 | 5.4 | 6 |
| 5000 | 3.5 | 8 |
| 7000 | 3.1 | 9 |

表7：不同进给速率条件下的工艺参数

The feed rates are chosen because of the non-linear increase in the force observed during some experiments explained in the previous chapter (Section 4.4 and Figure 45). The measured force at feed 5000 mm/min has a larger value than that of the feed rates of 3000 mm/min and 7000 mm/min. Further in the chapter, the dynamic forces of feed rates of 5000mm/min and 7000 mm/min are also compared, and it is shown that the cutting condition at the feed 5000 mm/min has a marginally stable profile( Figure 51). The results obtained in the previous chapter are compatible with the stability diagrams in Figure 66. The experiments are done with the same tool and the same radial depth, at 40000 rpm. The instability is easier to see with 5000 mm/min feed case comparing to the other feed rates at the spindle speed of 40000rpm. The limits of stability are around 7, 11 and 15 millimeters for the feed rates of 5000, 7000 and 3000 mm/min respectively. This is explained by why different force profiles are seen in these feed rates. The case with 5000mm/min in section 4.4 was marginally stable with 7mm depth of cut both for dynamic force simulations and experiments and this is also predicted with the stability graph given in Figure 66. Further investigation will be presented in the experimental results.

The effect of C numbers on different tools have also great impact on process parameters as they change the active grit number and cutting force coefficients. Two wheels are compared in terms of C numbers and different feed rates for stability diagrams.  The results are given in Figure 67, with respect to the radial depth of cut limit.



进给速率的选择是因为在前一章（第4.4节和图45）中的一些实验中观察到的力呈非线性增加。进给速率为5000 mm/min时的测量力值大于3000 mm/min和7000 mm/min的进给速率。在章节的后续部分，还比较了5000mm/min和7000mm/min进给速率的动态力，结果表明，进给速率为5000 mm/min时的切削条件具有边缘稳定轮廓（图51）。前一章获得的结果与图66中的稳定性图一致。实验使用相同的刀具和相同的径向深度，转速为40000 rpm。在40000 rpm的主轴转速下，与其他进给速率相比，5000 mm/min进给情况更容易观察到不稳定性。稳定性极限分别为5000、7000和3000 mm/min进给速率的7、11和15毫米。这解释了为什么在这些进给速率下观察到不同的力分布。第4.4节中5000mm/min的情况在动态力模拟和实验中都具有边缘稳定性，切削深度为7mm，这也与图66中给出的稳定性图预测一致。后续实验结果将进行进一步研究。

C数值对不同工具的影响也很大，会改变加工参数，如影响有效砂粒数和切削力系数。在稳定性图中，比较了两个砂轮的C数值和不同的进给速率。结果如图67所示，与径向切深限制相关。

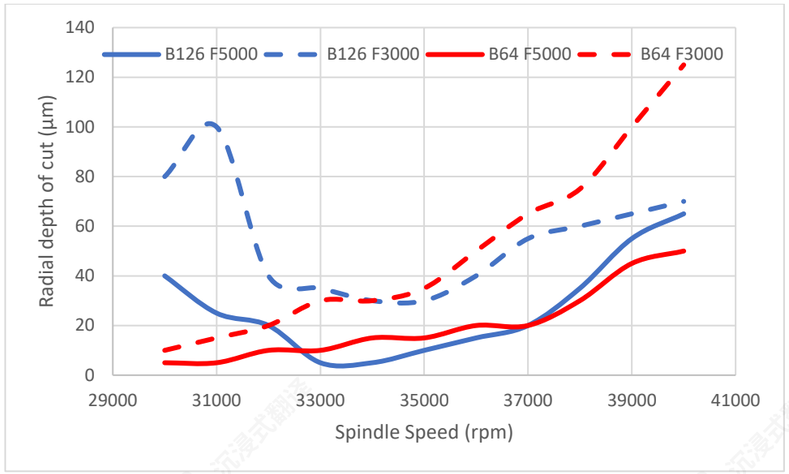



Figure 67: Stability charts with respect to radial depth of cut for different tools and feeds.

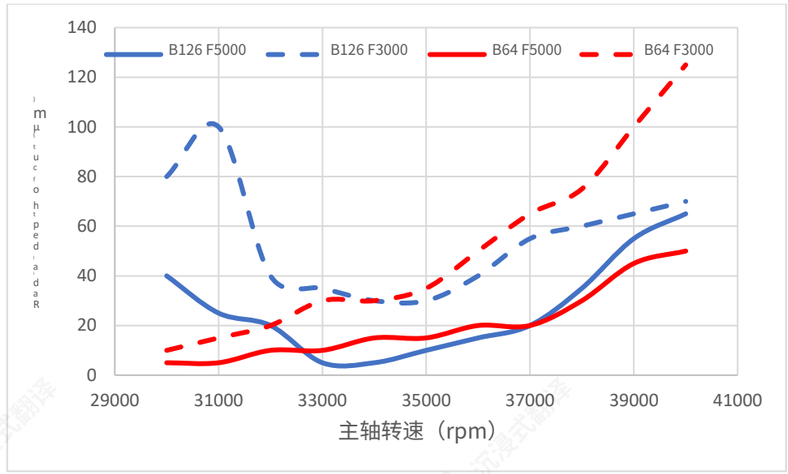



图67：不同刀具和进给下的径向切削深度稳定性图表。

| Wheel type | Feed rate (mm/min) | Cutting Coefficient (GPa) | Active grit number per element |
|---|---|---|---|
| B126 | 3000 | 5.4 | 6 |
| | 5000 | 3.5 | 8 |
| B64 | 3000 | 5 | 12 |
| | 5000 | 3.2 | 15 |

Table 8: Process parameters for different feed rate and wheel conditions

| 轮类型 | 进给速率 (mm/min) | 切削系数 (GPa) | 有效砂粒数量 每个元素 |
|---|---|---|---|
| B126 | 3000 | 5.4 | 6 |
| | 5000 | 3.5 | 8 |
| B64 | 3000 | 5 | 12 |
| | 5000 | 3.2 | 15 |

表8：不同进给速度和滚轮条件下的工艺参数

In Figure 66, the blue curves represent the B126 Electroplated CBN wheel named Wheel 1 in Figure 8 and Table 2. The axial depth of cut is fixed at 7mm. The red curve represents the B64 Electroplated CBN wheel named Wheel 2 in the same figure and the table. As it can be observed, the stability curves differ greatly just like the process parameters could change with the varying grit density (number of abrasive particles and the grinding force coefficients). The feeds of 5000mm/min offer worse stability condition as the limit is lower with the similar case of axial depth of cut limits given in Figure 66. The curves may be advantageous or disadvantageous over each other depending on the selection of the spindle speed. Its seems like if the second tool is selected with 3000 mm/min feed and 40000 rpm, the radial depth of cut limit can reach



在图66中，蓝色曲线代表B126电镀CBN砂轮，即图8和表2中的砂轮1。轴向切深固定为7毫米。红色曲线代表同一图和表格中的B64电镀CBN砂轮，即砂轮2。可以看出，稳定性曲线差异很大，就像加工参数会随着砂粒密度（磨粒数量和磨削力系数）的变化而变化一样。5000毫米/分钟的进给速率会导致稳定性条件更差，因为与图66中给出的轴向切深限制类似时，极限值更低。曲线的优劣取决于主轴转速的选择。如果选择第二工具，进给速率为3000毫米/分钟，转速为40000 rpm，那么径向切深限制可以达到

up to 120 μm which has the highest limit in the graph. Whereas if the spindle speed is chosen around 31000rpm, now it seems more advantageous to use the first wheel with feed rate of 3000mm/min in which the stability limit can reach up to 100 μm in terms of radial depth of cut.

The process parameters for the two wheels and two feed rates are given in Table 8. The cutting force coefficient in the normal direction is reduced when the feed rate is increased for both wheels. The change in cutting force coefficient and the number of active grits is the key factor to determine the stability limits.

### 5.4 Experimental results

A series of experiments were conducted to detect chatter in the tests. The same test setup in Figure 44 was used for the detection of chatter. The stability investigation was decided to be necessary because of the following reason. A large number of experiments were conducted with a variety of different process parameters and wheels in order to investigate abrasive processes. The occurrence of chatter was noted when using conventional wheels at regular speeds occasionally, but in the case of high speed grinding with small CBN tools, the occurrence of chatter was seen in a few cases. To identify the regions of instability, the following process parameters were spanned with the maximum spindle speed of 40m/s (with 20mm diameter wheel). As in the literature it is generally describe that one of the main observations on chatter was the increase in workpiece velocity [42,44,48] (workpiece velocity in cylindrical grinding corresponds to feed rate in surface grinding) the feed rates of 100 to 20000 mm/min is spanned with the radial depth of cut is varied between 10 μm to 50 μm. The workpiece width is selected to be 7mm which is the axial depth of cut. The force measurements of the tests are given in Figure 68.



高达 120 μ 米，这是图中最高的限制。而如果主轴转速选择在 31000rpm 左右，现在看来使用第一号轮，进给速度为 3000 毫米/分钟，其中径向切深稳定性极限可达到 100 μ 米，这样更有优势。

两种轮和两种进给速度的工艺参数见表 8。当进给速度增加时，两种轮的法向切削力系数都会降低。切削力系数的变化和有效磨粒数量是决定稳定性极限的关键因素。

### 5.4 实验结果

进行了一系列实验以检测测试中的颤振。检测颤振时使用了图44中相同的测试装置。稳定性调查被决定是必要的，原因如下。为了研究磨削工艺，进行了大量实验，使用了各种不同的工艺参数和砂轮。在常规砂轮和正常速度下偶尔会出现颤振，但在使用小CBN工具进行高速磨削时，颤振在少数情况下出现。为了识别不稳定性区域，以下工艺参数被覆盖，最大主轴转速为40m/s（使用直径20mm的砂轮）。正如文献中通常描述的那样，颤振的主要观察之一是工件速度
[42,44,48] 的增加（在圆柱磨削中，工件速度对应于平面磨削中的进给率），进给率在100到20000 mm/min之间变化，径向切深在 10 μm到 50 μm之间变化。工件宽度选择为7mm，这是轴向切深。测试的力测量数据如图68所示。

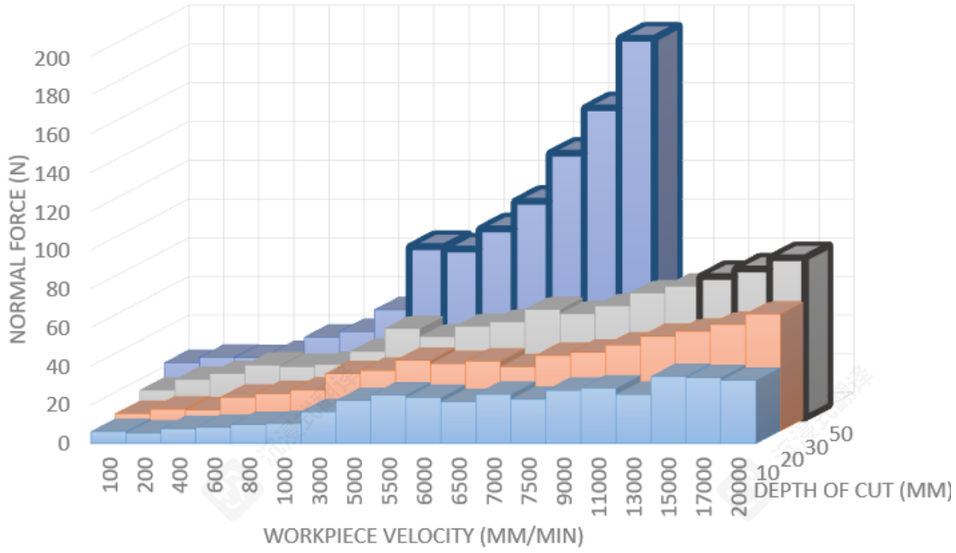



Figure 68: Force measurements of chatter investigation test

As seen from Figure 68 as the feed rate increases at some tests the force values are shown with bold figures (for example the last 3 tests of 30 μm depth of cut). These bold figures are marked because of their likelihood of being unstable. Those tests resulted in increasing force values in the dynamometer as well as a decrease in spindle speed during the grinding. The cases shown with bold figures are studied also because some frequencies other than the rotational frequency (between 1000-1500 Hz) of the spindle in the sound data were observed. This led to an idea that these cases of instability may be a sign of chatter. The force measurements of a stable and supposedly unstable cases are given in Figure 69. The shown forces are in Newton and in the normal direction. In Figure 69b, the force measurements have an increasing profile throughout the grinding process. The sound measurements of the stable and supposedly unstable cases are also shown in Figure 70.



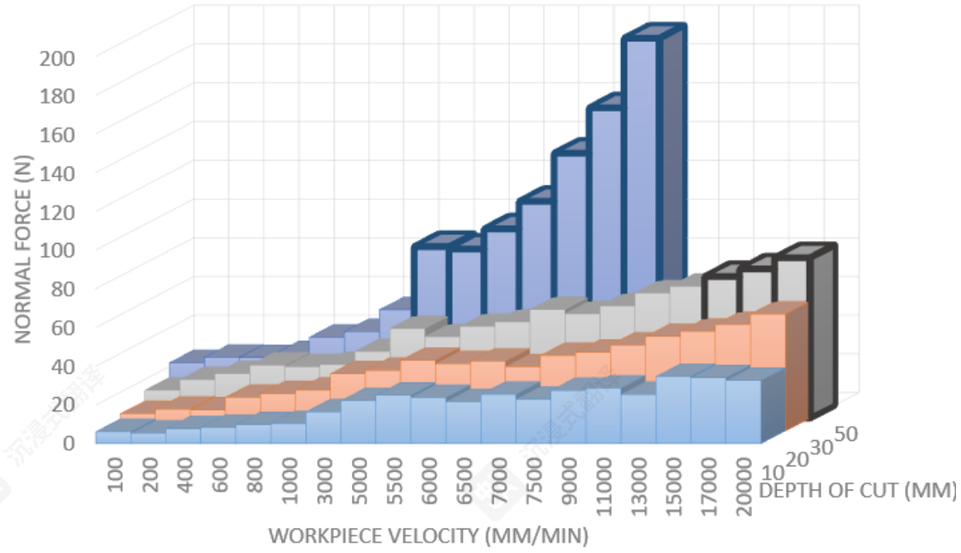



图68：颤振试验中的力测量

从图68可以看出，在一些测试中，随着进给率的增加，力值用粗体数字显示（例如，30 μm切深度的最后3个测试）。这些粗体数字被标记，因为它们可能是不稳定的。这些测试导致测力仪的力值增加，同时主轴转速在磨削过程中下降。用粗体数字显示的情况也进行了研究，因为在声音数据中观察到一些频率，除了主轴的旋转频率（1000-1500 Hz）之外。这导致了一个想法，即这些不稳定性情况可能是颤振的迹象。图69给出了稳定和假设不稳定情况的力测量数据。显示的力以牛顿为单位，方向为法向。在图69b中，力测量在整个磨削过程中呈现增加的趋势。图70也显示了稳定和假设不稳定情况的声学测量数据。

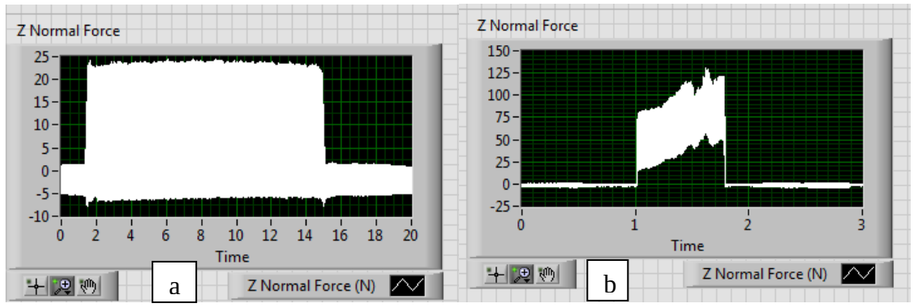



Figure 69: Comparison of the force measurement of the stable (a) and unstable (b) case

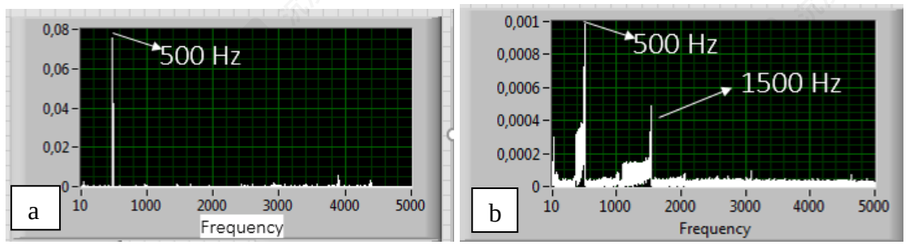
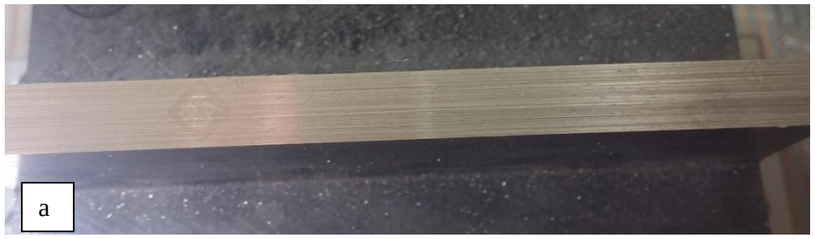



Figure 70: Comparison of the sound measurement of the stable (a) and unstable (b) case

The surface of the workpiece is shown for both cases in Figure 71. There are some marks seen on the surface and the quality has decreased dramatically in the cutting direction.





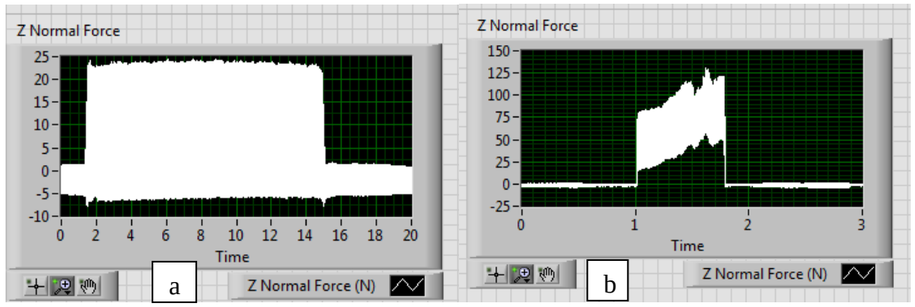




图69：稳定（a）和不稳定（b）情况下的力测量对比

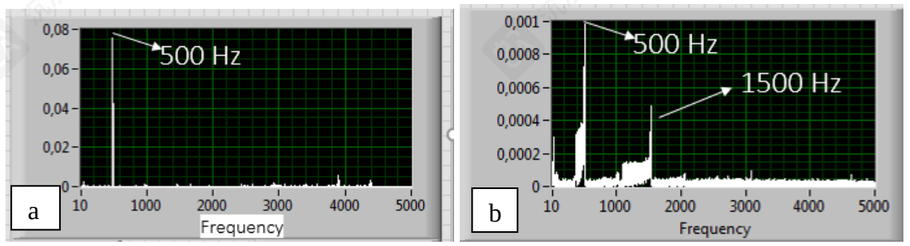
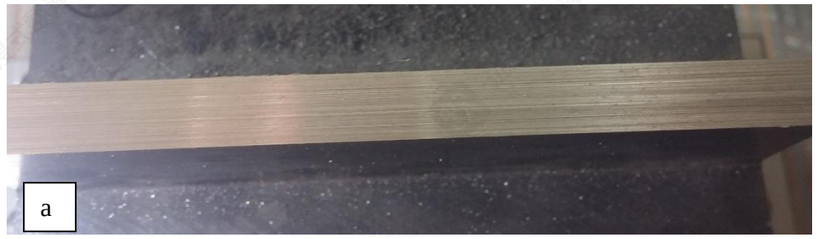



图70：稳定（a）和不稳定（b）情况下的声音测量对比

工件表面在图71中展示了两种情况。表面可见一些痕迹，且切削方向上的质量显著下降。



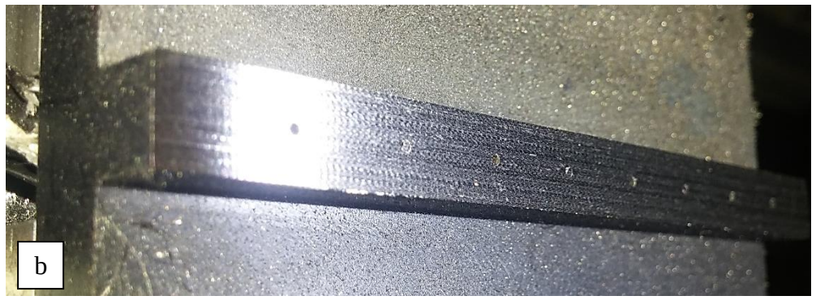



Figure 71: Comparison of surface of the workpiece of the stable (a) and unstable (b) case

The sound measurements on the unstable case show a frequency around 1500 Hz. Recall from Table 6 that the wheel used in these experiments has a natural frequency of 1900 Hz. It is known that the chatter occurs with a higher frequency than the natural frequency of the system. The output of 1500 Hz is a natural harmonic of the spindle speed frequency. Yet, a drop in the spindle speed can be easily observed in Figure 70b during the process which is a indication of insufficient power of the spindle.

To be sure, a surface roughness measurement was done on the surface of the workpiece as shown in Figure 72. The device MarSurf M400 Mobile Surface Measuring Instrument was used for the measurements. The comparison of the surfaces of the supposedly unstable and stable grinding cases is given in Figure 73.

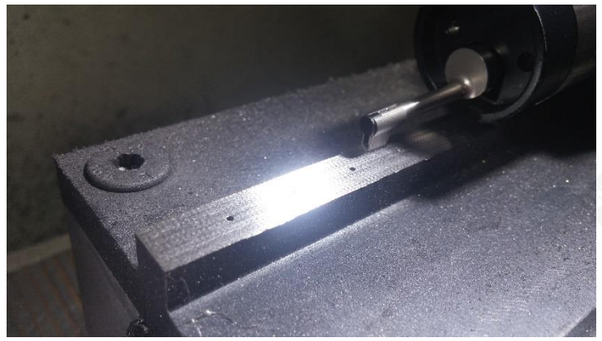


Figure 72: Surface roughness measurement of the workpiece



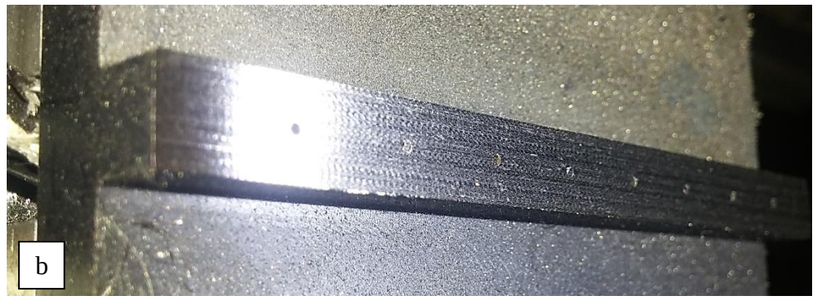



图71：稳定（a）和不稳定（b）情况下工件表面的对比

不稳定情况下的声学测量显示频率约为1500 Hz。如表6所示，这些实验中使用的砂轮具有1900 Hz的固有频率。已知颤振发生在系统固有频率以上的频率。1500 Hz的输出是主轴转速频率的倍频。然而，图70b中在加工过程中可以明显观察到主轴转速的下降，这表明主轴功率不足。

当然，如图72所示，在工件表面进行了表面粗糙度测量。测量使用了MarSurf M400移动式表面测量仪。图73给出了假设不稳定和稳定磨削情况下的表面对比。

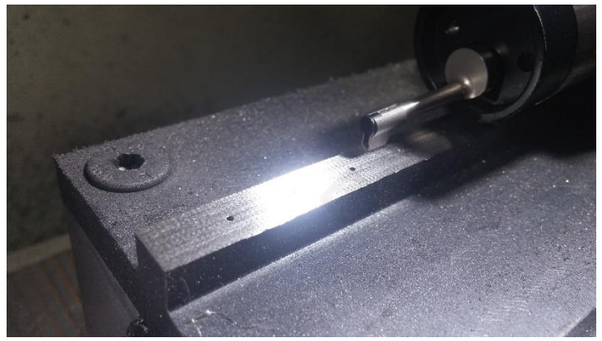


图72：工件表面粗糙度测量

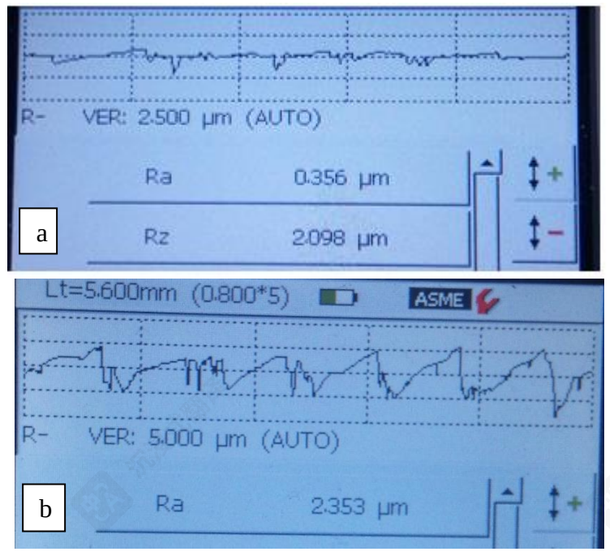



Figure 73: Comparison of surface profile of the workpiece of the stable (a) and unstable (b) case

In Figure 73, the difference between both cases can easily be observed. There are marks on the ground surface of the second case and the marks on the surface of the first case are not that visible. The surface roughness of first case is 0.356 µm and the second case is 2.353 µm (Average roughness Ra).

These signs were carefully studied in order to detect chatter during the experiments, but it is concluded that the supposedly unstable cases are not the case of chatter. A quick calculation of the wave length and the frequency of the marks in Figure 73b, show that they are just feed marks left on the surface. The amplitude of the vibrations may have increased significantly but it doesn't necessarily mean that this is a case of chatter. Therefore, it is decided that these cases are not chatter but an overloading case of the spindle.

The high-speed spindle installed on this grinding machine has the power limit of 2.5 kW at full speed. At full speed (40000rpm) with a 20mm wheel (corresponds to 40m/s), the power limit suggests that maximum power loading can be around 60 N force in the tangential direction which corresponds to around 150 N force or more in the normal direction (depends on the process parameters and the workpiece material). So, this



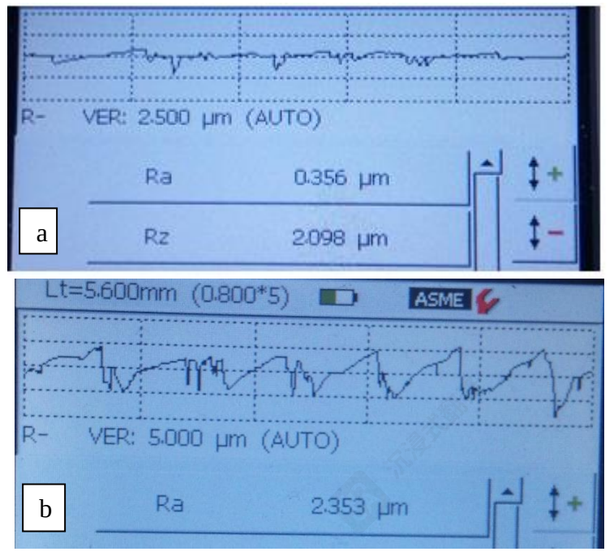



图73：稳定（a）和不稳定（b）情况下工件表面轮廓的比较

在图73中，两种情况下的差异很容易观察到。第二个案例的磨削表面有痕迹，而第一个案例表面的痕迹不太明显。第一个案例的表面粗糙度为0.356 µm，第二个案例为2.353 µm（平均粗糙度Ra）。

这些迹象在实验中得到了仔细研究，以便检测颤振，但结论是所谓的不稳定情况并非颤振。对图73b中标记的波长和频率进行快速计算表明，它们只是留在表面上的进给痕迹。振动的振幅可能显著增加，但这并不一定意味着这是颤振的情况。因此，决定这些情况并非颤振，而是主轴过载的情况。

这台磨床安装的高速主轴在最高转速时的功率限制为2.5 kW。在最高转速（40000rpm）下，使用20mm砂轮（对应40m/s），功率限制表明切向方向的最大功率负载约为60 N力，这对应于法向方向约为150 N力或更大（取决于工艺参数和工件材料）。所以，

amount of process forces will lead to a disturbance in the spindle that will change the spindle speed during grinding. In some cases, spindle nearly stopped especially at the last stages of the 50 μm depth of cut test set. At the feed rate of 7000mm/min the spindle came to a full stop and the tool was damaged. Bunks of grit pull-out was observed on the tool after the last operation, please see Figure 74.



Figure 74: Condition of the wheel after the last operation

In the light of the experimental procedure described above, it is concluded that the chatter could not be observed under the general grinding conditions that can be selected from the literature (high feed rate, low depth of cut) in this setup. The torque and power requirements of the process could not be sufficed by the high-speed spindle at full speed. The power curve of the spindle drops even more when the spindle speed is decreased. The advice of the company that provided the spindle was not to reduce the spindle speed lower than 30000 rpm. Therefore, the lower spindle speed tests also led to spindle stoppage. A possible solution to this situation is to change the spindle of the grinding machine to its regular spindle which has a normal speed and higher power capability.

Moreover, one may think that in order to see chatter, the stiffness and the damping ratio of the system can be lowered. This is possible by increasing the overhang length of the grinding wheel. For the above experimental sets, the overhang length of the wheel was selected as 5mm as it was the suggested value from the wheel manufacturer. If the overhang length is increased, apart from the amplified effect of the eccentricity of the



加工力的大小会导致主轴扰动，从而在磨削过程中改变主轴转速。在某些情况下，主轴几乎停止，尤其是在 50 μm 切削深度测试的最后阶段。在进给速度为 7000mm/min 时，主轴完全停止，刀具损坏。在最后一次操作后观察到刀具上有磨料拉出痕迹，请参见图 74。



图74：最后一次操作后的车轮状况

根据上述实验程序，可以得出结论：在此设置中，从文献中可选择的通用磨削条件（高进给率、低切削深度）下无法观察到振颤。该工艺的扭矩和功率需求无法由高速主轴在满速时满足。当主轴转速降低时，主轴的功率曲线下降得更明显。提供主轴的公司建议不要将主轴转速降低至30000 rpm以下。因此，较低主轴转速的测试也导致了主轴停机。针对这种情况的一种可能解决方案是更换磨床的主轴为其常规主轴，该主轴具有正常转速和更高的功率能力。

此外，人们可能会认为，为了看到 chatter 现象，可以降低系统的刚度和阻尼比。这可以通过增加砂轮的悬臂长度来实现。对于上述实验组，砂轮的悬臂长度被选为 5mm，因为这是砂轮制造商建议的值。如果增加悬臂长度，除了偏心效应的放大作用外，还会出现火花磨削现象。当砂轮无法去除预期的（标称）径向切削深度时，就会观察到火花磨削的必要性。在加工过程中，砂轮在法向的弹性变形会改变实际切削深度。因此，在第一次磨削后可能需要进行多次火花磨削。随着悬臂长度的增加，火花磨削会增多。此外，随着进给速度的增加（如实验中 {v1} 所示），火花磨削也会增加。增加的火花磨削会阻碍磨削的预测建模，因为磨削中的径向切削深度与铣削相比已经很低，而标称切削深度与实际切削深度之间的百分比差异会非常显著。

wheel, the spark-out phenomenon emerges. The need for spark-out is observed when the grinding wheel could not remove the intended (nominal) radial depth of cut. The elastic deformation of the wheel towards the normal direction changes the real depth of cut during the process. Therefore, many spark-out passes may be necessary after the first grinding pass. The spark-out increases as the overhang increases. as well as with increased feed rate as shown experimentally in [12]. An increased spark-out would hinder the predictive modeling in grinding because the radial depths of cut in abrasive operations are already low compared to milling operations and the difference between the nominal and real depth of cut would be significant in terms of percentage.

Hence, feed rate should be kept low in order to cope with the spark-out effect introduced by the increased overhang length and to compensate the power requirement that will be increased by higher depth of cut. A case of chatter is observed when the spindle speed is reduced from 40000 to 35000 rpm with a feed rate of 300 mm/min, radial depth of cut of 100 μm, axial depth of cut of 9mm and overhang distance of 21mm. The modal parameters of the new mounting of the wheel are given in Table 9.

| Frequency (Hz) | Damping Ratio (%) | Modal Stiffness (N/m) | Mass (kg) |
|---|---|---|---|
| 1129 | 0.534 | $1.62\times 10^6$ | 0.032 |

Table 9: Modal parameters of the 21mm overhang tool

The chatter was observed in accordance with the model predictions, both time domain and frequency domain. The comparison of the model and experiment results is given in Figure 75. The curves represent the stability limits calculated with both methods. The circles represent the experiments.



随着悬臂长度的增加，火花磨削会增多。此外，随着进给速度的增加（如实验中[12] 所示），火花磨削也会增加。增加的火花磨削会阻碍磨削的预测建模，因为磨削中的径向切削深度与铣削相比已经很低，而标称切削深度与实际切削深度之间的百分比差异会非常显著。

因此，进给速率应保持较低，以应对悬臂长度增加所引起的电弧效应，并补偿切削深度增加时功率需求的变化。当主轴转速从40000 rpm降低至35000 rpm，进给速率为300 mm/min，径向切削深度为100 μ米，轴向切削深度为9毫米，悬臂距离为21毫米时，观察到发生了颤振。新安装的轮子的模态参数见表9。

| 频率 (Hz) | 阻尼比 (%) | 模态刚度 (N/m) | 质量 (kg) |
|---|---|---|---|
| 1129 | 0.534 | $1.62\times 10^6$ | 0.032 |

表9：21mm悬臂工具的模态参数

根据模型预测观察到了颤振，包括时域和频域。模型与实验结果的对比见图75。曲线表示用两种方法计算的稳定性极限。圆圈表示实验数据。

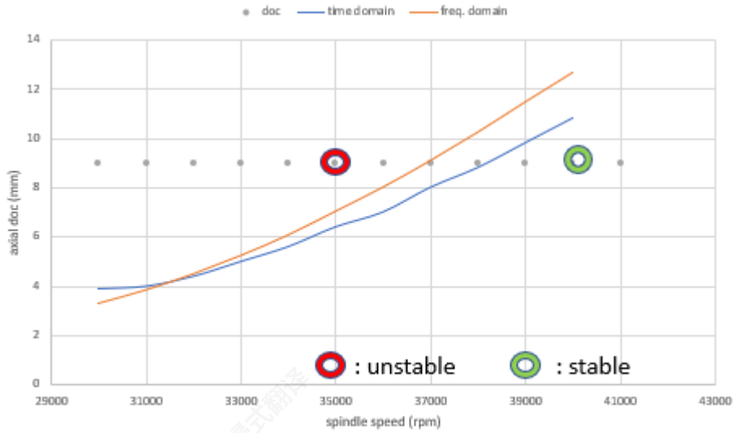



Figure 75: Model and experiment comparison

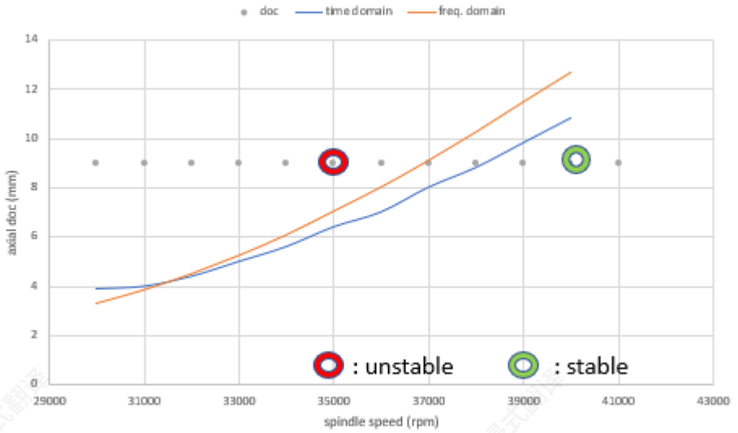



图75：模型与实验对比

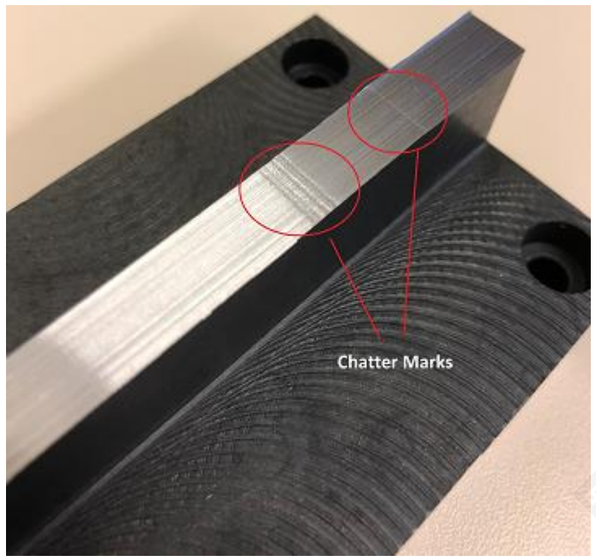



Figure 76: The chatter marks on the workpiece

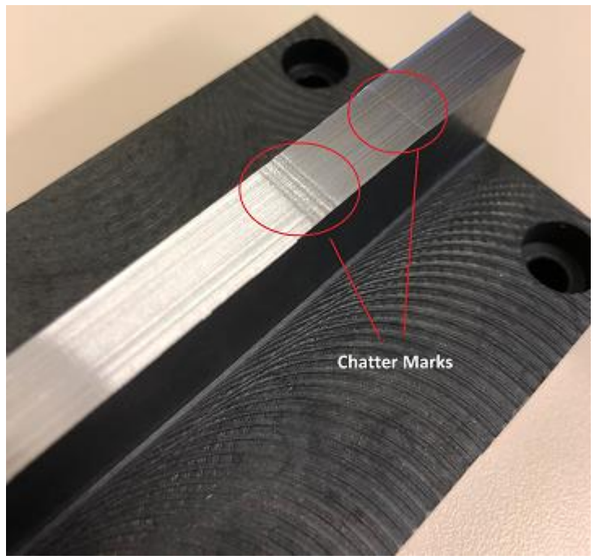



图76：工件上的振纹

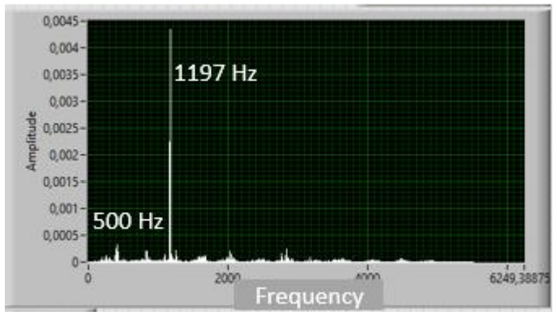



Figure 77: Detection of chatter sound

In Figure 76, the surface of the workpiece is shown. The chatter occurs at some parts of the workpiece along the cutting direction. During the experiments it was visually observed that the tool starts its unstable vibrations and then stops and become stable again. Later along the workpiece this phenomenon occurs one more time. This may be due to the process related damping factors that occur in the grinding operation. The process damping itself may have little effect, even though the spindle speed is high. Furthermore, it can be due to the fact that the active grit number constantly changes during the vibrations and where chatter occurs indicating so called active grit damping affect during the onset of chatter. In Figure 77, the sound measurement of the test is shown. The chatter vibration creates the sound frequency of 1197 Hz, which is slightly above the natural frequency of the wheel. The force measurements are shown in Figure 78. The forces are increased at the point of the chatter marks that are shown on workpiece.

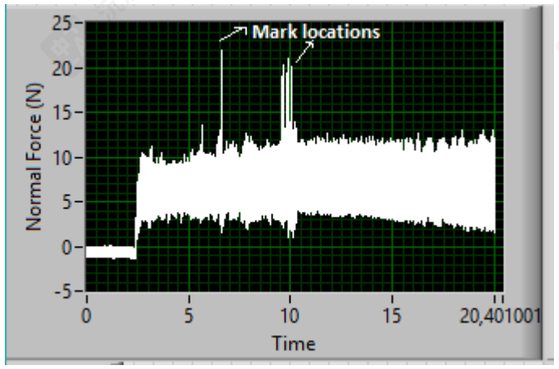



Figure 78: The force measurements of chatter case



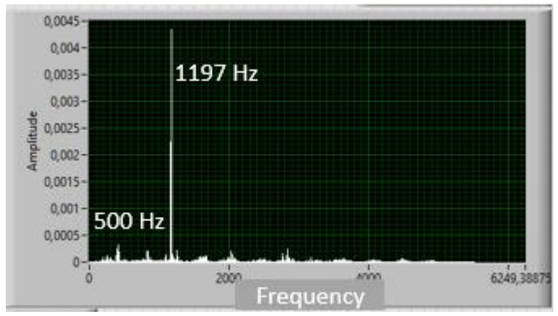



图77：颤振声检测

图76显示了工件表面。在切削方向上，工件的部分区域出现颤振。实验中观察到，刀具开始不稳定振动，然后停止并恢复稳定。在工件上，这种现象再次发生。这可能是由于磨削操作中出现的与阻尼相关的因素。尽管主轴转速很高，但过程阻尼本身可能影响不大。此外，这可能是由于在振动过程中，有效磨粒数量不断变化，而颤振发生的位置表明在颤振开始时存在所谓的有效磨粒阻尼效应。图77显示了测试的声音测量结果。颤振振动产生了1197 Hz的声音频率，略高于砂轮的固有频率。力测量结果如图78所示。在工件上显示的颤振标记处，力增加了。

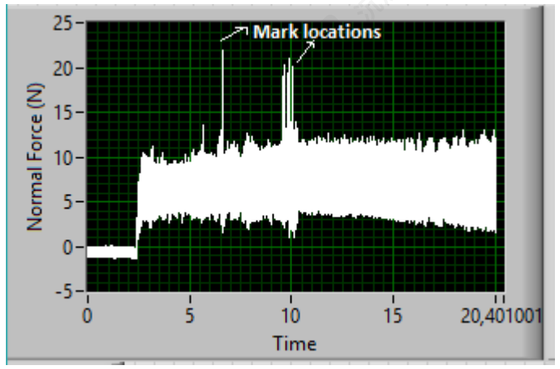



图78：颤振工况下的力测量

## 5.5 Summary

Chatter phenomena in grinding operations are investigated with the proposed dynamic-kinematic model. Two chatter methods are used, the frequency and the time domain solution models. The frequency domain solution model is an adaptation of the model that is proposed in [11] to the mounted point electroplated CBN wheels which is an adaptation of milling stability model to abrasive operations. The time domain model offers a closer perspective to grinding. Taking the stochastic nature of the process into account, the model is capable of predicting the active grit number which is dependent on the process parameters and the random delays that is produced by the random distribution of the grits on the wheel surface. The stability diagrams are obtained by calculating a series of time domain solutions. The unstable cases are detected by taking the Fast Fourier Transform of the vibrations of the tool and checking if the vibrations have the frequency of the chatter frequency that is generally slightly higher than the natural frequency of the tool. The simulations state an interesting point. In milling operations, the stability diagrams would not change if the feed rate is varied but it is not the case in grinding. It is shown that the diagrams would shift if the feed rate is changed in grinding operations as the active grit number and the cutting force coefficients are affected significantly. It was observed that the effect of feed rate can create differences in the stability conditions of the process. A series of experiments are conducted in order to scan the conventional process parameters for the grinding operations to see where the chatter would happen. But using electroplated mounted point (small) wheels and applying it to the high-speed surface grinding operation it was seen that chatter could not be observed due to the spindle power of high speed spindle and the spark-out effect within the conventional process parameter range. Instead, a creep feed grinding (CFG) process condition is observed in separate set of experiments and chatter could be observed with an increased overhang distance of the wheel (CFG conditions are generally low feed high depth of cut parameters). The chatter case could be predicted by the models.



## 5.5 总结

使用提出的动态-运动学模型研究了磨削操作中的颤振现象。使用了两种颤振方法，频率域和时间域解模型。频率域解模型是[11] 中提出的模型的改进，适用于电镀CBN砂轮，这是将铣削稳定性模型应用于磨削操作。时间域模型更接近磨削。考虑到过程的随机性，该模型能够预测有效磨粒数量，该数量取决于过程参数和由砂轮表面上磨粒随机分布产生的随机延迟。通过计算一系列时间域解获得稳定性图。通过计算刀具振动的快速傅里叶变换来检测不稳定情况，并检查振动是否具有颤振频率（通常略高于刀具的固有频率）。模拟提出了一个有趣的观点。在铣削操作中，如果进给率变化，稳定性图不会改变，但在磨削中则不是这样。结果表明，如果磨削操作中改变进给率，图会移动，因为有效磨粒数量和切削力系数会显著影响。观察到进给率的影响会导致过程稳定性条件的变化。进行了一系列实验，以扫描磨削操作的常规过程参数，以查看颤振会在哪里发生。但使用电镀安装点（小）砂轮并将其应用于高速表面磨削操作时，由于高速主轴的功率和常规过程参数范围内的火花消除效应，无法观察到颤振。相反，在另一组实验中观察到爬行进给磨削（CFG）工艺条件，并且随着砂轮悬臂长度的增加（CFG条件通常是低进给高切削深度参数），可以观察到颤振。该模型可以预测颤振情况。

## 6 SUGGESTIONS FOR THE FUTURE RESEARCH

A set of suggestions for the future studies are proposed below in the light of this research:

- The wheel modelling in grinding is crucial. If the wheel is modelled in an enhanced way, the process modelling will be more accurate. The wheel modelling is done my optical measurements however measurement of all the grits is not possible. Thus, a 3D scan measurement method can be developed to truly represent the wheel topography
- The high negative rake angle results in a dead zone during chip forming which leads to material to material contact. This has been observed during the SEM measurements of the wheel. The kinematics of the grinding operation can be enhanced by modelling the third deformation zones during grinding.
- The active grit calculation is validated through indirect methods in this thesis (with force and surface roughness prediction). A direct method to validate the active grit number can be developed.
- The time domain solution to the grinding stability problem is a versatile algorithm and involves several non-linearities that are existent in abrasive operations such as random delay and active grit number change during vibrations. The solution can be enhanced if the wear mechanisms can be integrated.
- The frequency domain solution to chatter problem is developed by using an averaging method in which the wheel is treated as a complete milling tool. The frequency domain solution can be enhanced by offering a random delay solution.
- The developed models can be implemented on cylindrical grinding operations where the chatter is studied more often in the literature.



## 6 未来研究的建议

基于本研究，以下提出了一些建议用于未来研究：

- 研磨中的砂轮建模至关重要。如果砂轮以增强的方式进行建模，过程建模将更加准确。砂轮建模是通过光学测量完成的，然而不可能测量所有磨料。因此，可以开发一种3D扫描测量方法来真实地表示砂轮形貌

- 高负前角导致芯片形成过程中出现死区，从而导致材料与材料接触。这在过程中已被观察到。

  轮子的SEM测量。通过在磨削过程中建模第三变形区，可以增强磨削运动的运动学。

- 本论文通过间接方法（包括力和表面粗糙度预测）验证了主动磨粒计算，可以直接开发一种验证主动磨粒数的方法。

- 研磨稳定性问题的时域解是一种通用的算法，并涉及研磨操作中存在的多个非线性因素，例如振动过程中的随机延迟和主动磨粒数变化。如果能够集成磨损机制，该解可以进一步增强。

- 使用一种将轮子视为完整铣削工具的均方根法，开发了颤振问题的频域解法。

  频域解法可以通过提供随机延迟解决方案来增强。

- 开发的模型可以在圆柱磨削操作中实现，而颤振在文献中通常被更频繁地研究。

# 7 DISCUSSIONS AND CONCLUSIONS

In this thesis, analytical methods including kinematic and dynamic models which represent the mechanical, kinematic and the dynamic behavior of the abrasive processes are presented. The developed models and algorithms are tested with experimental data. The identification of the wheel and grinding parameters such as the geometric properties of the abrasive grits, the grinding force coefficients and the modal parameters of the wheel are presented by proposing measurement and calibration methods. The wheel modelling is done by the grit-wise optical measurements, identifying the process related parameters of the grits and applying them to the geometric-kinematic model which makes the normal distribution of the grits on the wheel surface possible. The models are shown to be accurate to simulate the grinding operations. Specific contributions that are presented in this thesis are listed as:

- A method for grinding wheel measurement and modelling is presented. A necessity of modelling grinding wheel is crucial as a result of the stochastic nature of the grinding processes. The basic process parameters of the grinding wheel are not provided by the manufacturers and this makes modelling difficult or prone to over simplified methods. Instead, the abrasive operations are modeled with milling analogy and the abrasive grits are virtually put on the periphery of the wheel under certain assumptions.

- After the milling analogy is applied, the analogous process parameters such as number of cutting teeth (grit concentration), grit dimensions and process related angles are found by optic measurements using a light scatter microscope. 2D and 3D measurements enable identification of key parameters that will be used in the process modelling. For more accurate edge radius measurements of the CBN wheels, electron microscopy (SEM) is used.

- For the representation of the whole wheel individual grit measurements cannot be repeated for all of the grits on the wheel surface as there are tens of thousands of grits on the wheel. Instead, a normal distribution method is



# 7 讨论与结论

本论文中，介绍了包括运动学和动力学模型在内的分析方法，这些模型能够表征磨削过程的机械、运动学和动力学行为。所开发的模型和算法通过实验数据进行了测试。通过提出测量和校准方法，展示了砂轮和磨削参数（如磨料颗粒的几何特性、磨削力系数以及砂轮的模态参数）的识别。砂轮建模是通过逐颗粒的光学测量完成的，识别砂轮颗粒的过程相关参数，并将这些参数应用于几何-运动学模型，从而实现砂轮表面颗粒的分布。研究表明，这些模型能够准确模拟磨削操作。本论文中提出的具体贡献如下：

- 提出了一种砂轮测量与建模的方法。由于磨削过程具有随机性，砂轮建模的必要性至关重要。砂轮的基本工艺参数由制造商不提供，这使得建模变得困难或容易采用过度简化的方法。相反，采用铣削类比来模拟磨削操作，并在一定假设下将磨料颗粒虚拟地放置在砂轮的圆周上。

- 应用铣削类比后，通过使用光散射显微镜进行光学测量，可以找到类似工艺参数，如切削齿数（磨料浓度）、磨料颗粒尺寸和工艺相关角度。2D和3D测量能够识别将在工艺建模中使用的关键参数。为了更准确地测量CBN砂轮的边缘半径，使用电子显微镜（SEM）。

- 为了表示整个砂轮，由于砂轮上有数万个磨料颗粒，无法对砂轮表面的所有磨料颗粒进行重复的个体测量。相反，采用正态分布方法

applied on a smaller set of samples to create a virtual wheel to mimic the real wheel. This way, the properties of all of the grits can be stored in an array for modelling purposes.

应用于更小的样本集以创建虚拟轮来模拟真实轮。这样，所有磨粒的特性都可以存储在数组中用于建模目的。

- It is known in the literature that not all of the grits on the wheel surface participate in grinding process. Because of the stochastic nature of the process, some of the grits are active and some or not because during the chip removal process the active grits became dominant over the passive grits. This is mainly due to the height difference that exists among the abrasive grains. In this thesis, a simulation method for determining the number of active grits is developed namely the geometric-kinematic model.

- 文献中已知，轮面并非所有磨粒都参与磨削过程。由于过程的随机性，一些磨粒是活跃的，而另一些则不是，因为在切屑去除过程中，活跃磨粒变得比被动磨粒占主导地位。这主要是由于磨粒之间存在的高度差异造成的。在本论文中，开发了一种用于确定活跃磨粒数量的模拟方法，即几何-运动学模型。

- The geometric-kinematic model works after the individual grits are put on the periphery of the wheel. The virtual wheel is turned over the workpiece and the interactions of the workpiece and the individual grits are analyzed. The kinematic trajectories of the individual grits are printed over the trajectories of each other. The intersections of the trajectories are well studied in order to determine which grits are active or not. The grits that produce a chip thickness that is larger than the minimum chip thickness are flagged as active. The virtual wheel is then recreated with the passive grits are omitted.

- 几何-运动学模型在将单个磨粒放置在轮的圆周上后开始工作。虚拟轮旋转加工工件，并分析工件与单个磨粒的相互作用。单个磨粒的运动学轨迹被打印在其他轨迹之上。轨迹的交点得到了深入研究，以确定哪些磨粒是活跃的或非活跃的。产生大于最小切屑厚度的切屑厚度的磨粒被标记为活跃的。然后使用被动磨粒被省略的方式重新创建虚拟轮。

- The prediction of the surface roughness of the workpiece is a highly valuable prediction as the abrasive processes are usually applied for finishing operations. The surface roughness prediction can inherently be done with the developed geometric-kinematic model as the trajectories have the last surface data intersected with the workpiece. The last surfaces obtained from the trajectories will be printed for each element. Each last surface of each element along the axial direction of the wheel is 3D projected over each other. The result of the projection is the surface roughness of the workpiece surface along perpendicular to the cutting direction.

- 预测工件表面粗糙度是一项极具价值的预测，因为磨削工艺通常用于精加工操作。表面粗糙度预测可以基于开发的几何-运动学模型进行，因为轨迹包含了与工件相交的最后表面数据。从轨迹中获得的最后表面将为每个元素打印。每个元素沿轮轴方向的最后表面将在彼此之间进行3D投影。投影的结果是工件表面沿垂直于切削方向的粗糙度。



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- The force prediction using the geometric-kinematic model is based on the chip thickness calculation. Using the analogy to the milling tool run-out mechanism, radial differences between the trajectories of consecutive active grits will provide the chip thickness. This way, the chip thickness will be identified by analytic measures instead of empirical models.

- 基于几何-运动学模型预测切削力，其原理是进行切削厚度计算。通过类比铣刀跑合机制，相邻有效磨粒轨迹之间的径向差异将提供切削厚度。这样，切削厚度将通过解析方法识别，而非经验模型。

- In this thesis, two methods for predicting the grinding force is presented. Once the chip thickness can be identified through the geometric-kinematic model either mechanistic or analytic force model can be applied on the chip thickness to calculate the grinding force. Mechanistic model need calibration methods with a series of experiments whereas the analytic models are more applicable to wider range as the analytical relations between the process inputs and outputs can be established.

- 本论文提出了两种预测磨削力的方法。一旦通过几何-运动学模型识别出切削厚度，无论是机理模型还是解析模型，均可应用于切削厚度以计算磨削力。机理模型需要通过一系列实验进行标定，而解析模型更适用于更广泛的范围，因为可以建立工艺输入与输出之间的解析关系。

- The geometric-kinematic model is enhanced by a dynamic model to build the time-domain solution to the dynamics of the grinding processes. The milling analogy is also deployed for this model. The time domain solution is able tackle with the some linearities of the grinding process such as the random delay caused by the random placements of the grits over the wheel and the change in the number of active grits due to the vibrations.

- 几何-运动学模型通过动态模型进行增强，以构建磨削过程动力学的时域解。该模型也采用了铣削类比。时域解能够处理磨削过程中的某些线性特性，例如磨粒在砂轮上随机分布导致的随机延迟，以及因振动导致的有效磨粒数量变化。

- The stability diagrams are proposed by using the time-domain kinematic solution to the grinding problem and it is compared with both analytical averaging solutions and the experiments.

- 通过使用磨削问题的时域运动学解来提出稳定性图，并将其与解析平均解和实验结果进行了比较。

- The case of chatter was reported within the prediction boundaries. The chatter for surface grinding with mounted point wheels observed to happen different from the cases in milling or turning. The chatter may occur in cycles where it may start and stop several times. This is the reason of the stochastic nature of the process. A process damping effect is applied on the vibratory tool as the number of engaged grits will change if the chatter is propagated. This way, the chatter may be damped and then start again.

- 在预测边界内报告了颤振情况。带安装点轮的表面磨削颤振现象与铣削或车削中的情况不同。颤振可能以周期性方式发生，它可能开始和停止多次。这就是该过程随机性的原因。当颤振传播时，振动工具上会施加过程阻尼效应，因为参与接触的磨粒数量会改变。通过这种方式，颤振可能会被抑制，然后再次开始。


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